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AI-BASED INTELLIGENT MULTI-STAGE AND USER-CENTRIC RANSOMWARE ATTACK DETECTION

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AI-BASED INTELLIGENT MULTI-STAGE AND USER-CENTRIC RANSOMWARE ATTACK DETECTION

HENRY KABUYE

A thesis submitted in partial fulfilment of the requirements of the University of Northumbria at Newcastle for the degree of Doctor of Philosophy

Faculty of Faculty of Environment and Engineering

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Abstract

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Preface

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Declaration

I declare that the work contained in this thesis has not been submitted for any other award and that it is all my own work. I also confirm that this work fully acknowledges opinions, ideas and contributions from the work of others. The work was done in collaboration with the [collaborators].

Any ethical clearance for the research presented in this thesis has been approved. Approval has been sought and granted by the [Faculty Ethics Committee / University Ethics Committee / external committee] on [date].

I declare that the Word Count of this thesis is 83,564 words.

Name: Henry Kabuye

Signature:

Date: 9 July 2026

Chapter 1

Introduction

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Chapter 2

Literature Review

Ransomware has emerged as one of the most operationally disruptive and economically damaging categories of malicious software confronting organisations, governments, and critical infrastructure operators worldwide. Unlike earlier forms of financial malware that sought to exfiltrate credentials or conduct fraudulent transactions silently, ransomware explicitly attacks data availability: it encrypts or exfiltrates victim files and demands a ransom payment in exchange for the decryption key or the promise of non-publication Beaman, Barkworth, Akande, Hakak and Khan (2021); Greenberg (2018). The detection challenge it poses is correspondingly severe: the encryption payload must be identified and contained within a narrow window before irrecoverable damage is done, yet the early-stage behavioural and structural signals that precede encryption overlap substantially with those of legitimate software, making high-precision early detection technically demanding.

This chapter surveys the body of knowledge on which the empirical investigations in this thesis are built. Section 2.1 characterises the ransomware threat landscape — its taxonomy, attack lifecycle, and documented impact. Section 2.2 reviews machine learning approaches to malware and ransomware detection from classical methods through to contemporary deep learning architectures. Section 2.3 covers the theory and practice of uncertainty quantification in deep neural networks, with particular attention to Monte Carlo Dropout and probability calibration. Section 2.4 addresses explainable artificial

intelligence for cybersecurity, covering feature attribution, counterfactual methods, and their application to malware analysis. Section 2.5 reviews multi-agent architectures and modality fusion strategies relevant to multi-signal detection. Section 2.6 surveys neuro-symbolic AI and the operationalisation of threat-intelligence knowledge as a detection signal. Section 2.7 reviews concept drift and continual learning in the context of evolving malware populations. Section 2.8 synthesises the identified gaps and maps them to the three empirical contributions of this thesis.

2.1 The Ransomware Threat Landscape

2.1.1 Taxonomy and Attack Lifecycle

The term ransomware encompasses a heterogeneous family of malicious programs united by the common goal of extorting payment from victims in exchange for restoring access to their own data or systems Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Early academic taxonomies distinguished two primary categories: locker ransomware, which denies access to the system interface or desktop without modifying file contents, and crypto-ransomware, which selectively or exhaustively encrypts user files using asymmetric or hybrid cryptographic schemes Moussaileb, Cuppens, Lanet and Bouder (2022); Sikorski and Honig (2012). Contemporary ransomware campaigns have largely converged on the crypto variant, augmented by double-extortion tactics in which data is exfiltrated prior to encryption so that public release of sensitive materials constitutes a second leverage mechanism independent of the decryption key Beaman, Barkworth, Akande, Hakak and Khan (2021); Meland, Bayoumy and Sindre (2020).

The attack lifecycle of a ransomware campaign maps closely to the MITRE ATT&CK Enterprise framework Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). An attack typically begins with initial access via phishing, remote desktop protocol exploitation, or compromised credentials, followed by execution of a dropper or loader component. The ransomware then performs reconnaissance — enumerating network shares, user directories, and backup locations — before moving laterally to maximise the encrypted surface area. Privilege escalation and defence evasion tactics are employed to

disable endpoint protection and shadow copy services. Finally, the encryption payload is deployed and a ransom note delivered. The window between initial execution and the completion of encryption can be as short as minutes for opportunistic campaigns, making real-time or near-real-time detection essential Sgandurra, Muñoz-González, Mohsen and Lupu (2016); Hirano, Hodota and Kobayashi (2022).

A number of taxonomies have been proposed specifically to characterise the technical diversity of ransomware families. Sgandurra et al. Sgandurra, Muñoz-González, Mohsen and Lupu (2016) distinguish families by the cryptographic scheme employed (symmetric, asymmetric, or hybrid), the file targeting strategy (by extension, by directory, or exhaustive), and the command-and-control communication protocol. More recent surveys Yumlembam, Issac, Jacob and Yang (2024); Ozkan-Okay, Akin, Aslan, Kosunalp, Iliev, Stoyanov and Beloev (2024) extend these taxonomies to cover ransomware-as-a-service (RaaS) distribution models, which have substantially lowered the technical barrier to launching campaigns and accelerated the proliferation of novel variants. Polymorphic and metamorphic ransomware families that mutate their code at each deployment to evade signature-based detection represent a particularly acute challenge for static analysis systems Urooj, Al-Rimy, Zainal and Saeed (2024).

2.1.2 Economic and Operational Impact

The economic cost of ransomware is substantial and escalating. Europol's Internet Organised Crime Threat Assessment documents ransomware as the dominant cyber-extortion threat facing European organisations, with critical infrastructure sectors including healthcare, energy, and transportation identified as priority targets Europol (2020). The 2017 WannaCry campaign — estimated to have infected over 300,000 systems across 150 countries in a single day — caused disruption to the United Kingdom's National Health Service that resulted in the cancellation of nearly 20,000 outpatient appointments and cost an estimated £92 million in remediation and lost output Greenberg (2018). The 2021 Colonial Pipeline attack demonstrated that ransomware could directly impact physical infrastructure, forcing a major fuel supply pipeline operator to halt operations for nearly a week and triggering regional fuel shortages across the eastern United States Mudaliar (2024).

Beyond direct financial losses, ransomware imposes substantial indirect costs through reputational damage, regulatory fines for inadequate data protection, increased insurance premiums, and the allocation of significant analyst time to incident response and forensic investigation Beaman, Barkworth, Akande, Hakak and Khan (2021); Europol (2020). The operational asymmetry is stark: a successful ransomware campaign may be executed with modest technical resources by a RaaS affiliate, yet its remediation may require weeks of specialist effort and expenditure far exceeding the initial ransom demand. This asymmetry directly motivates the development of automated, high-throughput detection frameworks capable of operating with minimal analyst involvement under normal conditions while directing human attention precisely toward the cases in which it is most needed.

2.2 Machine Learning for Malware and Ransomware Detection

2.2.1 Classical Machine Learning Approaches

The application of machine learning to malware detection predates the deep learning era by over a decade. Early systems employed classical discriminative models — decision trees, naïve Bayes classifiers, support vector machines, and random forests — trained on handcrafted feature vectors derived from static program analysis Sarker, Kayes, Badsha, AlQahtani, Watters and Ng (2020). Static features commonly used in this period included import address table n-gram profiles, section entropy values, printable string statistics, and PE header field values; dynamic features derived from sandbox execution logs included system call sequences, registry modification patterns, and network traffic summaries Sikorski and Honig (2012).

Random forests proved particularly effective for static malware classification due to their robustness to high-dimensional sparse feature spaces and their natural support for feature importance estimation Fernando, Komninos and Chen (2020). Gradient-boosted ensembles such as XGBoost Chen and Guestrin (2016) extended this performance further on structured tabular data by capturing non-linear interactions between features through sequential tree residual fitting. Studies specifically targeting ransomware classification demonstrated that dynamic behavioural features — in particular, file system access pat-

terns and entropy measurements on written data — are strongly discriminative of ransomware activity when evaluated against legitimate software Sgandurra, Muñoz-González, Mohsen and Lupu (2016); Krishna (2020).

However, classical machine learning approaches exhibit two well-documented limitations in the ransomware context. First, they require extensive feature engineering expertise to achieve competitive performance: effective feature selection depends on domain knowledge about which program behaviours are indicative, and this knowledge must be continuously updated as adversaries adapt their tactics Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023); Gibert, Mateu and Planes (2020). Second, classical models are particularly susceptible to concept drift: as ransomware families evolve and novel variants are deployed, the feature distribution of the malicious class shifts away from the distribution seen during training, and model accuracy degrades measurably Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023); Fernando and Komninos (2024). These limitations motivate the transition to representation-learning approaches that can acquire task-relevant features automatically and be updated as the threat landscape evolves.

2.2.2 Deep Learning Approaches

Deep neural networks offer two principal advantages over classical models for malware detection. First, they can learn hierarchical feature representations directly from raw or minimally preprocessed inputs, reducing dependence on expert feature engineering Gibert, Mateu and Planes (2020). Second, architectures such as convolutional neural networks (CNNs) and recurrent networks can exploit spatial and temporal structure in the input data that is ignored by conventional flat feature vectors.

CNNs have been applied to malware classification by treating the byte stream of an executable file as a one-dimensional or two-dimensional signal and learning local pattern detectors via convolutional filters Kolosnjaji, Zarras, Webster and Eckert (2016). This approach achieves state-of-the-art accuracy on family classification benchmarks without any hand-specified features, and has been extended to ransomware-specific tasks with strong results Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022). Recurrent architec-

tures, and particularly Long Short-Term Memory (LSTM) networks, have been applied to system call sequence data to exploit the temporal ordering of API calls during execution — capturing the characteristic enumeration-then-encryption sequence that precedes ransomware payload delivery Hirano, Hodota and Kobayashi (2022); Hirano and Kobayashi (2022).

More recently, attention mechanisms and transformer-based architectures Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin (2017) have been explored for security log analysis and network intrusion detection, leveraging self-attention to model long-range dependencies in event sequences that may span hours of activity before an attack payload executes Tian, Zhang, Liu, Wang, Wu, Zhu, Zhang, Zhang, Yuan, Mao and Kim (2025); Zhang, Li, Zhang, Ye, Hu and Yan (2025). LLM-derived embeddings of log data have also been investigated as input representations for downstream classifiers Tian, Zhang, Liu, Wang, Wu, Zhu, Zhang, Zhang, Yuan, Mao and Kim (2025), though these systems are predominantly evaluated as standalone classifiers without uncertainty-aware fusion or structured triage logic.

Despite strong discriminative performance, deep learning classifiers for ransomware detection share a critical limitation: they produce overconfident, poorly calibrated posterior probability estimates, particularly for inputs near or between the training class distributions Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023); Guo, Pleiss, Sun and Weinberger (2017). A model that assigns 99% confidence to a borderline prediction provides the same output format as one making a prediction on a clear-cut case, giving the analyst no signal about when autonomous decisions are reliable and when human review should be sought. This limitation is the primary motivation for the uncertainty quantification methods reviewed in Section 2.3.

2.2.3 Feature Representations for Ransomware

A consistent finding across both classical and deep learning ransomware detection studies is that no single feature modality captures all discriminative signal, and that different modalities are most informative at different stages of the attack lifecycle.

Static or semantic features, derived from analysis of the executable without running it, provide rapid pre-execution screening but are susceptible to obfuscation: packing, encryption, and code mutation can substantially alter the static feature profile of a ransomware binary without changing its runtime behaviour Urooj, Al-Rimy, Zainal and Saeed (2024); Sikorski and Honig (2012). Studies using PE-format features — section entropy, import table statistics, and PE header fields — report high detection rates on known families but degraded performance on novel or obfuscated samples Gogineni, Derasari and Venkataramani (2022).

Dynamic or behavioural features, derived from monitoring the execution of the binary in a controlled environment, are considerably more robust to static obfuscation but introduce a detection latency: the sample must be executed, at least partially, before behavioural features can be collected Hirano, Hodota and Kobayashi (2022). System API call traces, file access patterns, network connection sequences, and registry modification logs all carry strong ransomware-indicative signal, with file I/O entropy elevation being particularly reliable as an indicator of encryption activity Sgandurra, Muñoz-González, Mohsen and Lupu (2016); Hirano, Hodota and Kobayashi (2022).

A substantial literature demonstrates that hybrid approaches that combine static and dynamic features outperform either modality alone Ferdous, Islam, Mahboubi and Islam (2024); Yumlembam, Issac, Jacob and Yang (2024). However, most hybrid approaches in the literature combine features from both modalities into a single flat vector fed to a single unified model, losing the modality-specific distributional characteristics and preventing the system from reasoning separately about confidence in the static and dynamic signals. The multi-agent architecture introduced in this thesis addresses this limitation by training independent agents on each modality and fusing their outputs at the score level, preserving modality-specific uncertainty estimates that are used to govern triage decisions.

2.3 Uncertainty Quantification in Deep Neural Networks

2.3.1 Sources and Types of Uncertainty

The uncertainty in the predictions of a deep neural network arises from two conceptually distinct sources Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023); Malinin and Gales (2018). Aleatoric uncertainty, also called data or irreducible uncertainty, reflects genuine ambiguity in the input: when a sample lies near or on the decision boundary between classes, the labelling itself is inherently uncertain and no model can resolve it without additional information. Epistemic uncertainty, also called model or reducible uncertainty, reflects the model's lack of knowledge: it arises when a sample falls outside or at the periphery of the training distribution, so the model has insufficient evidence to form a reliable estimate Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023).

In the ransomware detection context, both types of uncertainty carry operational significance. Aleatoric uncertainty indicates borderline samples that may require additional dynamic analysis or analyst review regardless of how much training data is available. Epistemic uncertainty indicates novel or out-of-distribution samples — potentially representing a new ransomware family or a benign program that has not previously been seen — where the model's prediction should be treated with scepticism Doan, Nguyen, Montague, Camtepe and Kanhere (2024). A detection framework that can distinguish between these two types of uncertainty provides more actionable intelligence than one that reports only a single scalar confidence score.

2.3.2 Monte Carlo Dropout

Gal and Ghahramani Gal and Ghahramani (2016) demonstrated that a neural network trained with dropout regularisation can be interpreted as an approximate Bayesian neural network: if dropout masks are sampled stochastically at inference time rather than fixed or averaged as in standard practice, the resulting distribution over predictions approximates the posterior predictive distribution of a Bayesian model over network weights.

This insight, known as Monte Carlo (MC) Dropout, enables epistemic uncertainty estimation from any standard dropout-regularised network without modification to the training procedure and with minimal computational overhead.

At inference time, T forward passes are performed through the network with independently sampled dropout masks, yielding T class probability estimates $\hat{p}_1, \ldots, \hat{p}_T$. The predictive mean $\bar{p} = \frac{1}{T}\sum_{t=1}^{T} \hat{p}_t$ serves as the point prediction, while the predictive standard deviation $\sigma = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(\hat{p}_t - \bar{p})^2}$ quantifies the epistemic uncertainty of that prediction. High values of σ on a given input indicate that the dropout ensemble disagrees substantially about the correct prediction, which is characteristic of out-of-distribution or borderline inputs Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023).

MC Dropout has been applied to a range of security tasks including anomaly detection, network intrusion classification, and, more recently, malware family identification Doan, Nguyen, Montague, Camtepe and Kanhere (2024). The number of Monte Carlo samples T governs the trade-off between uncertainty estimation quality and inference latency: empirical studies suggest that $T = 20$–50 passes are sufficient for stable uncertainty estimates in most classification settings, with diminishing returns beyond this range Gal and Ghahramani (2016). The method is particularly well suited to the multi-agent framework developed in this thesis, where uncertainty estimates produced by individual agents are used both for triage decisions and for inter-agent communication through the Peer Consultation Protocol.

2.3.3 Probability Calibration

A well-calibrated classifier is one whose posterior probability estimates are consistent with empirical frequencies: a batch of inputs to which the model assigns probability 0.8 should, on average, contain approximately 80% positives Guo, Pleiss, Sun and Weinberger (2017). The majority of deep neural networks trained with maximum likelihood objectives are poorly calibrated, exhibiting a systematic tendency toward overconfidence: the model assigns posterior probabilities close to 0 or 1 far more frequently than the actual frequency of correct predictions in those ranges would justify Guo, Pleiss, Sun and Weinberger

(2017); Malinin and Gales (2018). This overconfidence is exacerbated in the presence of class imbalance — a common condition in ransomware datasets — because the model is incentivised to minimise the cross-entropy loss on the dominant class at the expense of calibration on the minority class.

Guo et al. Guo, Pleiss, Sun and Weinberger (2017) demonstrated that temperature scaling, a post-hoc calibration method that divides the model's logit output by a scalar temperature parameter T_{cal} fitted on a held-out validation set, substantially reduces calibration error across a range of deep learning architectures with negligible impact on discriminative accuracy. The method is particularly attractive for multi-agent systems in which the probability outputs of independently trained agents must be combined, because it ensures that the fusion layer receives probability estimates that are comparable in their interpretive meaning across agents rather than reflecting idiosyncratic calibration errors specific to each agent's training regime.

2.4 Explainable Artificial Intelligence for Cybersecurity

2.4.1 Feature Attribution Methods

The opacity of deep neural network predictions — the so-called "black-box" property — is a significant practical barrier to the deployment of neural network-based malware detectors in operational security environments. Security analysts making containment or escalation decisions require not only a prediction but an explanation of the evidence underlying it: which features drove the ransomware classification, and how much would their values need to change before the classification reversed Lundberg and Lee (2017); Ribeiro, Singh and Guestrin (2016)? Explainable AI (XAI) methods address this requirement by associating a predicted label with an interpretable description of its drivers.

SHAP (SHapley Additive exPlanations) Lundberg and Lee (2017) grounds attribution in cooperative game theory: each feature's contribution to a prediction is defined as the Shapley value — the weighted average marginal contribution of that feature across all possible feature subsets. This attribution is guaranteed to satisfy a set of desirable axiomatic properties (efficiency, symmetry, dummy, and linearity) that make the resulting expla-

nations consistent and non-arbitrary. TreeSHAP, DeepSHAP, and KernelSHAP variants provide efficient computation for tree ensembles, deep networks, and model-agnostic settings respectively. For ransomware detection, SHAP feature importance plots can identify which PE-header fields or API call patterns are driving a detection, providing actionable intelligence for signature refinement and analyst triage.

LIME (Local Interpretable Model-agnostic Explanations) Ribeiro, Singh and Guestrin (2016) takes a complementary approach: for a specific input, LIME perturbs the input in the neighbourhood of its feature values, queries the black-box model on the perturbed samples, and fits a locally linear interpretable model to the resulting predictions. The coefficients of this local linear model constitute the explanation for that specific prediction. LIME is more computationally expensive than SHAP for repeated predictions but offers the advantage of being fully model-agnostic and producing explanations that are directly readable as weighted feature lists.

Both SHAP and LIME explanations have been applied to malware and ransomware detection tasks to validate model behaviour, identify spurious correlations in training data, and generate feature-level intelligence for analyst use Gülmez, Kakisim and Sogukpinar (2024); Ispahany, Islam, Islam and Khan (2024).

2.4.2 Counterfactual Explanations

While feature attribution methods identify which features drove a prediction, they do not directly answer the analyst's operational question: what would need to be different about this sample for it to be classified as benign? Counterfactual explanations address this question by finding the minimal perturbation of the input feature vector that would reverse the model's prediction, subject to the constraint that the perturbed input remains plausible given the known distribution of benign samples Farzadpour and Azghani (2024).

Counterfactual explanations are particularly valuable in the ransomware context because they translate model behaviour into analyst-actionable intelligence. A counterfactual that identifies two specific PE-header flag values as the minimal change needed to reclassify a suspicious executable as benign gives an analyst a targeted hypothesis to investigate; a counterfactual that requires altering 40 features by large amounts suggests that the sample

is confidently ransomware and can be contained with high precision. The distance from an input to its nearest counterfactual (typically measured in L_2 or L_0 norm) also provides an interpretable confidence proxy: inputs with close counterfactuals lie near the decision boundary and warrant greater analyst scrutiny.

2.4.3 Explainability in Malware Analysis

The application of XAI methods to malware analysis has grown substantially since 2020, driven by regulatory expectations around AI transparency and the practical demand for analyst-comprehensible outputs from automated security systems. Recent surveys Ispahany, Islam, Islam and Khan (2024); Yumlembam, Issac, Jacob and Yang (2024) identify feature attribution-based explanation as the most widely adopted approach in the malware detection literature, followed by rule extraction, attention visualisation, and counterfactual generation. XRan Gülmez, Kakisim and Sogukpinar (2024) specifically applies SHAP to explain ransomware classifications derived from API call profiles, demonstrating that attribution values correlate with known ransomware-indicative API families and can be used to cluster samples by behavioural similarity.

A consistent finding in this literature is that explanations produced by attribution methods are most useful when they are grounded in the analyst's existing knowledge framework: SHAP values over raw byte offsets are less actionable than SHAP values over named PE-format fields, even if both represent the same underlying model behaviour Lundberg and Lee (2017); Gülmez, Kakisim and Sogukpinar (2024). This observation motivates the feature-level explanation design choices in Chapter 3 of this thesis, where explanations are presented over named API call groups and PE-format categories rather than raw feature indices.

2.5 Multi-Agent Systems and Modality Fusion

2.5.1 Multi-Agent Architectures for Detection

Multi-agent systems distribute a complex detection or classification task across a collection of autonomous agents, each responsible for a subset of the problem, with coordination

mechanisms determining how individual agent outputs are combined into a system-level decision Soltani, Khajavi, Siavoshani and Jahangir (2023). In the cybersecurity domain, multi-agent approaches have been proposed for network intrusion detection, where individual agents monitor distinct network segments or protocol layers; for threat intelligence aggregation, where agents process feeds from different sources; and for adaptive response systems, where specialised agents handle different stages of the detection-containment-remediation pipeline Soltani, Khajavi, Siavoshani and Jahangir (2023); Chaithanya and Brahmananda (2022).

The case for decomposing ransomware detection into modality-specialised agents rests on a fundamental observation: the statistical properties of static PE-format features and dynamic runtime API traces are sufficiently different that a single unified model trained on a concatenation of both modalities is unlikely to learn optimally adapted representations for each Ferdous, Islam, Mahboubi and Islam (2024). Static features are high-dimensional and sparse, with strong category-level interactions; dynamic features are sequential and exhibit temporal autocorrelation within a single execution trace. Modality-specialised agents trained independently on each signal type can develop representations matched to the distributional properties of their respective inputs, with a fusion layer responsible for combining the resulting predictions Soltani, Khajavi, Siavoshani and Jahangir (2023).

2.5.2 Agent Fusion, Triage, and Escalation

The problem of combining the predictions of multiple specialised classifiers — ensemble fusion — has a substantial literature in pattern recognition and machine learning Ferdous, Islam, Mahboubi and Islam (2024). Simple fusion strategies include majority voting, averaging of posterior probabilities, and weighted averaging with weights proportional to individual agent accuracy. Learned fusion, in which a meta-learner (often a small neural network or logistic regression model) is trained on the outputs of the individual classifiers, generally outperforms fixed heuristic fusion by adapting the combination weights to the joint distribution of agent predictions on the training data Soltani, Khajavi, Siavoshani and Jahangir (2023).

An important dimension not adequately addressed in the existing fusion literature is the role of prediction uncertainty in the fusion process. Conventional fusion methods combine agent probability estimates as if they were equally reliable across all inputs; in practice, individual agents may be highly confident on inputs within their training distribution and substantially uncertain on novel or out-of-distribution inputs. A fusion scheme that ignores this variation treats a confident prediction and an uncertain prediction with equal weight, potentially allowing a highly uncertain agent to dominate the fused output in exactly those cases where its prediction is least reliable. The inverse-uncertainty-weighted Peer Consultation Protocol developed in Chapter 5 of this thesis addresses this gap directly, adjusting inter-agent fusion weights at inference time as a function of each agent's current epistemic uncertainty.

The concept of triage — routing detection outputs to different response pathways based on the system's confidence — is well established in clinical decision support literature but has received limited systematic treatment in the cybersecurity machine learning context. Farzadpour et al. Farzadpour and Azghani (2024) discuss threshold selection for security classifiers but focus exclusively on discriminative accuracy rather than on the operational trade-off between autonomous action and human escalation. Jointly optimising triage thresholds to maximise detection sensitivity while minimising unnecessary escalation — particularly using Bayesian optimisation over the combined objective — represents a methodological gap directly addressed in Chapter 4.

2.6 Knowledge-Augmented and Neuro-Symbolic Detection

2.6.1 Neuro-Symbolic Artificial Intelligence

Neuro-symbolic AI refers to architectures and frameworks that integrate neural learning components — capable of acquiring representations from raw data — with symbolic reasoning components that encode structured knowledge, logical constraints, or semantic relationships Besold et al. (2021). The motivation for such integration is well-established: neural networks generalise poorly to novel combinations of concepts not seen during training (the compositional generalisation problem), and they cannot readily incorporate do-

main knowledge expressed in symbolic form, such as logical rules, ontologies, or expert-curated decision criteria. Conversely, symbolic systems struggle with the uncertainty and noise inherent in real-world sensor data but can reason reliably over cleanly structured symbolic representations.

In the context of malware detection, the neuro-symbolic paradigm suggests augmenting purely data-driven detectors with structured representations of threat-intelligence knowledge: documented attack patterns, tool associations, MITRE ATT&CK tactic sequences, and platform-specific vulnerability profiles Besold et al. (2021); Tian, Zhang, Liu, Wang, Wu, Zhu, Zhang, Zhang, Yuan, Mao and Kim (2025). Such knowledge is already documented in practitioner-curated repositories — CVE databases, MITRE ATT&CK, threat actor profiles, DFIR investigation reports — but has not previously been operationalised as an independent agent-level signal within a multi-agent detection framework. Chapter 5 of this thesis introduces the Knowledge Agent as the component that bridges this gap, training a neural classifier on practitioner-sourced threat-intelligence features and integrating its output alongside the data-driven Semantic and Behavioural Agents.

2.6.2 Threat Intelligence as a Detection Signal

Threat intelligence — structured information about adversary capabilities, intentions, and tactics derived from observed incidents, practitioner analysis, and automated collection — constitutes a rich source of detection signal that is systematically under-exploited in the machine learning-based ransomware detection literature. The MITRE ATT&CK framework provides a standardised taxonomy of adversary techniques documented across hundreds of ransomware and advanced persistent threat campaigns, enabling quantitative characterisation of attack breadth and complexity Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Studies that have incorporated ATT&CK-derived features into detection pipelines report that tactic prevalence and tool association profiles are highly discriminative of ransomware deployments compared to benign or low-sophistication malicious software Alraizza, Algarni and Alrayzah (2025).

However, the integration of threat intelligence into automated detection systems faces two significant challenges. First, threat-intelligence data is typically expressed at a higher level

of abstraction than the raw feature vectors used in ML pipelines: bridging the gap between practitioner narratives and machine-readable feature representations requires structured annotation effort Ispahany, Islam, Islam and Khan (2024). Second, threat-intelligence sources are heterogeneous in quality and recency, and the mapping from observed tactic indicators to ground-truth maliciousness labels is inherently probabilistic. The RansKnow dataset constructed for Chapter 5 addresses these challenges by grounding features directly in the MITRE ATT&CK taxonomy and using practitioner-validated thresholds for label assignment, providing a principled knowledge-grounded training signal for the Knowledge Agent.

2.7 Concept Drift and Continual Learning

2.7.1 Concept Drift in Malware Detection

Concept drift occurs when the statistical relationship between input features and class labels changes over time, causing a classifier trained on a historical data distribution to degrade in performance on more recently collected samples Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023); Fernando and Komninos (2024). In the malware detection domain, concept drift is not a rare or exceptional event but an endemic and deliberate phenomenon: adversaries actively modify their malware to evade detection systems, introducing distributional shift into both the feature distributions of malicious samples and, through false positive reduction pressure, into the distribution of borderline benign software.

Ceschin et al. Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023) provide a systematic empirical analysis of concept drift in malware classifiers, demonstrating that accuracy degradation is measurable within months of model deployment and that models trained on older data distributions generalise poorly to current threats even when evaluated on samples from the same broad family taxonomy. Fernando et al. Fernando and Komninos (2024) propose semi-supervised adaptation mechanisms that update model representations continuously as new unlabelled samples arrive, reducing the frequency of model retraining. Both studies, however, focus on single-model classifiers without explicit

uncertainty quantification or the ability to route uncertain samples to human analysts for labelling.

2.7.2 Continual Learning and Experience Replay

Continual learning addresses the problem of updating a model's knowledge over time as new labelled examples become available, without access to the full historical training set and without catastrophically forgetting previously learned representations Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019). The catastrophic forgetting problem — the tendency of a neural network updated on new data to overwrite parameters encoding earlier knowledge — is the central challenge of continual learning: gradient updates that minimise loss on the current batch can simultaneously maximise loss on previously learned patterns if the two distributions are dissimilar Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019).

Experience replay addresses this problem by maintaining a buffer of previously observed examples and including a random sample from this buffer in each gradient update batch alongside the new examples. The mixed gradient update adjusts model parameters to simultaneously reduce loss on current and historical samples, mitigating forgetting without requiring access to the full historical dataset Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019). Reservoir sampling Vitter (1985) provides a space-efficient buffer maintenance strategy: the buffer maintains a fixed-size uniform random sample of all examples seen to date, with each new example replacing a randomly selected buffer entry with probability equal to the ratio of buffer size to examples seen. This ensures that the buffer is an unbiased representative sample of the entire historical distribution regardless of its temporal structure.

In the ransomware detection context, the Expert Feedback Loop in Chapter 5 operationalises experience replay in a specific and practically motivated way: human analyst labels on escalated cases are fed back into the system as the source of new labelled examples, and the Decision MLP (the only component updated) is trained on the mixed current-plus-replayed batch. This design ensures that analyst corrections to systematic model errors are incorporated incrementally without destabilising the learned representa-

tions of the upstream CNN agents.

2.8 Synthesis: Research Gaps and Thesis Positioning

The preceding review reveals several persistent limitations in the existing ransomware detection literature that collectively motivate the three empirical investigations in this thesis. These gaps are summarised individually before being tabulated in Table 1.

Gap 1: Absence of operational uncertainty quantification. The overwhelming majority of deep learning ransomware detectors produce scalar probability outputs without any characterisation of the reliability of those estimates on individual inputs Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023); Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022). The resulting systems are poorly calibrated and cannot distinguish between high-confidence predictions on clear-cut cases and low-confidence predictions on borderline or out-of-distribution samples. This prevents the construction of principled triage policies that balance autonomous action against human escalation based on model confidence.

Gap 2: Class-imbalance degradation without score-space augmentation. Public ransomware datasets are substantially imbalanced, with benign samples outnumbering ransomware samples in many collections. Standard augmentation methods such as SMOTE Chawla, Bowyer, Hall and Kegelmeyer (2002) operate in the raw feature space and may generate implausible synthetic samples in high-dimensional spaces. No prior work has applied score-space augmentation to the fused prediction vectors of a multi-agent system, where the dimensionality is low and the conditional distributions of agent scores are well-suited to generative modelling.

Gap 3: Single-modality or naïvely fused hybrid detection. Studies that combine static and dynamic features typically do so by concatenating feature vectors into a single input to a single unified model Ferdous, Islam, Mahboubi and Islam (2024), sacrificing modality-specific representation learning and producing a single aggregate confidence score that carries no information about which modality was driving the prediction. No prior work constructs modality-specialised agents with independent uncertainty estimates that are

fused through a learned decision component with uncertainty-aware weighting.

Gap 4: No uncertainty-driven triage with Bayesian threshold optimisation. While threshold selection for security classifiers is discussed in the literature Farzadpour and Azghani (2024), no prior work jointly optimises the detection threshold and the uncertainty threshold governing autonomous versus escalated decisions using Bayesian optimisation over a combined discriminative-operational objective function.

Gap 5: Exclusion of structured threat-intelligence from agent-level detection. The MITRE ATT&CK framework and practitioner-curated threat-intelligence sources encode substantial knowledge about ransomware deployment patterns that is not captured in measurement-derived static or dynamic features Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023); Alraizza, Algarni and Alrayzah (2025). No prior work operationalises this knowledge as an independent neural agent whose uncertainty output participates in a multi-agent triage framework alongside data-driven agents.

Gap 6: Static agent cooperation without uncertainty communication. In existing multi-agent detection systems, agents contribute their outputs to a fusion layer without any mechanism for communicating their current predictive reliability to one another Soltani, Khajavi, Siavoshani and Jahangir (2023). A high-uncertainty agent contributes the same weight to the fused prediction as a confident agent, regardless of distributional shift or feature availability.

Gap 7: No continual learning under analyst feedback in multi-agent ransomware detection. Continual learning methods have been applied to single-model malware classifiers Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023); Fernando and Komninos (2024), but no prior work incorporates an analyst-in-the-loop feedback mechanism into a multi-agent detection framework that updates the system's decision logic incrementally from escalated cases while preserving the learned representations of the upstream agent backbone networks.

Taken together, these gaps define a coherent research programme: the limitations of single-agent detection (G1, G2) motivate the single-agent uncertainty-aware baseline in Chapter 3; the limitations of single-modality and unfused hybrid detection (G3, G4) motivate

the two-agent multi-modal framework in Chapter 4; and the absence of knowledge integration, uncertainty communication, and continual learning (G5, G6, G7) motivate the three-agent extended architecture in Chapter 5. Each empirical chapter is therefore not an independent contribution but a principled step in a cumulative research programme, with each chapter's design directly addressing the limitations identified in the previous.

2.9 Summary

This chapter has reviewed the body of knowledge underpinning the three empirical investigations in this thesis. The ransomware threat landscape is characterised by growing operational and economic impact, rapid adversarial adaptation, and the deployment of multi-stage attack chains that span reconnaissance, lateral movement, and encryption payload delivery. Machine learning-based detection has progressed from classical handcrafted-feature methods through CNN and recurrent deep learning architectures, achieving strong in-distribution performance but exhibiting systematic overconfidence and limited multi-modal reasoning. Uncertainty quantification via Monte Carlo Dropout, calibration via temperature scaling, and explainability via SHAP and LIME attribution provide the technical foundations for analyst-interpretable, confidence-aware detection. Multi-agent architectures and neuro-symbolic knowledge integration offer principled pathways beyond single-model detection, and continual learning via experience replay addresses the endemic problem of concept drift in evolving malware populations. Seven specific research gaps have been identified and mapped to the empirical contributions of Chapters 3, 4, and 5, establishing the logical necessity and coherent progression of the investigations that follow.

Table 1: Identified research gaps and their correspondence to the empirical contributions of this thesis. Representative prior works that partially address each gap are listed in column 2; column 3 identifies the chapter that resolves it.

Research Gap	Representative Prior Work	Addressed In
G1: No operational uncertainty quantification in ransomware detection	Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022); Gal and Ghahramani (2016)	Chapter 3
G2: Class-imbalance augmentation without score-space GAN methods	Chawla, Bowyer, Hall and Kegelmeyer (2002); Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019)	Chapter 3, 4
G3: Single-modality or naïvely fused hybrid detection	Ferdous, Islam, Mahboubi and Islam (2024); Yumlembam, Issac, Jacob and Yang (2024)	Chapter 4
G4: No Bayesian uncertainty-threshold triage optimisation	Farzadpour and Azghani (2024); Akiba, Sano, Yanase, Ohta and Koyama (2019)	Chapter 4, 5
G5: Threat-intelligence excluded from agent-level detection	Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023); Alraizza, Algarni and Alrayzah (2025)	Chapter 5
G6: Static agent cooperation without uncertainty communication	Soltani, Khajavi, Siavoshani and Jahangir (2023); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023)	Chapter 5
G7: No analyst feedback loop in multi-agent ransomware detection	Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023); Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019)	Chapter 5

Chapter 3

Research Design

Establishing a coherent research design is foundational to any doctoral investigation. The methodological and philosophical choices made at the outset of a research programme determine the types of knowledge it produces, the criteria by which its findings are evaluated, and the degree to which its conclusions can be generalised beyond the specific experimental context in which they were obtained Saunders, Lewis and Thornhill (2019). This chapter articulates the philosophical stance and methodological framework underpinning the three empirical investigations presented in Chapters 3, 4, and 5. Each investigation builds incrementally upon its predecessor: the first establishes a single-agent uncertainty-aware detection baseline Kabuye, Issac, Yumlembam and Neera (2025); the second extends this to a two-agent multi-modal framework with uncertainty-driven triage Kabuye, Issac and Neera (2026); and the third introduces a knowledge-grounded third agent with inter-agent consultation and continual learning. All three studies share the same philosophical orientation, experimental strategy, and evaluation framework, which are described here to avoid repetition and to demonstrate the internal consistency of the research programme as a whole.

3.1 Research Position

The philosophical orientation of a research programme is most systematically articulated through the Research Onion framework proposed by Saunders, Lewis and Thornhill Saun-

ders, Lewis and Thornhill (2019). Working from the outermost philosophical layer inward through research approach, strategy, time horizon, and data collection and analysis techniques, the framework ensures that each methodological choice is coherent with the philosophical commitments from which it derives. Figure 1 illustrates the Research Onion and the position adopted within each layer for this thesis. The three innermost layers — strategy, time horizon, and techniques — are addressed in Section 3.2 and Section 3.3; the three outermost philosophical layers are addressed below.

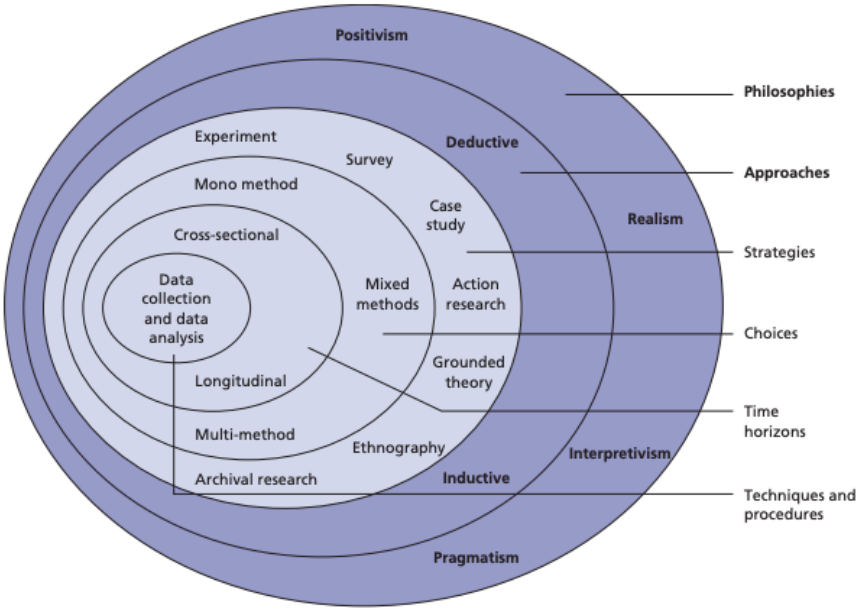


Figure 1: The Research Onion (adapted from Saunders, Lewis and Thornhill (2019)), illustrating the nested layers of philosophical and methodological choice. This thesis occupies a post-positivist realist position, adopts a deductive approach, employs an experimental mono-method quantitative strategy, and applies a cross-sectional time horizon with a simulated longitudinal component in Chapter 5.

3.1.1 Ontology

Ontology concerns the nature of reality and what entities exist in the world independently of an observer's perceptions of them. The ontological position adopted in this thesis is post-positivist realism: the view that ransomware constitutes an objective, observer-independent phenomenon with measurable behavioural and structural signatures. A ransomware binary either encrypts user data or it does not; it either contacts command-and-control infrastructure or it does not; it either exhibits the MITRE ATT&CK tactic patterns documented by practitioner communities or it does not. These are facts about

the world, not social constructions, and they are in principle discoverable through careful measurement Saunders, Lewis and Thornhill (2019).

This realist stance does not, however, imply that all knowledge about ransomware is complete or certain. The thesis departs from naïve positivism in acknowledging that measurements are inherently noisy, datasets are finite, and the classification boundary between malicious and benign software is probabilistic in marginal cases. Novel ransomware families can exhibit structural and behavioural signatures that overlap substantially with legitimate software, particularly during the reconnaissance and staging phases of an attack that precede the encryption payload Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023); Beaman, Barkworth, Akande, Hakak and Khan (2021). The precise boundary of the malicious class is therefore epistemically uncertain even if its existence is ontologically real.

This observation directly motivates the epistemic uncertainty quantification that is central to each empirical chapter in this thesis. Rather than merely acknowledging measurement uncertainty as a limitation to be minimised, the research operationalises it as an executive control variable embedded directly in the detection pipeline's triage logic. The position is consistent with what Saunders et al. Saunders, Lewis and Thornhill (2019) term critical realism: an ontological stance that accepts the independent existence of a real world but holds that knowledge of it is always mediated by observation, theory, and the instruments of measurement. In applied security research, this position has practical consequences: a model that assigns overconfident posterior probabilities to borderline cases provides less actionable intelligence to an analyst than one that correctly identifies and communicates the limits of its own knowledge Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023); Doan, Nguyen, Montague, Camtepe and Kanhere (2024).

3.1.2 Epistemology

Epistemology concerns how knowledge is acquired, validated, and distinguished from belief or conjecture. The epistemological stance adopted here is broadly empiricist and quantitative: knowledge about ransomware detection is generated by designing controlled

experiments, collecting labelled data from authoritative sources, training models on that data, and evaluating their predictions against held-out ground truth. The primary epistemological commitment is to falsifiability: every empirical claim in this thesis is, in principle, refutable by experiment. The proposition that an uncertainty-aware triage policy reduces false containment events compared to a threshold-only policy, for example, is directly testable by comparing the two policies' decisions against ground-truth labels on a held-out evaluation partition.

The thesis operates within a post-positivist paradigm rather than a strictly positivist one. Strict positivism holds that scientific knowledge is derived from direct, value-free observation and that measurement produces certain knowledge about the world. Post-positivism acknowledges that all observations are theory-laden, that all measurement is imperfect, and that probabilistic rather than deterministic claims are appropriate when working with complex natural phenomena Saunders, Lewis and Thornhill (2019). This is directly reflected in the methodology: each model is evaluated not only by point-estimate accuracy metrics but also by calibration diagnostics that assess the reliability of probability estimates across the full confidence distribution Guo, Pleiss, Sun and Weinberger (2017), and by escalation-rate analyses that quantify the operational cost of residual uncertainty.

Ground truth for this research is established through expert-consensus mechanisms. The RanSAP and RDset benchmarks provide labels derived from multi-scanner analysis and controlled dynamic sandbox execution; the RansKnow dataset provides labels derived from practitioner-validated thresholds on ATT&CK tactic prevalence. These are accepted proxies for ground truth rather than perfect knowledge of it — consistent with the post-positivist acknowledgement that all empirical knowledge is provisional and subject to revision as better data and methods become available.

3.1.3 Axiology

Axiology concerns the role of values in the design and conduct of research, including both the values of the investigator and the values embedded in the choice of research problem. The axiological position adopted in this thesis is broadly objectivist: the investigator aims to maintain value-neutrality in data collection, model training, and the reporting of eval-

uation results, disclosing performance honestly regardless of whether findings support or contradict prior hypotheses. All evaluation metrics — including cases where the proposed frameworks fall short of stated targets or produce results that require qualification — are reported in full in the empirical chapters.

However, the choice of research problem is itself a value-laden act. This thesis is motivated by the measurable harm that ransomware causes to public-sector organisations, critical infrastructure, and individual users Europol (2020); Beaman, Barkworth, Akande, Hakak and Khan (2021). The research is therefore oriented toward the development of defensive detection capabilities rather than toward understanding ransomware for its own sake or producing knowledge that could assist adversarial actors. This orientation does not compromise methodological objectivity but it does delimit the scope of the investigation: techniques that might yield detection insights but require active deployment of malicious payloads in non-sandboxed environments, reverse engineering of proprietary production systems, or collection of personal data from affected individuals fall outside the scope of this research.

All datasets used are either publicly available research benchmarks or researcher-constructed from open-source public materials. No personally identifiable information was collected or processed at any stage of the investigation. These commitments are addressed in greater detail in Section 3.4.

3.2 Research Methodology

Having established the philosophical position of the research, this section addresses the three inner layers of the Research Onion: the research approach, the research strategy, and the time horizon. Together these choices define the logical structure of each empirical investigation and ensure that the methods employed in Chapters 3–5 are coherent with the post-positivist realist stance described above.

3.2.1 Research Approach

The research approach adopted throughout this thesis is deductive. A deductive approach begins with general theory and established prior knowledge, derives specific testable hypotheses from that theory, and then subjects those hypotheses to empirical evaluation Saunders, Lewis and Thornhill (2019). The general theoretical premises of this thesis are drawn from the established literature on machine learning-based malware detection Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023); Gibert, Mateu and Planes (2020), uncertainty quantification in deep neural networks Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023), neuro-symbolic artificial intelligence Besold et al. (2021), and multi-agent detection systems Soltani, Khajavi, Siavoshani and Jahangir (2023). From these premises, the thesis derives and tests a progression of three hypotheses:

1. That integrating data augmentation, Monte Carlo Dropout uncertainty quantification, and post-hoc explainability within a single-agent CNN framework yields detection accuracy and analyst interpretability superior to baseline classifiers on the RanSAP and RDset benchmarks.

2. That decomposing detection across modality-specialised agents fused through a learned decision layer, with an uncertainty-driven triage policy governing the balance between autonomous containment and human escalation, yields superior operational control compared to a single unified model under both safety-optimal and low-escalation operating conditions.

3. That augmenting the two-agent framework with a knowledge-grounded third agent trained on threat-intelligence data, a Peer Consultation Protocol enabling inter-agent uncertainty communication, and an Expert Feedback Loop for continual learning yields robust detection performance under distributional shift introduced by novel ransomware family variants.

Each hypothesis is tested in a separate empirical chapter using controlled experiments on the corresponding datasets, with performance assessed against the evaluation framework

described in Section 3.3.3.

3.2.2 Research Strategy

The research strategy employed is experimental. Controlled experiments on benchmark datasets with pre-defined and fixed training, validation, and test partitions allow direct, reproducible comparison between the proposed frameworks and established baseline approaches Saunders, Lewis and Thornhill (2019). Each empirical chapter follows a consistent experimental protocol: the full dataset is partitioned before any model training; hyperparameter selection is performed exclusively on the validation partition through Bayesian optimisation Akiba, Sano, Yanase, Ohta and Koyama (2019); and all final performance figures are computed on the held-out test partition only, which is never used for model selection or threshold tuning.

The experimental strategy supports two complementary forms of analysis. Confirmatory analysis tests the specific performance hypotheses stated above using pre-specified evaluation metrics. Exploratory analysis examines uncertainty distributions, calibration curves, counterfactual explanations, and the effects of the peer consultation mechanism in order to characterise model behaviour and generate insights that motivate the extensions presented in subsequent chapters. Ablation studies are employed in each chapter to isolate the contribution of individual framework components — data augmentation, MC Dropout, the triage policy thresholds, peer consultation, and continual learning — from the performance of the system as a whole.

A mono-method quantitative methodological choice is adopted throughout. All data are numeric, all hypotheses are stated in quantitative terms, and all findings are expressed as measurable performance metrics. Mixed-methods or qualitative approaches are not employed, as the research questions are amenable to precise quantitative formulation and the available datasets provide sufficient quantitative evidence to answer them.

3.2.3 Time Horizon

The primary time horizon of the research is cross-sectional: all three empirical investigations evaluate their respective frameworks on static, fixed datasets at a single point in

time, reflecting the standard practice in machine learning benchmarking research Saunders, Lewis and Thornhill (2019). Cross-sectional evaluation on held-out test sets provides the most direct and reproducible measure of a model's generalisation ability, and is the standard adopted by the security machine learning community.

The third empirical study (Chapter 5) introduces a controlled longitudinal element through the Expert Feedback Loop mechanism, which models the behaviour of a detection framework receiving labelled analyst feedback over successive deployment cycles and updating its decision logic incrementally through experience replay Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019). This element is evaluated through a simulated sequential replay protocol on the RansKnow dataset rather than through live deployment, preserving the experimental control of a cross-sectional design while demonstrating the system's capacity for temporal adaptation under concept drift. The distinction between this simulated longitudinal evaluation and genuine longitudinal study in a production environment is discussed in Chapter 5.

3.3 Research Methods and Techniques

The preceding sections established the philosophical and strategic orientation of the research. This section describes the specific datasets, model architectures, and evaluation procedures through which the three research hypotheses are pursued.

3.3.1 Datasets and Data Sources

Three datasets provide the empirical basis for the investigations in this thesis. Each captures a distinct dimension of ransomware activity, reflecting the thesis's core argument that ransomware exhibits multiple observable signatures most effectively detected by specialised models trained on different data modalities. The key characteristics of all three datasets are summarised in Table 2.

RanSAP (Ransomware System API behaviour dataset) Hirano, Hodota and Kobayashi (2022) captures dynamic runtime behaviour through system API call sequences recorded during controlled sandbox execution of ransomware samples and benign applications. The

dataset records 30-second time-window API traces across multiple ransomware families, providing a temporal signal that reflects the characteristic I/O access patterns, file enumeration sequences, and encryption activity exhibited during a live ransomware attack. In this thesis, RanSAP serves as the primary dataset for training the Behavioural Agent in Chapters 4 and 5, and also contributes to the multi-class classification task in Chapter 3.

RDset provides PE-format static analysis features derived from portable executable file headers, including section entropy profiles, import address table contents, and PE flag configurations. These features encode the semantic structure of executable files without requiring dynamic execution, enabling rapid pre-execution screening of samples at the point of entry. The Semantic Agent in Chapters 4 and 5 is trained on the RDset, and Chapter 3 employs these features in the CNN-based binary classifier.

RansKnow is a researcher-constructed threat-intelligence dataset comprising 307 samples sourced from publicly available YouTube security briefings, conference presentations, and practitioner walkthroughs. Each sample is represented by 23 features spanning ten MITRE ATT&CK tactic occurrence indicators, eight ransomware-associated tool count features, three platform-specificity signals, and two aggregate totals. Binary labels are derived from the operational threshold condition (Tactic_Total $>$ 10) OR (Tool_Total $>$ 2), which reflects practitioner consensus on the minimum tactical breadth and tooling complexity that distinguishes a documented ransomware deployment from benign or marginal activity. RansKnow is used in Chapter 5 to train the Knowledge Agent.

Table 2: Summary characteristics of the three datasets used in this thesis.

Dataset	Modality	Primary Use	Chapter
RanSAP	Runtime API traces	Behavioural Agent training	3, 4, 5
RDset	PE / static file features	Semantic Agent training	3, 4, 5
RansKnow	Threat-intelligence	Knowledge Agent training	5

3.3.2 Machine Learning Architecture and Training

The primary classification architecture employed across all empirical chapters is a Convolutional Neural Network (CNN) adapted for structured tabular feature representations.

CNNs have been demonstrated to extract effective local feature interactions from security data Kolosnjaji, Zarras, Webster and Eckert (2016); Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022), and their convolutional structure provides a natural basis for applying Monte Carlo Dropout uncertainty estimation without modification to the standard training procedure Gal and Ghahramani (2016). Each CNN backbone consists of stacked convolutional blocks followed by fully connected layers; specific architectural hyperparameters are reported within each empirical chapter.

Uncertainty quantification is implemented via Monte Carlo (MC) Dropout throughout all three studies. At inference time, dropout masks are sampled T times per input, producing a distribution over class probabilities from which the mean prediction $\bar{p}$ and predictive standard deviation σ are computed. This approach yields calibrated epistemic uncertainty estimates that reflect genuine uncertainty about the decision boundary, particularly on inputs near or between the training classes Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023). The value of T is tuned per study to balance uncertainty quality against inference latency, with values of $T = 30$ used in Chapter 3 and Chapter 4, and an adaptive mechanism ($T_{\min} = 5$, $T_{\max} = 50$) employed by the Knowledge Agent in Chapter 5.

Score-space data augmentation via Conditional Tabular GAN (CTGAN) Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019) is applied in Chapters 4 and 5 to augment the training distribution of the Decision MLP on the fused agent score vectors. CTGAN learns the conditional distribution of features given class labels through an adversarial training procedure and generates high-fidelity synthetic tabular samples that preserve feature correlations without overfitting to the artefacts of a finite dataset. This is a departure from the raw-feature augmentation employed in Chapter 3 and is justified by the substantially lower dimensionality of the score space (four dimensions in Chapter 4, six dimensions in Chapter 5) compared to the original feature representations.

Explainability is provided through SHAP (SHapley Additive exPlanations) feature attribution values Lundberg and Lee (2017) and, in Chapter 4, through a novel counterfactual explanation protocol that identifies the minimal feature perturbation necessary to reverse a ransomware prediction. These mechanisms address the practical analyst requirement

for interpretable explanations that accompany automated detection decisions and can be acted upon without specialist knowledge of the underlying model architecture.

Temperature calibration is applied to all agent posterior probabilities following Guo, Pleiss, Sun and Weinberger (2017) before fusion, ensuring that each agent's confidence score reflects a reliable frequency-of-correctness estimate rather than an uncalibrated logit output. The calibration temperature is fitted by minimising the negative log-likelihood on a held-out validation set and is fixed thereafter.

Continual learning in Chapter 5 employs the Expert Feedback Loop, which updates the Decision MLP weights after each labelled feedback batch through reservoir-sampled experience replay Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019), preventing catastrophic forgetting of previously seen distributions while incorporating new information from analyst corrections. Only the Decision MLP (approximately 4,000 parameters) is updated; all agent backbone CNNs remain frozen throughout the feedback process.

3.3.3 Evaluation Framework

Model performance is quantified using a complementary set of metrics chosen to characterise different aspects of operational utility. AUC (Area Under the ROC Curve) provides a threshold-independent measure of discriminative ability. Balanced accuracy (the arithmetic mean of per-class recall) accounts for class imbalance without requiring threshold selection and is the primary metric used for Bayesian hyperparameter optimisation. Precision, recall, and F1-score characterise the precision–recall trade-off at the selected operating threshold.

Calibration quality is assessed via the Expected Calibration Error (ECE) and reliability diagrams following the temperature scaling protocol of Guo et al. Guo, Pleiss, Sun and Weinberger (2017), which decompose the gap between model confidence and empirical accuracy across equal-width confidence bins.

The operational performance of the uncertainty-aware triage policy in Chapters 4 and 5 is assessed through two complementary metrics: the escalation rate (the fraction of test-set predictions forwarded to human review) and the containment precision (the fraction

of auto-contained predictions that correspond to genuine ransomware). Together these metrics characterise the practical utility of the framework for a human analyst who must act on the system's decisions under time pressure.

Hyperparameter selection in all three studies employs Bayesian optimisation with Tree-structured Parzen Estimator (TPE) sampling as implemented in the Optuna library Akiba, Sano, Yanase, Ohta and Koyama (2019), which builds probabilistic surrogate models of the objective function and efficiently directs the search toward regions of the hyperparameter space likely to yield high-performing configurations.

3.4 Ethical Considerations

The research presented in this thesis raises a number of ethical considerations that were addressed proactively at the design stage.

Data ethics and privacy. All datasets used in this thesis are either publicly available research benchmarks or researcher-constructed from openly accessible public materials. The RanSAP and RDset benchmarks are distributed by their originators for academic use without restriction on non-commercial research. The RansKnow dataset was constructed exclusively from publicly available YouTube videos, conference proceedings, practitioner blog posts, and security briefing transcripts; no private communications, organisational records, or personally identifiable information were collected or processed at any stage. The research does not interact with any production systems, operational networks, or live user data.

Malware handling. The thesis does not require direct interaction with live malware payloads or command-and-control infrastructure. The RanSAP dataset provides pre-collected API execution traces from controlled sandbox environments operated by the dataset's originators; this thesis uses these traces as static feature matrices without re-execution of any malicious code. The RDset provides pre-extracted PE structural features; no executable binaries are loaded or analysed dynamically within this research. All experiments were conducted on isolated research computing infrastructure with no external network connectivity during model training or evaluation.

No human subjects. The research involves no human participants, surveys, interviews, user studies, or any collection of personal data. No ethical approval for human-subjects research was required or sought. The research does not profile any individual and does not produce outputs that could be used to identify, monitor, or discriminate against any person.

Defensive orientation and responsible disclosure. The multi-agent framework described in this thesis is a prototype research system evaluated on benchmark datasets. It is designed exclusively to detect ransomware activity and produce analyst-interpretable explanations, and it does not produce techniques, tools, or knowledge that would provide operational advantages to adversarial actors beyond what is already available in the public security research literature. Any future deployment of these methods in an operational security environment would require independent security auditing, institutional oversight, and appropriate governance to ensure that automated containment decisions are exercised responsibly, with suitable human review mechanisms in place for escalated or uncertain cases.

3.5 Thesis Chapter Structure

The following provides a concise orientation to the chapters of this thesis and their relationship to the research objectives stated above.

Chapter 1: Introduction provides the motivating context for the research, documents the scale and impact of ransomware attacks on modern organisations and critical infrastructure Europol (2020); Beaman, Barkworth, Akande, Hakak and Khan (2021), identifies the principal limitations of existing detection approaches with respect to uncertainty, interpretability, and multi-modal reasoning, and states the specific research objectives and contributions of the thesis.

Chapter 2: Research Design (this chapter) establishes the philosophical position, methodological framework, dataset characteristics, evaluation approach, and ethical commitments that underpin all three empirical investigations, ensuring that their findings are interpreted within a coherent epistemological context.

Chapter 3: Explainable and Uncertainty-Aware AI-based Ransomware Detection presents the first empirical study, published as a journal article in IEEE Access Kabuye, Issac, Yumlembam and Neera (2025). This chapter develops a single-agent CNN framework that integrates MC Dropout uncertainty quantification, CTGAN-based class-imbalance augmentation, and SHAP/LIME post-hoc explainability. The study establishes the performance baseline upon which the subsequent multi-agent architectures build, and demonstrates that operationalising uncertainty alongside accuracy-based metrics provides a richer characterisation of model behaviour than accuracy metrics alone.

Chapter 4: Agentic SABRE — An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection presents the second empirical study (accepted for publication in Expert Systems with Applications) Kabuye, Issac and Neera (2026). This chapter decomposes ransomware detection into two modality-specialised agents (Semantic and Behavioural) fused through a learned Decision Agent. The uncertainty-aware triage policy is introduced as the primary operational contribution, enabling the system to balance autonomous containment against human escalation through learnable threshold parameters optimised with Bayesian TPE.

Chapter 5: Extended Agentic SABRE — Knowledge-Augmented Three-Agent Ransomware Detection with Peer Consultation and Continual Learning presents the third empirical study (currently under review at IEEE Transactions on Information Forensics and Security). This chapter introduces the Knowledge Agent trained on the RansKnow threat-intelligence dataset, the Peer Consultation Protocol through which agents communicate epistemic uncertainty before fusion, and the Expert Feedback Loop for analyst-guided continual learning. Together these three extensions address the limitations of purely data-driven detection and bring the framework closer to the neuro-symbolic ideal of combining learned statistical representations with structured domain knowledge.

3.6 Summary

This chapter has established the philosophical and methodological foundations of the research programme. A post-positivist realist ontology justifies the treatment of ransomware behaviour as objectively measurable while acknowledging the inherent probabilistic uncer-

tainty of measurements near the decision boundary. An empiricist epistemology grounds knowledge claims in controlled, reproducible experiments evaluated against authoritative ground truth. A deductive research approach derives testable hypotheses from the established literature and tests them through mono-method quantitative experiments under a cross-sectional (with limited longitudinal) time horizon. Three purpose-selected datasets covering runtime API behaviour, static PE-format features, and practitioner-curated threat intelligence provide the empirical material; CNNs with MC Dropout uncertainty quantification, CTGAN score-space augmentation, and SHAP/LIME explainability constitute the core technical framework; and the evaluation combines discriminative accuracy, calibration quality, and operational triage metrics to assess each system against both technical performance and practical deployment criteria. The following three chapters present the empirical investigations conducted within this framework, building from the single-agent baseline through the two-agent triage system to the full three-agent knowledge-augmented architecture.

Chapter 4

Explainable and Uncertainty Aware AI-based Ransomware Detection

4.1 Introduction

Ransomware attacks have emerged as one of the most serious cybersecurity threats, leading to major operational, financial, and reputational damage to organisations and individuals worldwide Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Over the past few years, these attacks have grown in complexity and scale, with predicted damages projected to exceed $265 billion by 2031 due to the lucrative nature of ransom payments Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). According to the NCC (National Computing Centre) Group Cyber Threat Intelligence Monthly Reports NCC Group (n.d.), there has been a steady increase in ransomware incidents, as illustrated in Figure 2. Unlike traditional malware such as viruses, backdoors, and trojan horses, ransomware encrypts critical data or locks systems, demanding ransom payments (often in cryptocurrency) to restore access Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Its growing sophistication and financial impact call for adaptive detection methods that go beyond conventional approaches. Figure 1 presents a time series plot that shows the global monthly trends in ransomware attacks. The data comes from NCC Group's Cyber Threat Intelligence reports. The graph highlights the steady increase in ransomware incidents over time,

demonstrating that ransomware remains a growing global threat. This visual serves as motivation for why advanced detection methods are necessary, especially ones that can adapt to evolving ransomware behaviours.

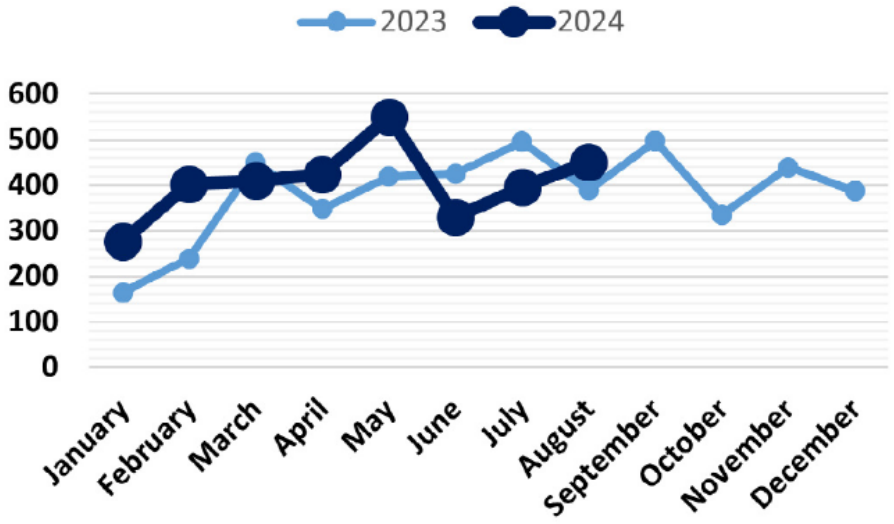


Figure 2: Global Monthly Ransomware report NCC Group (n.d.)

The financial implications of ransomware can be substantial. Ransom payments vary from a few hundred to millions of dollars. However, recovery expenses, such as data restoration, system rebuilding, and incident response, often exceed the ransom Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Operational disruptions, including downtime and productivity losses, frequently result in significant financial losses for affected organisations Conti, Dehghantanha, Franke and Watson (2018). For instance, on June 19, 2024, a cyberattack on CDK Global triggered widespread software outages in car dealerships nationwide Mudaliar (2024). The company reportedly paid $25 million to restore operations, highlighting the magnitude of the financial consequences posed by ransomware.

Existing detection systems largely rely on signature-based methodologies, which match malware samples to predefined patterns. However, such systems struggle against rapidly evolving ransomware and zero-day attacks Yadron (2014); Beaman, Barkworth, Akande, Hakak and Khan (2021); Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). They also demand continuous updates and specialised expertise Yadron (2014); Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023), un-

derscoring the need for methods that can adapt to unknown ransomware threats in real time Yumlembam, Issac, Jacob and Yang (2024).

Artificial Intelligence (AI), particularly machine learning (ML) and deep learning (DL), offers dynamic and adaptive capabilities for detecting and classifying emerging ransomware variants Sikorski and Honig (2012); Beaman, Barkworth, Akande, Hakak and Khan (2021). However, AI-based detection systems often lack clarity in how decisions are made, and they may struggle to manage predictive uncertainties Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023). These limitations can erode confidence in AI-driven models, particularly in high-stakes environments where understanding the rationale behind classifications is essential. Recent studies Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023) emphasise the importance of reliable uncertainty estimation in cybersecurity, where overconfident predictions can lead to high risks.

This paper addresses these issues by introducing a holistic framework that combines advanced data augmentation, explainable AI (XAI), and uncertainty quantification. Specifically, the approach integrates SMOTE and a conditional TabGAN (CTGAN), Monte Carlo dropout for uncertainty, and SHAP (SHapley Additive exPlanations) to enhance model interpretability. Using both the RanSAP Hirano, Hodota and Kobayashi (2022) dataset and Mathur's Mathur (2020) dataset (named "RDset" in this study), the framework achieves high F1-scores of up to 0.9978 and maintains a low false positive rate (0.0024), underscoring the effectiveness of combining diverse datasets and modern AI techniques for robust ransomware detection.

In summary, this paper makes the following contributions:

- Comprehensive Detection Framework: Proposes a unified approach integrating data augmentation, uncertainty quantification, and explainability to enhance ransomware detection.
- Improved Generalization: Demonstrates how blending RanSAP and RDset datasets improves adaptability across diverse ransomware families.

- Uncertainty Aware: Utilizes Monte Carlo dropout to quantify prediction uncertainties, improving management of false positives and false negatives in production settings.
- Explainable AI for Transparency: Employs SHAP to make model decisions more understandable and instil greater trust among cybersecurity professionals.
- Empirical Evaluation and Analysis: Provides comprehensive evaluations of Random Forest, Support Vector Machine (SVM), KNN, and CNN, illustrating the framework's robustness.

The rest of this paper is structured as follows: Section 4.2 examines prior work in ransomware detection and explainable AI, Section 4.3 describes the proposed framework, Section 4.4 details our experimental setup and findings, and Section 4.5 discusses key insights, implications, and future research directions. Finally, Section 4.6 concludes the paper.

4.2 Literature Review

Ransomware is classified into various types, each exhibiting distinct attack vectors and features Beaman, Barkworth, Akande, Hakak and Khan (2021); Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Scareware deceives users through fake malware infection notifications, prompting them to pay for unnecessary services. Doxware, or leakware, threatens to publicly expose stolen data unless a ransom is paid. Lockers restrict device access entirely, rendering the system unusable until payment is made. Encryptors, the most prevalent type, encrypt data and then demand a ransom for the decryption key Beaman, Barkworth, Akande, Hakak and Khan (2021). Over time, these variants have evolved from simple to highly sophisticated encryption techniques, such as double extortion, where attackers both encrypt and exfiltrate data, threatening to release it publicly if unpaid Europol (2020), and fileless ransomware, which leverages legitimate applications and processes like PowerShell to encrypt data in memory Krishna (2020). This evolution has led to the Ransomware-as-a-Service (RaaS) model, where criminals lease ransomware tools and services to affiliates for shared profits Meland, Bayoumy and

Sindre (2020).

High-profile attacks on critical infrastructure and healthcare institutions underscore the seriousness of the threat. For instance, in 2017, WannaCry affected roughly 200,000 systems in 150 countries, exploiting a Windows OS vulnerability Huang, Siegel and Madnick (2018). Similarly, NotPetya, initially disguised as ransomware, caused widespread damage among multinational firms Greenberg (2018). In response, Lee et al. Lee, Lee, Park, Kim and Lee (2023) proposed hiding files with certain extensions and creating link files to raise the attackers' operational costs. This camouflage approach aims to protect critical files by complicating efforts to classify and locate valuable data.

Due to the rising frequency of ransomware, organisations have adopted stronger defensive measures. Regular data backups, staff awareness programs for phishing and social engineering, and advanced detection techniques are all necessary strategies Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Static and dynamic analyses remain widely used methods for detecting malware Beaman, Barkworth, Akande, Hakak and Khan (2021). Static analysis evaluates features in the binary file before execution, whereas dynamic analysis executes suspicious files in a controlled environment, capturing behavioural patterns for detection. Hybrid analysis merges both static and dynamic approaches to expand malware detection coverage Kolosnjaji, Zarras, Webster and Eckert (2016).

Governments and regulatory bodies have enacted stringent data protection legislation and reporting requirements to curb the impact of ransomware. For example, under the General Data Protection Regulation (GDPR), organisations must report data breaches within 72 hours European Commission (2020). Additionally, more businesses now seek cyber insurance to offset potential ransomware costs Mott, Turner, Nurse, MacColl, Sullivan, Cartwright and Cartwright (2023). However, the rising cyber insurance premiums reflect the increasing frequency of attacks Mott, Turner, Nurse, MacColl, Sullivan, Cartwright and Cartwright (2023).

Similarly, AI-based techniques have become integral to ransomware detection and classification Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023);

Beaman, Barkworth, Akande, Hakak and Khan (2021). Such systems rely heavily on machine learning algorithms that analyse large volumes of data for malicious patterns, often more efficiently than manual rule creation Fernando, Komninos and Chen (2020); Gibert, Mateu and Planes (2020); Ferdous, Islam, Mahboubi and Islam (2024). Yet, AI models still face challenges including optimisation issues, data drift, classification issues, data imbalances, and model uncertainty.

Farzadpour and Azghani Farzadpour and Azghani (2024) proposed an adaptive thresholding technique for fingerprint forgery detection, demonstrating that advanced feature extraction and classification methods can boost system robustness. Though focused on biometrics, this approach holds potential for ransomware detection, particularly where data might be incomplete or noisy.

Building on adversarial resilience, Yumlembam et al. Yumlembam, Issac, Jacob and Yang (2024) investigated Genetic Algorithms (GA) and Particle Swarm Optimisation (PSO) to fine-tune hyperparameters, thereby enhancing detection against adversarial samples. They also integrated conformal prediction layers to reduce incorrect predictions, reinforcing the model's defense against adversarial perturbations.

Dynamic analysis, supported by AI, observes ransomware behaviours in a controlled environment, detecting unauthorised file encryption or suspicious network activity in real time Beaman, Barkworth, Akande, Hakak and Khan (2021); Amiri, Heidari, Navimipour, Unal and Mousavi (2024). This approach can improve detection accuracy and shorten the time needed to identify threats. However, adversarial attacks remain a concern, as they alter specific features to evade detection Yumlembam, Issac, Jacob and Yang (2024). Integrating AI into hybrid analysis fuses static and dynamic methods, yielding a more holistic view of ransomware operations Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). Deep learning, for example, enhances feature extraction and classification, enabling the detection of ransomware throughout multiple attack stages.

Uncertainty estimation in deep learning has recently gained traction in mission-critical scenarios, such as autonomous systems and medical diagnostics. Gawlikowski et al. Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang,

Bamler and Zhu (2023) surveyed various techniques, including ensemble models and Bayesian Neural Networks, which can help minimise overconfident false positives in model development. Similarly, conformal prediction, as highlighted by Yumlembam et al. Yumlembam, Issac, Jacob and Yang (2024), offers a more reliable and trustworthy prediction framework, allowing models to reject uncertain instances.

Data drift (where underlying distributions shift over time), poses another key challenge. Ashrapov (2020) demonstrated the use of Generative Adversarial Networks (GANs) to generate synthetic data that captures distribution shifts, enabling retraining for sustained effectiveness. In ransomware detection, this approach adapts to evolving attack patterns by continuously updating model parameters. Such continuous learning is crucial as new ransomware strains emerge, ensuring detection capabilities remain aligned with the threat landscape Sarker, Kayes, Badsha, AlQahtani, Watters and Ng (2020).

4.3 Proposed Framework

The proposed holistic framework, shown in Figure 3 and described in Algorithms 1 and 2, addresses limitations in the existing ransomware detection approaches. It merges four essential components: data augmentation, uncertainty awareness, explainable AI, and adversarial resilience. This section provides a comprehensive overview of the framework's architecture and its underlying methodologies.

4.3.1 Overview of the Framework

The framework employed combines multiple techniques to address common challenges in AI-based ransomware detection, including predictive uncertainty, interpretability, and robustness against adversarial attacks. Specifically, it employs Monte Carlo Dropout and Bayesian Neural Networks (BNNs) to quantify uncertainty, drawing on established best practices Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023). These tools produce reliable uncertainty metrics, enabling the system to flag or discard ambiguous predictions before they become false positives (benign software misclassified as ransomware) or false negatives (missed ransomware).

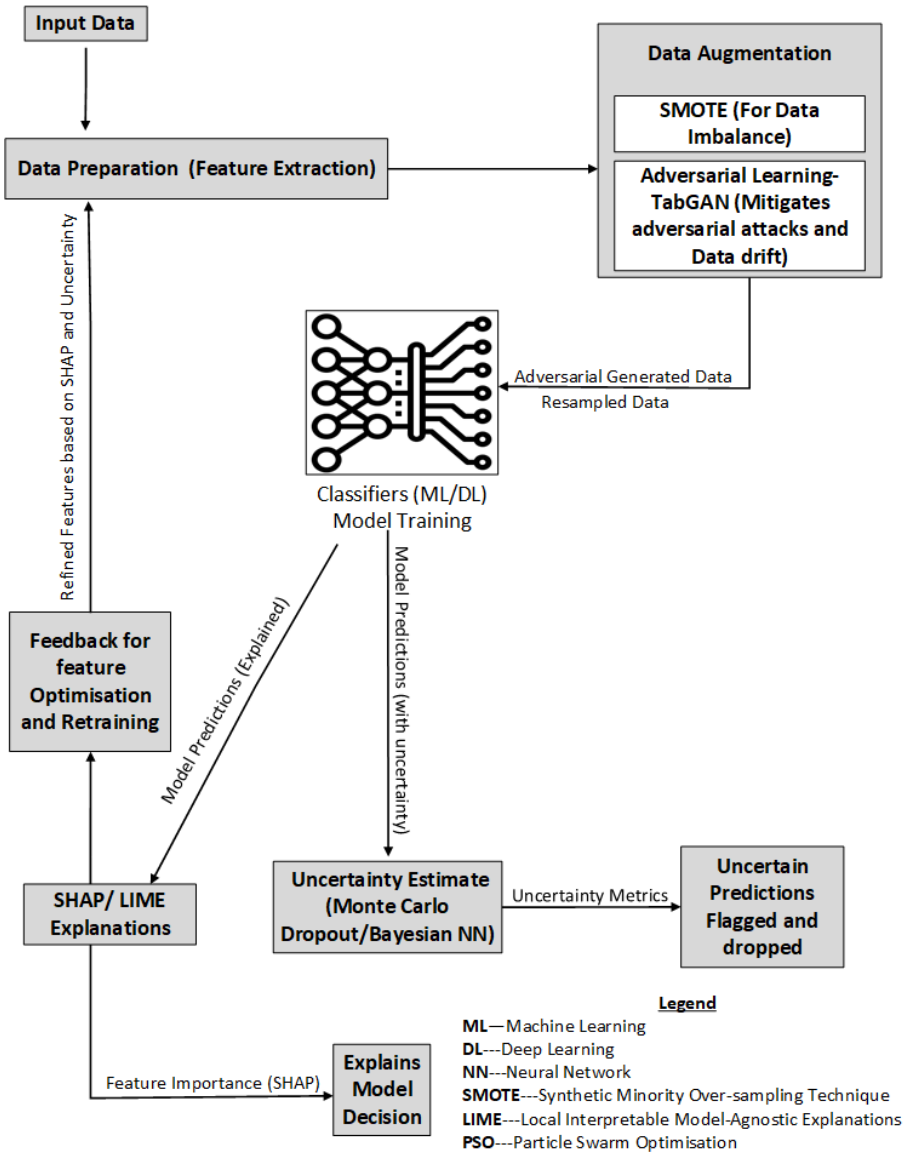


Figure 3: Proposed Detection Framework

Figure 3 illustrates the entire proposed detection framework. This framework integrates multiple advanced techniques to enhance ransomware detection. The components include data augmentation methods such as SMOTE and CTGAN for balancing the dataset and simulating adversarial data, uncertainty quantification using Monte Carlo Dropout and Bayesian Neural Networks to capture prediction uncertainties, explainability modules using SHAP and LIME to interpret the model's decisions, supervised learning algorithms such as Random Forest, SVM, KNN, and CNN for classification, and optimization strategies using Particle Swarm Optimization to fine-tune hyperparameters and improve performance. This integrated pipeline enables the system to handle class imbalance, improve

model robustness, and provide interpretable and trustworthy outputs.

To address both data imbalance and adversarial threats, the framework incorporates SMOTE and CTGAN. SMOTE balances the dataset by generating synthetic samples for under-represented classes, while CTGAN (Figure 4) generates adversarial examples to adapt models to data drift and potential evasion tactics. Integrating these synthetic datasets into the training pipeline enhances robustness across diverse settings. Figure 3 presents the internal working of the CTGAN model. CTGAN, or Conditional Tabular GAN, generates synthetic data for tabular datasets. Unlike traditional GANs that work well for image data, CTGAN handles mixed data types, such as categorical and continuous features common in ransomware datasets. This synthetic data generation helps augment underrepresented classes and simulate potential adversarial ransomware behaviours that the model may encounter in real-world scenarios.

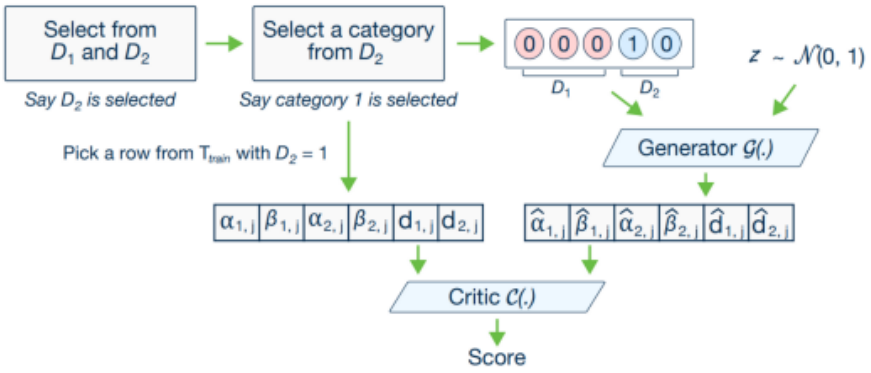


Figure 4: CTGAN model Ashrapov (2020)

Model interpretability is achieved through XAI techniques, including SHAP values, LIME, and feature importance analysis. These tools clarify the model's decision-making process, enabling cybersecurity professionals to understand and trust its outputs. SHAP and LIME explain individual predictions, while feature importance outlines the top contributing features.

An optimisation-and-retraining loop forms a critical pillar of this framework. Through supervised learning methods and CTGAN-generated adversarial samples, the model continuously adapts to new threat vectors. Particle Swarm Optimisation (PSO) refines feature selection and hyperparameters, boosting model performance over successive itera-

tions.

Uncertainty estimation via Monte Carlo Dropout and BNNs yields predictive distributions that highlight cases of high uncertainty. By integrating uncertainty quantification, data augmentation, and explainability, the framework delivers robust, interpretable, and flexible ransomware detection.

Algorithm 1 : Data Augmentation with SMOTE and CTGAN

1: Input: Raw dataset D containing system behaviors and process activities.

2: Data Preparation:

3: Extract representative features $X = \{x_1, \ldots, x_n\}$ from D.

4: Preprocess features (normalization, encoding) as required.

5: Handle Imbalance with SMOTE:

6: Identify minority classes and generate synthetic samples:

$$x_{\text{new}} = x_i + \lambda(x_j - x_i), \quad \lambda \in [0, 1]$$

7: Form an intermediate balanced dataset D_{SMOTE}.

8: Adversarial Data Generation with CTGAN:

9: Use CTGAN to produce adversarial and drift-mitigated samples:

$$\{X_{\text{adv}}, y_{\text{adv}}\} = \text{GANGenerator}(X_{\text{train}}, y_{\text{train}}, X_{\text{test}}, \text{ad params})$$

10: Combine these samples with D_{SMOTE} to form:

$$D_{\text{aug}} = D_{\text{SMOTE}} \cup \{X_{\text{adv}}, y_{\text{adv}}\}$$

11: Output: Augmented dataset D_{aug}.

The framework also leverages supervised learning models, such as Random Forest, SVM, CNNs, and RNNs, which excel at learning intricate data patterns. Trained on D_{aug}, these models classify ransomware vs. benign files with high accuracy. Clustering methods (e.g., K-means, hierarchical clustering) assist in detecting zero-day threats by revealing anomalies in unlabelled samples.

Adversarial training further fortifies the system by leveraging CTGAN to simulate adversarial scenarios and data drifts, ensuring resilience against evolving ransomware strains.

Algorithms 1 and 2 are as follows. Algorithm 1 outlines the process of data augmentation using a combination of SMOTE and CTGAN to address class imbalance and generate adversarial examples. The procedure begins with a raw dataset denoted as D, which contains samples extracted from system behaviors such as file I/O patterns, entropy levels, and storage access metrics. Feature extraction and preprocessing are applied first, involving normalization or encoding of features depending on their type. SMOTE is then used to synthetically generate new samples for the minority classes by interpolating between existing samples, ensuring the new points lie within the same feature space. This results in a balanced intermediate dataset referred to as DSMOTE. Following this, CTGAN is employed to generate synthetic samples that capture more complex patterns and adversarial characteristics. CTGAN is particularly suited for handling both categorical and continuous features and generates synthetic data by learning the conditional distributions in the original data. The outputs from SMOTE and CTGAN are merged to produce the final augmented dataset, Daug, which contains both balanced and adversarially enriched samples and is used in the subsequent stages of training.

Algorithm 2 describes the model training pipeline, incorporating advanced optimisation, uncertainty quantification, and explainability mechanisms. The process begins with Daug, the augmented dataset from Algorithm 1. Particle Swarm Optimisation (PSO) is applied to identify optimal feature subsets and hyperparameters for the chosen classifier. PSO simulates a swarm of particles searching the solution space, updating their velocities and positions based on their own best-known positions and the global best. Once optimised, the selected machine learning model—be it Random Forest, CNN, SVM, or KNN—is trained by minimising a loss function over all samples in the dataset. During training, uncertainty is estimated using Monte Carlo Dropout or Bayesian Neural Networks. This involves running the model multiple times with stochastic dropout layers activated and averaging the predictions to obtain an uncertainty score for each sample. Samples with high uncertainty are flagged for review. Explainability is introduced via SHAP, which computes feature attribution values using a game-theoretic framework to determine the

contribution of each feature to a given prediction. These values are critical in interpreting model decisions, especially in security-sensitive contexts. The model may be retrained or fine-tuned on uncertain or misclassified samples to further enhance performance. The algorithm outputs trained predictions, uncertainty metrics, and interpretability scores.

4.4 Experimental Setup and Results

This section details the experimental setup and presents the results of various machine-learning approaches tested on the RanSAP and RDset datasets. We evaluate the models by prioritizing F1-score, precision, and recall over accuracy. These metrics provide a balanced view of performance in imbalanced scenarios, such as ransomware detection. Hasarat Hasarat (2024) supports this focus by noting that accuracy can be misleading in imbalanced datasets, whereas the F1-score captures the trade-off between recall and precision, placing emphasis on minimising false positives and negatives.

4.4.1 Datasets and Preprocessing

We use two main datasets, RanSAP Hirano, Hodota and Kobayashi (2022) and Mathur's RDset Mathur (2020), each offering a substantial collection of ransomware and benign samples. To prepare for both traditional and deep learning models, we apply data cleaning, normalisation, and feature engineering.

The RanSAP dataset contains approximately 200 million storage access patterns from ransomware families (TeslaCrypt, WannaCry, Cerber, Sodinokibi, GandCrab-v4, Darkside and Ryuk) and benign applications (Microsoft Excel, Firefox, AESCrypt, SDelete and Zip). It includes both read and write operations, covering timestamps (seconds and milliseconds), logical block addresses (LBA), block sizes, and normalised Shannon entropy Shannon (1948) of written sectors. Meanwhile, the RDset dataset consists of 138,047 samples with 57 features, comprising 70% ransomware families and 30% legitimate software (see Figure 7). Both datasets are labelled for supervised learning, and each adheres to domain-specific feature extraction approaches.

Algorithm 2 : Model Training, Explainability, and Uncertainty Estimation

Input: Augmented dataset D_{aug}.

Feature and Hyperparameter Optimisation (PSO):

Use PSO to select optimal features and hyperparameters:

$$v_i(t+1) = w\, v_i(t) + c_1 r_1 (p_{best} - x_i) + c_2 r_2 (g_{best} - x_i)$$

$$x_i(t+1) = x_i(t) + v_i(t+1)$$

Model Training:

Train the classifier f_θ on D_{aug}:

$$\theta = \arg\min_{\theta} \sum_{(x,y)\in D_{\text{aug}}} J(\theta, x, y)$$

Uncertainty Estimation:

Apply Monte Carlo Dropout (or BNN) to obtain predictive distributions:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} f_\theta^{(t)}(x)$$

Flag predictions with uncertainty above a specified threshold.

Explainable AI (XAI) with SHAP:

Compute SHAP values:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \big[f(S \cup \{i\}) - f(S) \big]$$

Identify key features and refine the model if necessary.

Retraining:

Incorporate updated data D_{new}, addressing flagged uncertainties:

$$\theta_{\text{new}} = \arg\min_{\theta} \sum_{(x,y)\in D_{\text{new}}} J(\theta, x, y)$$

Output: Final model predictions, uncertainty metrics, and SHAP-based explanations.

4.4.2 Criteria for Feature Extraction and Engineering

We identified key features distinguishing ransomware from benign software based on previous studies and domain expertise. Algorithms 3 and 4 illustrate how features are extracted from the RanSAP and RDset datasets, respectively. These features inform the detection framework (Algorithms 1 and 2) by proactively separating ransomware from benign behaviour.

Algorithm 3 RanSAP Feature Extraction

Require: DataFrame df with columns ['Timestamp_s', 'Timestamp_ms', 'LBA', 'Size', 'Entropy', 'Normalize', 'Type']
Ensure: Array of extracted features
  Initialise $features \leftarrow []$
  $time_window \leftarrow 10$ ▷ Time window set to 10 seconds
  for $i \leftarrow 0$ to $len(df) - time_window$ step $time_window$ do
    $window \leftarrow df.iloc[i : i + time_window]$
    $avg_write_throughput \leftarrow \text{mean}(window[window['Type'] ==' write']['Size'])$
    $avg_read_throughput \leftarrow \text{mean}(window[window['Type'] ==' read']['Size'])$
    $var_lba_write \leftarrow \text{var}(window[window['Type'] ==' write']['LBA'])$
    $var_lba_read \leftarrow \text{var}(window[window['Type'] ==' read']['LBA'])$
    $avg_normalized_entropy \leftarrow \text{mean}(window[window['Type'] ==' write']['Normalize'])$
    $features$.append([
      $avg_write_throughput$,
      $avg_read_throughput$,
      var_lba_write,
      var_lba_read,
      $avg_normalized_entropy$])
  end for
  return np.array($features$)

Algorithm 3 outlines RanSAP feature extraction, capturing the sequence and type of operations (creation, modification, deletion) that often differ from benign software. Ransomware often employs distinct encryption algorithms, observable through their type and parameters. Network-related features, such as unusual connections or data transfers patterns, also help pinpoint ransomware's communication channels. The frequency and types of system calls further differentiate ransomware from benign applications. As mentioned, Algorithm 3 focuses on extracting meaningful statistical features from the RanSAP dataset, which records disk-level behaviour over time. The input is a log file containing timestamped entries of read and write operations along with associated metadata like Logical Block Addresses (LBA), operation type, and entropy. The log is segmented into

fixed 10-second time windows to facilitate temporal analysis. Within each time window, the algorithm computes several statistical features: average write throughput, average read throughput, variance in write LBAs, variance in read LBAs, and average normalised entropy. These features provide a compact yet expressive summary of storage activity and are highly indicative of ransomware behaviour, which often involves high-entropy writes and irregular access patterns. The result is a structured feature array for each time window that can be directly used as input to machine learning models.

Algorithm 4 deals with feature extraction and dimensionality reduction for the RDset dataset, which contains static attributes from Windows Portable Executable (PE) headers and related metadata. The process begins by dropping identifier columns such as file names and hashes, which do not contribute to classification. The dataset is split into input features X and target labels y, and all feature columns are standardised using Z-score normalisation to ensure consistency in scale. Features with low variance are removed using a predefined threshold, as they are unlikely to contribute meaningfully to classification decisions. To further refine the feature set, the Variance Inflation Factor (VIF) is calculated for each remaining feature to identify and remove multicollinearity. Features with a VIF score exceeding a certain threshold, typically 10, are eliminated to reduce redundancy and improve model interpretability. The resulting set of features, Xfinal, is a reduced and optimised representation of the original RDset, containing only the most relevant and independent features for effective ransomware detection.

For RanSAP, we follow Hirano et al. Hirano, Hodota and Kobayashi (2022), while for RDset, we adopt Masum et al. Masum, Faruk, Shahriar, Qian, Lo and Adnan (2022) feature calculation techniques. Only a subset of RanSAP focusing on Win2008r2_250GB_SSD access patterns is used (Figures 5 and 6). Figure 4 shows the binary class distribution in the RanSAP dataset after applying a 30-second time window during feature extraction. The chart demonstrates that there is an imbalance between benign samples and ransomware samples. This imbalance justifies the need for data augmentation methods like SMOTE and CTGAN to ensure that models are not biased towards the majority class. Figure 5 shows the multi-class distribution for RanSAP after feature extraction with the same 30-second window. It displays data across 12 classes, representing multiple

ransomware families and benign applications. The class imbalance is even more significant here, making it more challenging to train models that can generalise well across all ransomware families.

We apply Equations 4.1–4.5 to obtain metrics like average write/read throughput, variance of write/read LBAs, and normalised Shannon entropy.

Algorithm 4 RDset Feature Extraction using Variance Threshold and VIF

Require: DataFrame D with columns $['Name', 'md5', 'legitimate', \ldots]$
Ensure: Array of extracted features X_{final}
  Load and preprocess data
  Load dataset D
  Drop identity variables: $D \leftarrow D \setminus \{Name, md5\}$
  Separate features and target variable:
  $X \leftarrow D \setminus \{legitimate\}$
  $y \leftarrow D[legitimate]$
  Standardize the features
  $X_{standardized} \leftarrow$ StandardScaler().fit_transform(X)
  $X_{standardized_df} \leftarrow$ DataFrame($X_{standardized}$, columns = X.columns)
  Apply variance threshold
  selector $\leftarrow$ VarianceThreshold(1)
  $X_{high_variance} \leftarrow$ selector.fit_transform($X_{standardized_df}$)
  selected_features $\leftarrow$
  $X_{standardized_df}$.columns[selector.get_support(indices=True)]
  $X_{high_variance_df} \leftarrow$ DataFrame($X_{high_variance}$, columns = selected_features)
  Calculate Variance Inflation Factor (VIF)
  vif_data $\leftarrow$ DataFrame()
  vif_data["$Feature$"] $\leftarrow$ $X_{high_variance_df}$.columns
  vif_data["VIF"] $\leftarrow$
  [variance_inflation_factor($X_{high_variance_df}$.values, i)
  $\forall i \in 0 \ldots (X_{high_variance_df}$.shape[1] $- 1)$]
  Filter out features with high VIF
  final_features $\leftarrow$ vif_data[vif_data["VIF"] ≤ 10]["$Feature$"]
  $X_{final} \leftarrow X_{high_variance_df}$[final_features]
  Output the final set of selected features
  Print "Selected features after applying variance threshold and VIF:"
  Print X_{final}.columns
  return X_{final}

Average Write Throughput, $T_{write}(t)$, is calculated as:

$$T_{write}(t) = \frac{\sum_{i=1}^{n_{write}} \text{Size}_i}{\text{time_window}} \tag{4.1}$$

where n_{write} is the number of write operations in the time window, and Size_i is the size

of the i-th write operation.

Average Read Throughput, $T_{read}(t)$, is calculated as:

$$T_{read}(t) = \frac{\sum_{i=1}^{n_{read}} \text{Size}_i}{\text{time_window}} \tag{4.2}$$

where n_{read} is the number of read operations in the time window, and Size_i is the size of the i-th read operation.

Variance of Write LBAs, $V_{write}(t)$, is calculated as:

$$V_{write}(t) = \text{Var}(\{\text{LBA}_i\}_{i=1}^{n_{write}}) \tag{4.3}$$

where LBA_i is the logical block address of the i-th write operation.

Variance of Read LBAs, $V_{read}(t)$, is calculated as:

$$V_{read}(t) = \text{Var}(\{\text{LBA}_i\}_{i=1}^{n_{read}}) \tag{4.4}$$

where LBA_i is the logical block address of the i-th read operation.

Average Normalised Shannon Entropy of written sectors, $H_{write}(t)$, is calculated as:

$$H_{write}(t) = \frac{\sum_{i=1}^{n_{write}} \text{Normalise}_i}{n_{write}} \tag{4.5}$$

where n_{write} is the number of write operations in the time window, and Normalise_i is the normalised entropy of the i-th written sector.

For RDset, variables “Md5” and “Name” are dropped to avoid confounding model development. The target variable “legitimate” equals 1 for safe software and 0 for ransomware (Figure 7). We use Equations 4.6–4.8 to apply Z-score standardisation, variance thresholding, and variance inflation factor (VIF). Features failing these criteria are excluded to reduce noise and multicollinearity. Figure 6 displays the class distribution for the RDset dataset, which consists of about 70% ransomware samples and 30% benign samples. This figure again confirms that class imbalance is a significant issue in both datasets used in this study.

Z-score Standardisation is used to standardise the RDset features by removing the mean and scaling to unit variance.

$$z_i = \frac{x_i - \mu_i}{\sigma_i} \tag{4.6}$$

where z_i is the standardised value of feature i, x_i is the original value of feature i, μ_i is the mean of feature i and σ_i is the standard deviation of feature i.

Variance Threshold is applied to remove RDset features with low variance. This ensures that only features with variability are retained. The threshold value is set to 1 in this context.

$$\sigma_i^2 \geq \tau \tag{4.7}$$

where σ_i^2 is the variance of feature i and τ is the variance threshold, set to 1.

Variance Inflation Factor (VIF) is used to detect multicollinearity among the features. Features with VIF greater than 10 are considered to have high multicollinearity and are removed.

$$\text{VIF}_i = \frac{1}{1 - R_i^2} \tag{4.8}$$

where VIF_i is the Variance Inflation Factor for feature i and R_i^2 is the coefficient of determination of the regression of feature i on all other features.

Input: D: Original dataset, τ: Variance threshold (default is 1) and $VIF_{threshold}$: Variance Inflation Factor threshold (default is 10)

Output:X_{final}: Final set of selected features

4.4.3 Initial Experimental Results

We initially conducted a comprehensive evaluation of several machine learning models to establish baseline performance on the RanSAP and RDset datasets. Specifically, we

Figure 5: 2-Class distribution of 30-second time window for RanSAP

assessed Random Forest, k-Nearest Neighbours (k-NN), and Support Vector Machines (SVM), while initially excluding convolutional neural networks (CNN) from these early comparisons. The models were evaluated based on a variety of standard classification metrics, including accuracy, precision, recall, F1-score, as well as cross-validation to ensure robustness across different data splits. All models were tested without applying any data augmentation, uncertainty quantification, or hyperparameter optimisation techniques in order to establish their natural baseline performance under raw input conditions.

Tables 3 and 4 present the results for binary and multi-class classification respectively on the RanSAP dataset. Table 5 shows analogous results for RDset under similar experimental settings. These tables provide valuable insight into the strengths and weaknesses of each classifier across both binary and multi-class formulations.

Table 3 shows that, for binary classification on RanSAP, Random Forest achieves the strongest performance, yielding an F1-score of 0.9613. This suggests that the ensemble nature of Random Forest, which combines multiple decision trees, allows it to capture complex decision boundaries effectively even without additional model tuning or augmentation. In comparison, k-Nearest Neighbours achieves an F1-score of 0.8612, while Support

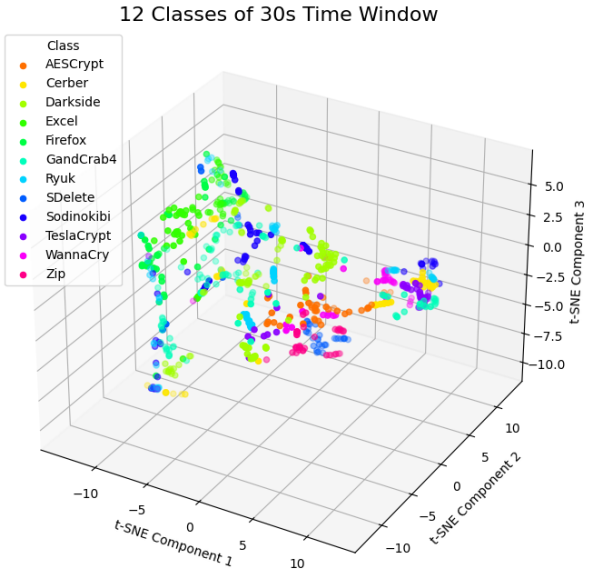


Figure 6: 12-Class distribution of 30-second time window for RanSAP

Vector Machines yield an F1-score of 0.8029. Although both k-NN and SVM exhibit reasonable performance, their lower F1-scores suggest a reduced ability to balance precision and recall compared to Random Forest in this specific binary classification scenario.

Expanding to the more challenging multi-class classification problem on RanSAP, Table 4 highlights a significant performance drop for k-NN and SVM. Random Forest remains the strongest performer, achieving an F1-score of 0.8755. However, k-NN and SVM struggle more with the increased class complexity, reflecting the inherent challenges of correctly separating multiple classes with limited data representation. This result reinforces the robustness of Random Forest as an initial baseline model across both binary and multi-class settings on the RanSAP dataset, while highlighting the greater sensitivity of k-NN and SVM to the increased classification complexity.

In parallel, Table 5 presents the corresponding baseline results for the RDset dataset, which features a larger and more diverse data distribution. Once again, Random Forest outperforms all other classical machine learning models, achieving an impressive F1-score of 0.9864, demonstrating near-perfect classification performance. Notably, deep learning begins to demonstrate its strengths on this dataset: the CNN model achieves an F1-score

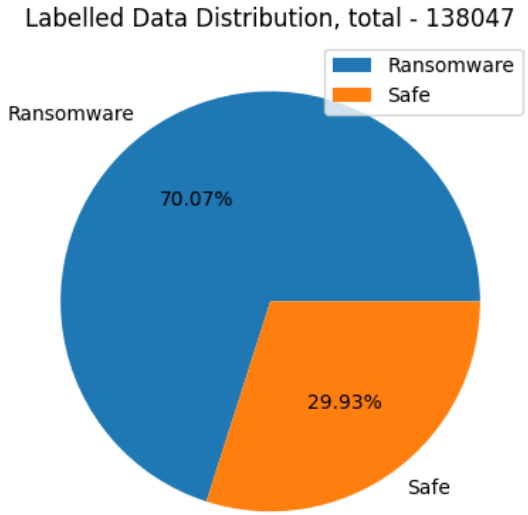


Figure 7: Distribution of Labelled Data for RDset

of 0.9733, indicating that neural network-based approaches can capture complex, non-linear patterns in RDset very effectively. This strong CNN performance provides early evidence that more advanced deep learning architectures may offer additional benefits, especially as data complexity increases.

To further investigate the potential of deep learning approaches on RDset, we implemented a convolutional neural network (CNN) architecture using PyTorch. The network was designed as a fully connected CNN rather than one using traditional convolutional layers, due to the nature of the input data. The architecture consisted of two hidden layers with 128 and 64 neurons respectively, followed by an output layer whose size matched the number of output classes. Rectified Linear Unit (ReLU) activation functions were used throughout the network to introduce non-linearity and improve learning dynamics. The model was optimised using the Adam optimiser with a learning rate of 0.001, and trained using CrossEntropyLoss to handle the multi-class classification objective. The RDset dataset was partitioned into training and testing subsets using a 70/30 split, and training was conducted over 10 epochs with a batch size of 32.

The CNN demonstrated strong generalisation capability across both binary and multi-

class scenarios. After training, the model achieved an accuracy of 0.9838, a precision of 0.9684, a recall of 0.9782, and an F1-score of 0.9733 on the held-out test set. These results confirm that the CNN was able to capture complex and subtle patterns in the RDset data, yielding robust performance that closely rivals and, in some metrics, approaches the strong baseline set by Random Forest. This also suggests that CNN-based deep learning models may offer greater scalability and adaptability as the dataset size and complexity increase.

Table 3: Initial Model Performance Metrics (30s) - Binary Classification Models for RanSAP

Model	Accuracy	Precision	Recall	F1 Score	Cross-Validation Scores
Random Forest	0.9593	0.9600	0.9625	0.9613	[0.9632, 0.9632, 0.9354, 0.9815, 0.9531]
k-NN	0.8281	0.8400	0.8800	0.8612	[0.7918, 0.8396, 0.8409, 0.8148, 0.8543]
SVM	0.7301	0.7800	0.8300	0.8029	[0.7152, 0.7503, 0.7033, 0.7302, 0.7517]

Table 4: Initial Model Performance Metrics (30s) - Multi-Class (12) Classification Models for RanSAP

Model	Accuracy	Precision	Recall	F1 Score	Cross-Validation Scores
Random Forest	0.8508	0.8700	0.8800	0.8755	[0.8680, 0.7819, 0.8574, 0.8415, 0.9054]
k-NN	0.5114	0.5900	0.6400	0.6140	[0.5281, 0.5377, 0.5055, 0.4776, 0.5083]
SVM	0.2723	0.2500	0.3000	0.2715	[0.2801, 0.2823, 0.2234, 0.2834, 0.2923]

4.4.4 Data Augmentation

Data augmentation plays a crucial role in addressing class imbalance and enhancing model performance, particularly in domains such as ransomware detection where the number of benign samples often significantly exceeds the number of malicious instances. In imbalanced datasets, models tend to become biased toward the majority class, leading to poor detection rates for minority classes, such as rare or newly emerging ransomware variants. By artificially generating additional samples for underrepresented classes, data augmentation helps create a more balanced dataset that allows machine learning algorithms to better

Table 5: Initial Model Performance Metrics for RDset

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.9918	0.9834	0.9894	0.9864
SVM	0.9760	0.9710	0.9486	0.9597
KNN	0.9856	0.9735	0.9790	0.9762
CNN	0.9838	0.9684	0.9782	0.9733

learn the underlying patterns associated with both benign and malicious behaviours.

Through augmentation, the model is exposed to a wider variety of scenarios, which improves its ability to generalise to unseen data and reduces the risk of overfitting to the dominant class. Techniques such as Synthetic Minority Oversampling Technique (SMOTE) generate synthetic samples by interpolating between existing minority class samples, producing realistic data points that strengthen the decision boundaries between classes. More advanced methods like Conditional Tabular Generative Adversarial Networks (CTGAN) further enhance the diversity of synthetic data by modelling complex distributions within the minority class, capturing intricate relationships between features that simpler oversampling methods may overlook.

In ransomware detection specifically, augmentation not only helps balance the dataset but also exposes the model to potential variations in ransomware behaviour, including adversarial patterns that may arise in real-world attacks. This results in models that are more robust to sophisticated and evolving threats. Additionally, augmented data enables better training of deep learning models, such as Convolutional Neural Networks (CNNs), which often require large and diverse datasets to fully leverage their representation learning capabilities.

Here, we examine SMOTE and CTGAN for ransomware detection across RanSAP and RDset.

Performance on RanSAP Dataset: For RanSAP, SMOTE and PSO balanced classes and tuned hyperparameters. Figure 8 and Table 6 show marginal gains in precision and recall for binary classification, but limited multi-class improvements. While SMOTE and PSO lowered false positives in binary classification, their effects may diminish in more complex scenarios. Figure 7 compares model performance metrics with and without applying SMOTE and PSO for the RanSAP dataset. The bar graph shows that applying SMOTE and PSO leads to slight improvements in precision and recall for binary classification. However, for multi-class classification, these improvements are negligible, suggesting that the RanSAP dataset's rich feature set may already allow models to perform well without extensive preprocessing. Table 4 compares performance metrics with and without PSO

and SMOTE for RanSAP. The results show minor improvements in binary classification after applying PSO and SMOTE, while multi-class classification sees virtually no change, suggesting that the intrinsic feature quality in RanSAP already supports strong performance.

Performance on RDset Dataset: We first assessed RDset without augmentation (Table 5). Random Forest led at 0.9918 accuracy, 0.9834 precision, 0.9894 recall, and 0.9864 F1. KNN, CNN, and SVM also performed well. Applying SMOTE (Table 7) yielded steady improvements across all models. Random Forest soared to 0.9961 accuracy (F1 = 0.9962). KNN and CNN also improved, while SVM maintained a solid 0.9732 F1. Hence SMOTE effectively addresses class imbalance, boosting generalisation.

Conversely, CTGAN (Table 8) produced mixed outcomes. Random Forest reached 0.9965 accuracy (F1 = 0.9955), but SVM, KNN, and CNN declined, indicating possible noise or outliers from CTGAN. Fine-tuning the CTGAN process may be necessary for broader compatibility. Table 6 presents performance after applying CTGAN to RDset. While Random Forest improves slightly to an F1-score of 0.9955, SVM's performance collapses entirely, achieving an F1-score of zero. KNN and CNN also experience reduced performance. This shows that CTGAN can introduce noisy synthetic samples that harm certain models if not properly tuned.

Combined RanSAP and RDset Results: Table 11 shows we tested both datasets together. Under SMOTE, Random Forest led with a 0.9963 F1 and 99.63% accuracy, with KNN and CNN also scoring 0.9925 and 0.9906 F1, respectively. SVM achieved 0.9884. With CTGAN, CNN + Monte Carlo Dropout topped at 0.9978 F1 and 99.77% accuracy, while Random Forest excelled at 0.9976 F1 and 99.75% accuracy. SVM gained ground (0.9970 F1), and KNN remained consistent. Overall, SMOTE consistently boosts results; CTGAN offers large improvements for robust models, however, but may hamper others with added noise. Tailoring these methods to the model and dataset is crucial for optimal performance.

Table 6: Comparison of Performance Metrics with and without PSO/SMOTE for RanSAP

Classification Type	PSO/SMOTE	Accuracy	Precision	Recall	F1 Score
Binary Classification	With PSO/SMOTE	0.972	0.976	0.969	0.970
	Without PSO/SMOTE	0.974	0.974	0.974	0.974
Multi-Class Classification	With PSO/SMOTE	0.895	0.903	0.887	0.885
	Without PSO/SMOTE	0.896	0.905	0.887	0.885

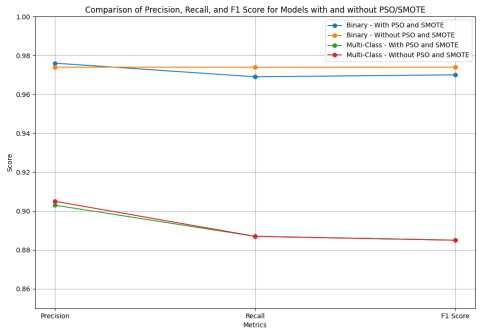


Figure 8: Visual comparison metrics with and without PSO/SMOTE for RanSAP

4.4.5 Uncertainty Quantification

We employed Monte Carlo-based uncertainty estimation to examine its effect on Random Forest, SVM, k-NN, and CNN models (Figure 9). Table 10 compares precision with and without uncertainty. Moreover, a CNN with Monte Carlo Dropout combined with CTGAN reached 0.9978 F1 and 99.77% accuracy (Table 11). Hence, combining uncertainty-aware strategies with advanced augmentation can greatly improve ransomware detection.

Random Forest also benefited from uncertainty estimation, especially with SMOTE, showing improved precision and recall in binary and multi-class tasks (Table 9). Monte Carlo Dropout curbed overconfident predictions, boosting reliability for real-world ransomware scenarios. Conformal prediction added another layer of interpretability, identifying ambiguous samples for manual review. Collectively, these methods enhance both robustness

Table 7: Model Performance Metrics with SMOTE for RDset

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.9961	0.9907	0.9969	0.9962
SVM	0.9733	0.9798	0.9666	0.9732
KNN	0.9887	0.9839	0.9937	0.9888
CNN	0.9845	0.9850	0.9840	0.9845

Table 8: Model Performance Metrics with CTGAN for RDset

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.9965	0.9948	0.9961	0.9955
SVM	0.6101	0.0000	0.0000	0.0000
KNN	0.7503	0.6810	0.6762	0.6786
CNN	0.6555	0.5360	0.8669	0.6624

and explainability.

Table 9: Optimised Features as per Random Forest Model for RanSAP (10s)

Metric	Binary Classification	Multi-Class Classification
Conformal Prediction Coverage	0.9698	0.8922
Optimised Random Forest F1 Score	0.9700	0.8894
Best Parameters (n_estimators, max_depth)	[78.18, 9.44]	
Best Cross-Validation Score	0.9740	

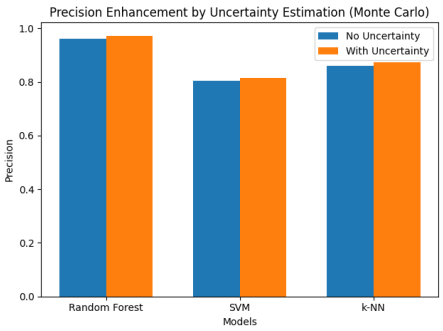


Figure 9: Precision Enhancement by Uncertainty Estimation (Monte Carlo) for RanSAP.

Table 10: Comparison metrics with and without Uncertainty Estimation (Monte Carlo) for RanSAP

Model	Without Uncertainty				With Uncertainty			
	Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score
Random Forest	0.9598	0.9613	0.9582	0.9597	0.9689	0.9725	0.9653	0.9689
SVM	0.7976	0.8029	0.7923	0.7976	0.8094	0.8153	0.8034	0.8093
k-NN	0.8569	0.8612	0.8525	0.8568	0.8686	0.8734	0.8637	0.8685

4.4.6 SHAP Analysis for Explainable AI

To interpret model predictions, we computed SHAP values for both binary and multi-class models. SHAP reveals each feature's contribution, thus clarifying AI decision processes Gülmez, Kakisim and Sogukpinar (2024). Figures 10 and 11 show feature importances for RDset and RanSAP, respectively. Tables 12 and 13 provide tabular views of these impacts. Figure 9 presents feature importance results from the Random Forest model trained on

Table 11: Performance Metrics for Ransomware Detection Models Using Combined RanSAP and RDset Datasets

Technique	Model	Accuracy	Precision	Recall	F1-Score
SMOTE	Random Forest	0.9963	0.9976	0.9950	0.9963
	SVM	0.9885	0.9881	0.9888	0.9884
	KNN	0.9926	0.9970	0.9881	0.9925
	CNN with MC Dropout	0.9906	0.9948	0.9864	0.9906
CTGAN	Random Forest	0.9975	0.9968	0.9984	0.9976
	SVM	0.9968	0.9967	0.9973	0.9970
	KNN	0.9922	0.9902	0.9947	0.9925
	CNN with MC Dropout	0.9977	0.9972	0.9983	0.9978

RDset. The figure identifies 'SizeOfStackReserve' as the most critical feature, followed by entropy-based features such as 'SectionsMaxEntropy' and 'ResourcesMinEntropy'. This indicates that ransomware detection in this dataset heavily depends on features derived from the Portable Executable (PE) header and entropy measures. Figure 10 shows the SHAP feature importance summary for the RanSAP dataset. The mean absolute SHAP values indicate that 'Hwrite' (the normalized Shannon entropy of write operations) is the most influential feature, followed by 'Twrite' (write throughput), 'Vwrite' (variance of write LBAs), 'Tread' (read throughput), and 'Vread' (variance of read LBAs). This demonstrates that disk-level storage behaviour, especially write patterns, is highly predictive for ransomware detection. Table 10 lists the feature importance rankings for RDset as calculated by the Random Forest model. 'SizeOfStackReserve' ranks highest, followed by entropy-related features such as 'SectionsMaxEntropy' and 'ResourcesMinEntropy'. These features are effective in distinguishing ransomware from benign files in RDset. Table 11 summarises the SHAP values for RanSAP, confirming that 'Hwrite', 'Twrite', and 'Vwrite' are the most critical features for classification, aligning with the findings from the SHAP visualisation.

For binary classification on RanSAP, Figure 11 highlights "Hwrite" and "Vwrite" as top influences in identifying ransomware from benign. For multi-class, SHAP analyses identify "Twrite" and "Tread" as critical in distinguishing families like Darkside and GandCrab4. Such insights guide enhanced feature selection and model refinements.

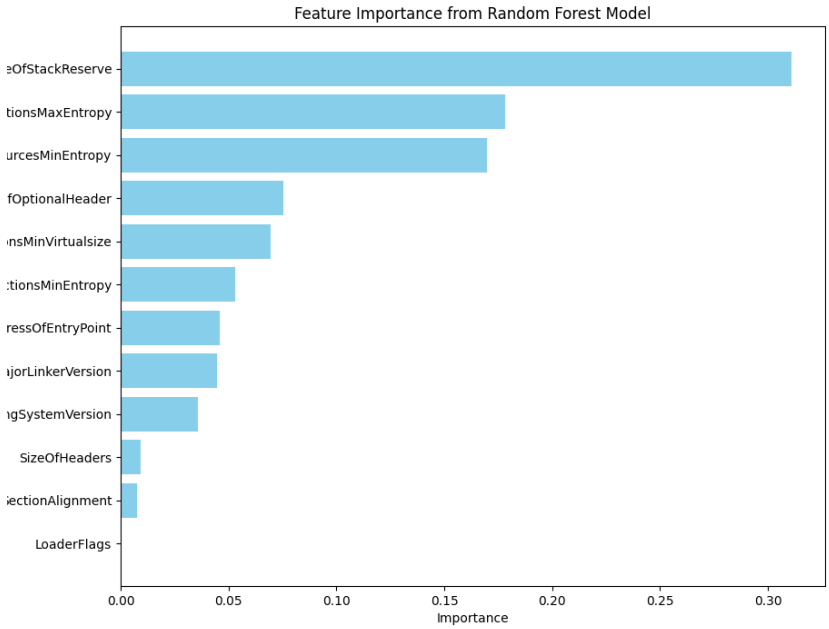


Figure 10: Feature Importance on Random Forest Model for RDset

Table 12: Feature Importance in RDset

Feature	Importance
SizeOfStackReserve	0.30
SectionsMaxEntropy	0.18
ResourcesMinEntropy	0.15
SizeOfOptionalHeader	0.08
SectionsMinVirtualsize	0.06
SectionsMinEntropy	0.05
AddressOfEntryPoint	0.04
MajorLinkerVersion	0.03
OperatingSystemVersion	0.02
SizeOfHeaders	0.01
SectionAlignment	0.01
LoaderFlags	0.01

Table 13: SHAP Values for Important Features in RanSAP

Feature	Mean(|SHAP Value|)
Hwrite	0.70
Twrite	0.50
Vwrite	0.40
Tread	0.30
Vread	0.20

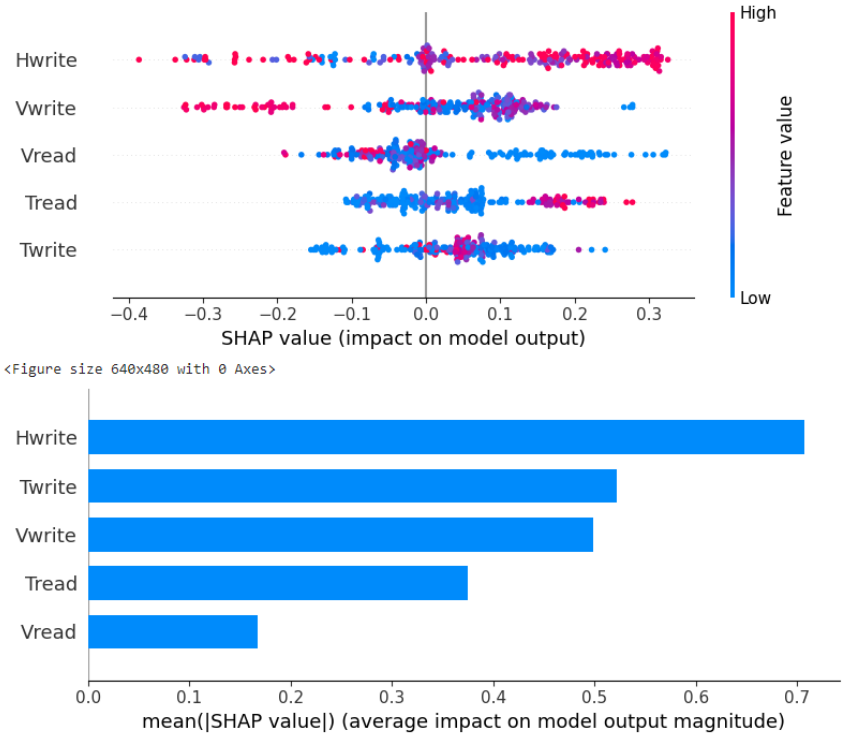


Figure 11: SHAP Feature value and mean for RanSAP

4.4.7 LIME for Local Explanations

Local Interpretable Model-Agnostic Explanations (LIME) further clarify individual predictions by approximating local model behaviour. Figures 12 and 13 show how LIME highlights key features driving legitimate or ransomware classifications. For instance, “ImageBase,” “Subsystem,” and “SectionsMaxEntropy” dominantly shape predictions in RDset, offering practical insight into feature relevance and potential attack surfaces. Figure 11 presents a LIME explanation for a single benign file classification in RDset. The explanation highlights that features such as ‘Subsystem’, ‘ImageBase’, and ‘SectionsMaxEntropy’ contributed most strongly to the model classifying this sample as benign. This provides transparency into why the model made a certain prediction.

Figure 12 shows a LIME explanation for a ransomware file in RDset. In this case, features like ‘SectionsMaxEntropy’ and ‘ResourcesMinEntropy’ were most influential in classifying the file as ransomware. This local interpretability helps security analysts understand what aspects of the file’s structure signalled malicious behaviour.

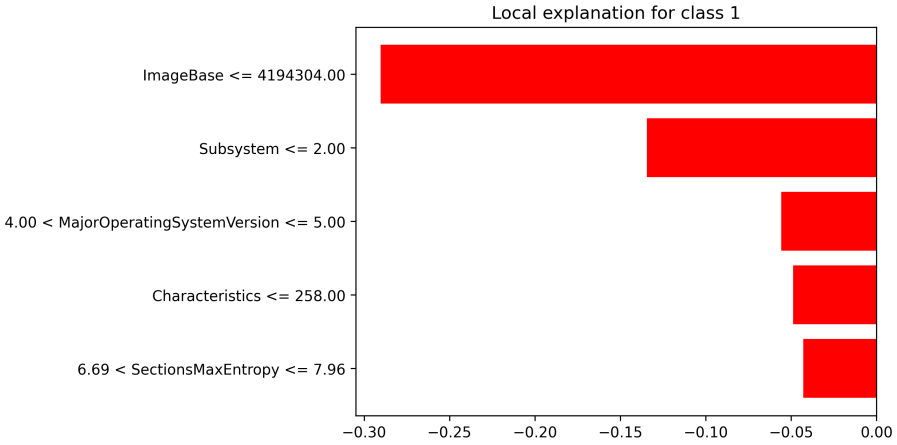


Figure 12: LIME explanation for a legitimate file, showing the top contributing features for RDset.

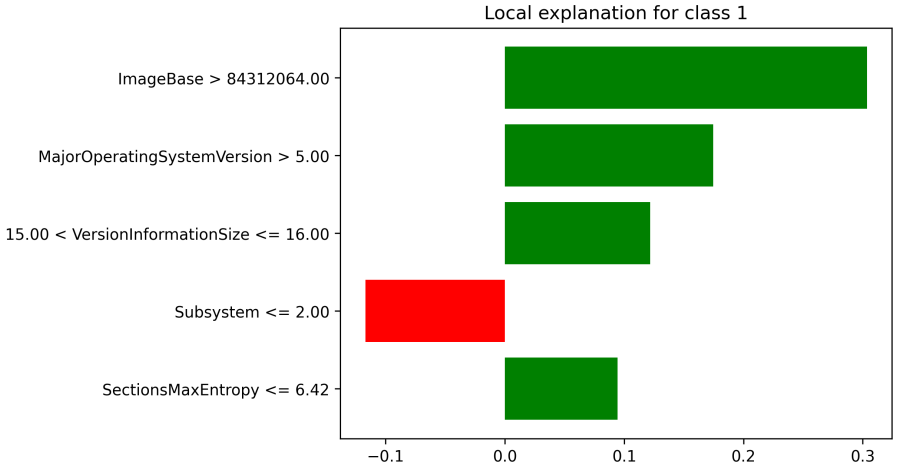


Figure 13: LIME explanation for a ransomware file, showing the top contributing features for RDset.

4.4.8 Model Evaluation through Confusion Matrices

Confusion matrices (Figures 14 and 15) for Random Forest, SVM, and k-NN reveal performance nuances in binary and multi-class tasks. Although Random Forest achieves a balanced trade-off between true positives and false negatives, its accuracy varies by class in multi-class classification, pointing to possible improvements in feature engineering or hybrid modelling. Figure 13 depicts the multi-class confusion matrix for RanSAP using 12 classes. The matrix shows how frequently each ransomware family and benign class were correctly or incorrectly classified. Random Forest performs well overall, but some confusion exists between ransomware families such as Darkside and GandCrab4, which

share overlapping features and behaviours.

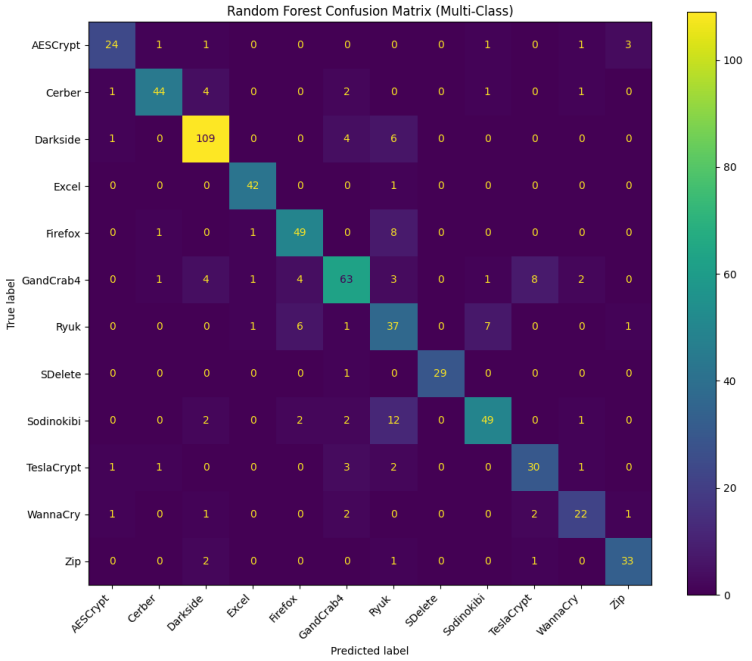


Figure 14: 12 Classes Confusion Matrix of 30-second time window for RanSAP

Figure 14 presents the binary confusion matrix for RanSAP, classifying data simply as ransomware or benign. The matrix indicates a very high true positive rate and very low false positive and false negative rates, showcasing strong performance in the simpler binary classification task.

4.4.9 Negative Predictive Value (NPV) Analysis for Multi-Class Classification

NPV indicates how effectively the model identifies true negatives. Figure 16 shows NPV per class for multi-class RanSAP detection, with "Excel," "SDelete," and "Zip" yielding high NPVs, while "Darkside" and "GandCrab4" are occasionally misclassified. An average NPV of 0.9627 underscores overall effectiveness but pinpoints classes needing further calibration. Figure 15 displays the Negative Predictive Value (NPV) for each class in multi-class classification for RanSAP. The benign classes such as Excel, SDelete, and Zip have near-perfect NPV, meaning they are rarely misclassified as ransomware. Conversely, ransomware families like Darkside and GandCrab4 have lower NPVs, suggesting that these are occasionally misclassified as benign, warranting further model refinement.

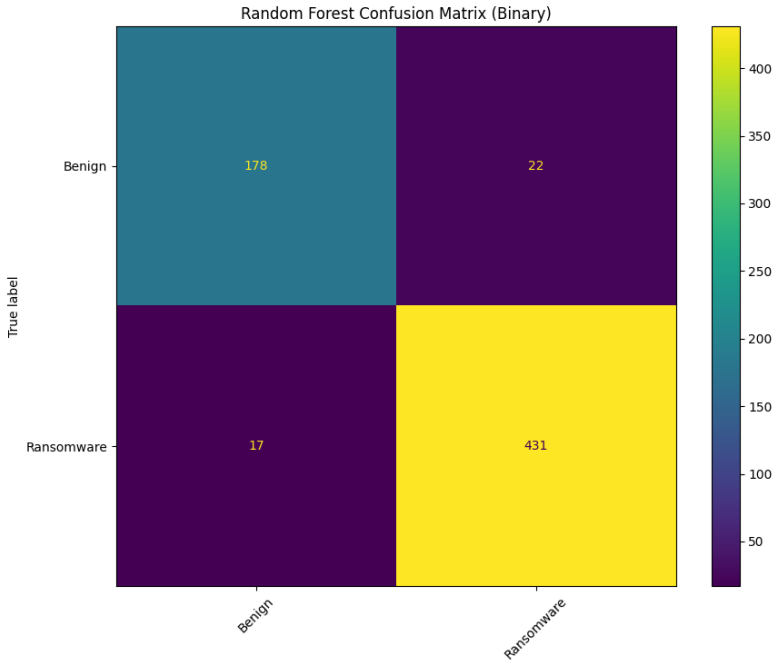


Figure 15: 2 Classes Confusion Matrix of 30-second time window for RanSAP

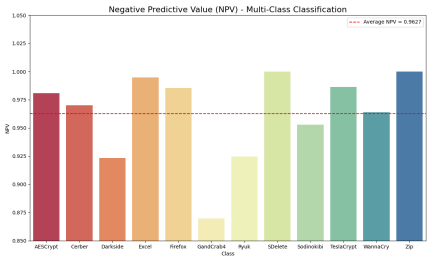


Figure 16: Negative Predictive Value (NPV) - Multi-Class Classification for RanSAP

4.4.10 Loss Analysis in Deep Learning Models

We examined loss curves (Figure 17) for binary vs. multi-class CNN training on RanSAP. Binary loss converged steadily, showing minimal over fitting. Multi-class loss fluctuated more, suggesting difficulties in handling complex class boundaries. Potential solutions include advanced hyper parameter tuning or stronger data augmentation. Figure 17 compares the training loss curves for CNN models on RanSAP for binary and multi-class classification. The binary classification model shows a steady decrease in loss and stable convergence, while the multi-class model's loss curve fluctuates more, reflecting the increased difficulty of distinguishing between many ransomware families.

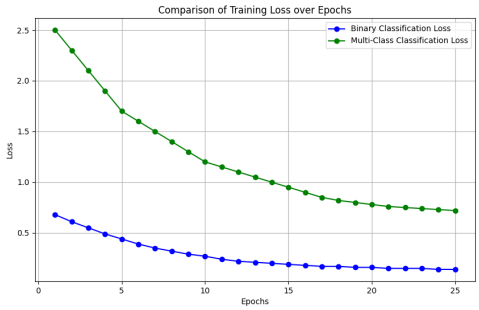


Figure 17: CNN Comparison of Training Loss over Epochs for RanSAP

4.4.11 Classification Results Comparison

Table 15 provides a comprehensive performance summary across various preprocessing conditions for RDset. It shows that models consistently benefit from applying Z-score standardisation, variance thresholding, and VIF-based feature selection. Random Forest with SMOTE consistently delivers excellent results, while SVM continues to struggle with CTGAN-generated data unless feature selection is carefully applied. It compiles results for Random Forest, SVM, KNN, and CNN under various preprocessing steps (Z-score, variance threshold, Monte Carlo Dropout). Masum et al. Masum, Faruk, Shahriar, Qian, Lo and Adnan (2022) also employed a feature selection approach, achieving 0.99 ± 0.01 accuracy with Random Forest—similar to our 99.61% using SMOTE. Meanwhile, Hirano et al. Hirano, Hodota and Kobayashi (2022) introduced the RanSAP dataset and attained a 0.959 F1 in binary classification using Random Forest. Our CTGAN-based approach reached 99.54% accuracy, illustrating the impact of synthetic oversampling.

Table 14: Comparison of Techniques/Aspect with Related Works

Technique/Aspect	Current	Masum et al.Masum, Faruk, Shahriar, Qian, Lo and Adnan (2022)	Hirano et al.Hirano, Hodota and Kobayashi (2022)
Adversarial Handling	✓	×	×
Uncertainty Handling	✓	×	×
Explainable AI (SHAP/LIME)	✓	×	×
Continuous Learning	✓	×	×
RanSAP Dataset	✓	×	✓
RDset Dataset	✓	✓	×
Feature Calculation	✓	✓	✓
SMOTE for Imbalance	✓	×	×
PSO for Optimisation	✓	×	×
Accuracy	0.9977	0.99 ± 0.01	0.962
Precision	0.9972	0.99 ± 0.00	0.951
Recall	0.9983	0.97 ± 0.03	0.941
F1-Score	0.9978	0.97 ± 0.03	0.962
False Positive Rate	0.0024	xxx	0.059

Our framework surpasses the two compared works in Table 14 (where the check mark(✓) represents the technique/aspect used and crossmark (×) represents the opposite.), achieving 0.9977 accuracy and 0.9978 F1, reflecting more balanced detection (fewer false positives and negatives). We attribute this to adversarial training (CTGAN), uncertainty estimation, and enhanced explainability (SHAP/LIME), which collectively bolster reliability. Table 12 compares the proposed framework against two previous studies by Masum et al. and Hirano et al. The current framework outperforms both prior works across all metrics, including accuracy, precision, recall, and F1-score, largely due to the integration of advanced augmentation, uncertainty handling, and explainability techniques.

Section	Model	Accuracy	Precision	Recall	F1	Incorrect Predictions
Without Z-Score, Variance, High VIF	RF with SMOTE	0.9961	0.9949	0.9975	0.9962	-
	SVM with SMOTE	0.4989	1.0000	0.0001	0.0001	-
	KNN with SMOTE	0.9897	0.9854	0.9942	0.9898	-
	CNN with SMOTE	0.9687	0.9757	0.9614	0.9685	-
	RF with CTGAN	0.9954	0.9932	0.9976	0.9954	-
	SVM with CTGAN	0.5045	0.9999	0.0032	0.0064	-
	KNN with CTGAN	0.9796	0.9745	0.9875	0.9810	-
	CNN with CTGAN	0.9634	0.9702	0.9558	0.9630	-
	Monte Carlo CNN	0.9843	0.9848	0.9839	0.9843	912
With Z-Score, Variance, High VIF Removed	RF with SMOTE	0.9951	0.9935	0.9968	0.9951	-
	SVM with SMOTE	0.9809	0.9878	0.9738	0.9808	-
	KNN with SMOTE	0.9884	0.9850	0.9920	0.9885	-
	CNN with SMOTE	0.9877	0.9834	0.9923	0.9878	-
	RF with CTGAN	0.9962	0.9945	0.9979	0.9962	-
	SVM with CTGAN	0.5150	0.9870	0.0224	0.0438	-
	KNN with CTGAN	0.9891	0.9871	0.9910	0.9890	-
	CNN with CTGAN	0.9865	0.9820	0.9912	0.9865	-
	Monte Carlo CNN	0.9833	0.9852	0.9800	0.9826	691

Table 15: Performance Results of Different Models and Techniques on RDset

By blending feature engineering, data augmentation, optimisation, and explainability techniques (e.g., SHAP, LIME, PSO), our holistic framework sets a novel benchmark for ransomware detection. High accuracy, strong F1, and low false positives highlight its reliability in practical security contexts.

4.5 Discussion and Insights

This study explores the integration of uncertainty estimation, explainable AI, data augmentation, and advanced optimisation techniques—specifically, Monte Carlo Dropout, Bayesian Neural Networks (BNN), LIME, SHAP, SMOTE, CTGAN, and PSO—for ransomware detection, particularly in imbalanced datasets. The results underscore the necessity of tailoring these methods to the dataset's characteristics and the classification

objectives.

SMOTE and PSO moderately improved precision by reducing false positives in binary classification, but their impact on recall and F1 score was limited. This suggests that these methods primarily assist in addressing minority class representation without substantially enhancing the overall model balance. In multi-class tasks, SMOTE and PSO offered negligible improvements, indicating that RanSAP's intrinsic engineered feature quality already supports strong performance without requiring extensive preprocessing. Therefore, SMOTE and PSO should be applied selectively, depending on dataset traits and the specific classification challenges.

The SHAP analysis identified 'Hwrite' and 'Twrite' as key features for both binary and multi-class detection, highlighting their importance in distinguishing benign samples from ransomware. Additionally, SHAP revealed that certain features are uniquely significant for specific ransomware families. This insight allows for more refined detection strategies, enhancing the model's adaptability to various ransomware variants and enabling better targeting of model improvements for specific families.

When evaluating model performance with SMOTE and CTGAN, Random Forest, SVM, KNN, and CNN exhibited different responses. SMOTE consistently improved the models' performance by effectively mitigating class imbalance, with Random Forest achieving an impressive F1-score of 0.9963 on the combined dataset. In contrast, CTGAN produced mixed results; CNN benefitted greatly, achieving an F1-score of 0.9978, while SVM and KNN struggled, likely due to the presence of noise or outliers introduced by the synthetic data. These observations reaffirm the importance of model-specific tuning when employing augmentation techniques.

Uncertainty estimation also played a crucial role in enhancing model reliability. Monte Carlo Dropout produced an average uncertainty score of 0.15 in binary classification, which is particularly valuable in cybersecurity contexts where high-stakes decisions demand reliable confidence measures. Conformal prediction added an additional interpretability layer by quantifying uncertainty and identifying ambiguous cases that may require further review. Collectively, these approaches improved the reliability of predictions and provided

actionable insights into model confidence, supporting informed decision-making in practical deployments.

The analysis of Negative Predictive Value (NPV) revealed important real-world implications. NPV varied considerably across ransomware types in multi-class tasks. While classes such as 'Excel' and 'SDelete' achieved near-perfect NPV, ransomware families like 'Darkside' and 'GandCrab4' exhibited lower values, emphasising the need for per-class model refinements to minimise false negatives. Maintaining high NPV is essential in practical environments since misclassifying benign software as ransomware can lead to severe operational consequences.

These findings carry significant implications for both researchers and cybersecurity practitioners. In datasets like RanSAP, which are rich in distinctive features, simpler models may perform sufficiently well even without extensive data augmentation. However, datasets like RDset, which are characterised by strong class imbalance, benefit substantially from augmentation techniques such as SMOTE and CTGAN, particularly when these are combined with uncertainty-aware methods. SHAP analysis further underscores the importance of explainable AI by illuminating the contributions of individual features, thereby improving transparency and trust. Uncertainty quantification complements these explainability efforts by highlighting ambiguous predictions, ultimately establishing a solid foundation for robust, interpretable, and adaptive ransomware detection systems.

4.5.1 Recommendations

Across our experiments, a CNN with Monte Carlo Dropout and CTGAN consistently achieved the highest F1 score (0.9978) and accuracy (0.9977), making it a strong pick for scenarios requiring precise, reliable detection. For scalability, combining Random Forest with SMOTE proved effective (F1 = 0.9963) and is well-suited for faster training when computational resources are constrained. Hybridising CTGAN with dimensionality reduction or feature selection could further improve SVM and KNN.

We also recommend incorporating dynamic features (For example, storage access patterns) for added context, as combining static and dynamic features (RanSAP plus RDset can yield stronger results. Lastly, explainable AI tools like SHAP and LIME, used in tandem

with Monte Carlo dropout, enhance transparency and reliability —qualities that are vital for cybersecurity stakeholders in high-stakes settings. Prioritising these strategies can foster greater trust in AI-driven security solutions.

4.5.2 Limitations

One key limitation is the dataset scope. Although RanSAP spans multiple ransomware families and RDset includes varied legitimate samples, neither dataset may fully represent every emerging ransomware variant or real-world benign application. Newly discovered threats with different behaviours could require further testing and adaptation.

Another constraint arises from synthetic data noise. While CTGAN greatly aids adversarial training, it sometimes introduces outliers that complicate performance for certain algorithms (e.g., SVM, KNN). Consequently, synthetic data might not always perfectly capture the real-world distribution of ransomware, diminishing its effectiveness.

Computational overheads also pose a challenge. Methods involving Monte Carlo Dropout and iterative optimisations (e.g., PSO) can be resource-intensive, potentially limiting applicability in large-scale or real-time environments. Regular retraining or fine-tuning might become cumbersome under strict latency or resource constraints.

Additionally, class imbalance improvements from SMOTE and similar techniques were not uniformly beneficial across all classes. Ransomware families such as “Darkside” or “GandCrab4” showed relatively lower NPV and recall, indicating that certain features or representations might be lacking for more complex multi-class tasks.

Explainability remains partially limited, even though SHAP and LIME add transparency to model decisions. These post-hoc tools offer insights into predictions but do not make the underlying models intrinsically interpretable. Security analysts still need domain knowledge to interpret results and implement effective remediation measures.

Finally, real-time deployment has not been extensively tested. Although the models achieved high accuracy in offline experiments, sub-second detection or on-the-fly adaptation in production environments may require additional engineering, particularly in latency-sensitive scenarios.

4.5.3 Future Directions

Building on the current findings, several promising research avenues warrant exploration. Advanced optimisation techniques such as Bayesian Optimisation, Genetic Algorithms, or Grid Search could be employed to refine hyperparameters across a wide range of models and datasets. Moreover, the combination of Particle Swarm Optimisation (PSO) with CTGAN offers an opportunity to jointly optimise both feature selection and synthetic data generation, potentially leading to even more robust and well-calibrated models.

In addition to optimisation, enhanced resampling methods may provide further improvements in handling severe class imbalance. Techniques like Borderline-SMOTE, ADASYN, and SMOTE-ENN could surpass the performance of standard SMOTE, especially in scenarios where certain ransomware families are heavily underrepresented. By generating more informative synthetic samples, these methods can help models generalise better across minority classes.

Integrated explainable AI represents another important direction. The combination of SHAP and LIME with Counterfactual Explanations, alongside uncertainty quantification, can offer a more comprehensive understanding of model behaviour and decision-making. This integrated approach has the potential to significantly increase trust in AI-driven cybersecurity solutions by not only explaining why a prediction was made but also illustrating how alternative scenarios could have led to different outcomes.

Class-specific tuning remains a crucial area for improvement, particularly for ransomware families that currently exhibit lower Negative Predictive Value (NPV). Tailoring models through class-specific hyperparameter adjustments, targeted feature engineering, or specialised loss functions can help reduce false negatives and ensure that even the more challenging ransomware variants are detected reliably.

Expanding the framework to incorporate multiple datasets would also strengthen its generalisability. By integrating additional ransomware datasets and employing cross-dataset learning strategies, the framework could achieve greater robustness and adaptability when encountering new or evolving ransomware families.

Real-time detection capabilities offer another avenue for advancement. Future work could

involve developing lightweight, latency-optimised CNN architectures capable of generating rapid ransomware alerts. Complementing this, adaptive learning mechanisms may allow models to be updated dynamically as new ransomware strains emerge, ensuring that detection remains effective over time.

Finally, extending the framework's application beyond ransomware to include other types of malware such as rootkits, spyware, and advanced persistent threats (APTs) would demonstrate the broader feasibility of the techniques employed in this study. The combination of SMOTE, CTGAN, and Monte Carlo Dropout may prove equally valuable in these other cybersecurity domains, potentially leading to comprehensive AI-powered solutions for a wide array of cyber threats.

4.6 Conclusion

This study presents a unified ransomware detection framework that integrates data augmentation, explainable AI, and uncertainty quantification. By combining RanSAP and RDset datasets, the framework achieves marked performance improvements. When SMOTE was applied, Random Forest reached an F1-score of 0.9963, while a CNN with Monte Carlo Dropout attained 0.9906. Further incorporating CTGAN boosted the CNN's F1-score to 0.9978, underscoring the robustness of our approach.

Compared to existing methods, our framework delivers higher detection accuracy and a lower false positive rate, qualities vital for real-world deployment. SHAP-based explanations satisfy the demand for interpretability in high-stakes cybersecurity settings, while Monte Carlo Dropout safeguards against overconfident predictions by providing rigorous uncertainty estimates.

By merging diverse datasets with advanced modelling and augmentation techniques, this work advances the state of ransomware detection. Future research aims to refine real-time detection capabilities, explore dynamic behavioural features, and extend coverage to other forms of malware. Overall, our framework offers a solid foundation for scalable, interpretable, and reliable AI-based solutions in cybersecurity.

Chapter 5

Agentic SABRE: An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection

5.1 Introduction

The preceding chapter demonstrated that integrating data augmentation, explainable AI, and Monte Carlo Dropout uncertainty quantification yields a robust single-agent framework for ransomware detection, achieving F1-scores of up to 0.9978 on the RDset and RanSAP datasets Kabuye, Issac, Yumlembam and Neera (2025). However, that work employed a single unified model that processed all available feature signals together, without distinguishing between fundamentally different types of evidence. Ransomware exhibits two distinct behavioural signatures: a semantic signature encoded in the structural metadata of executable files (import tables, section entropy, PE flags), and a behavioural signature expressed through runtime I/O access patterns, memory allocation dynamics, and filesystem activity. These two signal types have different statistical properties, different distributional characteristics, and are most naturally captured by different model architectures operating on different datasets.

This chapter introduces Agentic SABRE (Semantic-Behavioural Arbitration for Ran-

somware Evaluation), a multi-agent framework that decomposes ransomware detection into two specialised agents — a Semantic Agent and a Behavioural Agent — each trained independently on its own modality-specific dataset. The framework fuses their outputs through a learned Decision Agent and applies an uncertainty-aware triage policy that governs whether the system may act autonomously or must escalate suspicious activity to a human analyst. Unlike the prior single-agent approach, which used uncertainty primarily as a post-hoc diagnostic signal, Agentic SABRE operationalises uncertainty as an executive control variable: it is embedded directly into the triage decision logic, determining whether high-risk predictions trigger autonomous containment or are held for human review.

The framework makes five principal contributions beyond the IEEE Access baseline:

- A multi-agent architecture that assigns semantic embeddings and behavioural telemetry to structurally heterogeneous specialised agents, replacing the single-model design with a modular pipeline.
- An uncertainty-aware inference mechanism using Monte Carlo Dropout at each agent, enabling both a predictive mean probability and an epistemic uncertainty estimate to be produced per agent per sample.
- A score-space CTGAN augmentation strategy that synthesises score vectors in the low-dimensional fusion space rather than augmenting raw features, preserving the uncertainty geometry of the agents while improving the Decision Agent's training distribution.
- A formal triage policy governed by a risk threshold τ and an uncertainty budget κ, with a stricter autonomous containment pair $(\tau_{\text{high}}, \kappa_{\text{low}})$, creating a calibrated computational contract between automated defence and analyst oversight.
- Post-hoc explainability via gradient saliency, permutation importance, and counterfactual analysis at both the agent and fusion levels, enabling local and global interpretation of all system decisions.

The remainder of this chapter is organised as follows. Section 5.2 provides background

on the limitations of monolithic detectors and the motivation for multi-agent uncertainty-aware design. Section 5.3 surveys the most relevant prior work. Section 5.4 presents the complete Agentic SABRE architecture. Section 5.5 formalises the triage policy and its formal algorithms. Section 5.6 analyses the triage geometry. Section 5.7 defines the threat model. Section 5.8 describes the explainability mechanisms. Section 5.9 details the experimental setup. Section 5.10 reports empirical results. Section 5.11 presents ablation studies. Section 5.12 discusses operational implications. Section 5.13 acknowledges limitations. Section 5.14 summarises the chapter's contributions.

5.2 Background and Motivation

5.2.1 Limitations of Monolithic Detectors

Ransomware has become a dominant form of cybercrime, driven by increasingly sophisticated techniques for evasion, polymorphism, and operational stealth. Traditional approaches to ransomware detection have relied on signature matching, heuristic rule sets, or simple statistical anomaly detection. Although effective for well-known threats, such approaches degrade rapidly in the presence of obfuscation, encryption, behavioural drift, and previously unseen attack families. Machine learning has emerged as a powerful tool, leveraging system call traces, API telemetry, filesystem access patterns, and related forensic signals Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022); Alraizza and Algarni (2023). However, purely model-based detection suffers from three persistent challenges.

First, data imbalance remains severe: benign activity dominates operational telemetry. Oversampling techniques such as SMOTE and generative models such as CTGAN have been proposed to mitigate minority-class scarcity Chawla, Bowyer, Hall and Kegelmeyer (2002); Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019). Second, semantic modelling using transformers and embedding-based architectures has improved contextual understanding of system logs and API sequences, yet these models often lack robustness under concept drift and out-of-distribution conditions Fernando, Komninos and Chen (2020). Third, most existing machine learning detectors provide only point predictions

and do not quantify predictive uncertainty, making it difficult to determine whether a model's output is reliable enough to drive autonomous containment decisions Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023).

The single-agent framework developed in the preceding chapter, like most prior ransomware detectors, processes all available features through a single learned model. This design imposes two structural limitations. First, it assumes that semantic and behavioural features can be jointly represented in a single feature space, which may not hold when the two modalities have fundamentally different statistical properties. Semantic PE-header embeddings are typically high-dimensional, sparse, and cluster strongly by malware family, whereas runtime telemetry features are low-dimensional, continuous, and exhibit heavy class overlap due to behavioural polymorphism and living-off-the-land techniques Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023). A single model must simultaneously learn both types of structure, potentially compromising its performance on each. Second, monolithic models provide a single point prediction or, at best, a single scalar uncertainty estimate for the joint input. This makes it impossible to determine which modality is driving uncertainty.

5.2.2 The Case for Multi-Agent Uncertainty-Aware Design

Recent work in cybersecurity has begun to explore uncertainty quantification as a means of addressing model unreliability in deployment Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023); Kabuye, Issac, Yumlembam and Neera (2025). Techniques such as Monte Carlo Dropout and deep ensemble variance estimation allow detectors to express confidence levels alongside predicted probabilities. Nevertheless, most existing cybersecurity systems use uncertainty only as an auxiliary diagnostic tool rather than integrating it into a structured triage or escalation policy. As a result, model uncertainty is rarely incorporated into actionable decision-making. Parallel to these developments, neuro-symbolic and multi-agent architectures have gained traction for their modularity and interpretability Besold et al. (2021). However, most prior systems treat multi-agent fusion as a simple

ensemble procedure, and few integrate uncertainty-aware reasoning or explicit policies for calibrated automation.

The multi-agent design in Agentic SABRE addresses both limitations by assigning each modality to a dedicated agent with its own uncertainty estimate. As noted in Ispahany, Islam, Islam and Khan (2024), most ransomware detection surveys identify the lack of uncertainty-aware decision mechanisms as a key gap; the literature overwhelmingly focuses on improving classification accuracy while leaving uncertainty unexploited at the system level. Agentic SABRE takes the next step by operationalising this uncertainty within a formal triage policy.

A critical distinction separates Agentic SABRE from conventional ensemble methods. Ensemble detectors aggregate predictions from multiple models trained on the same feature space, typically via majority vote or average probability. Agentic SABRE differs in three respects. First, its two agents operate on structurally heterogeneous input spaces — 384-dimensional PE embedding vectors versus 16-dimensional runtime telemetry statistics — making them complementary by design rather than redundant. Second, the triage decision conditions on per-agent epistemic uncertainty, not merely on the posterior probability; a standard ensemble produces no such per-model uncertainty signal. Third, the triage policy is an executable control mechanism that can formally withhold autonomous action under ambiguity — a property that soft label aggregation does not possess.

5.3 Related Work

5.3.1 Machine Learning for Ransomware Detection

Machine learning approaches to ransomware detection range from classical models operating on handcrafted features to deep learning systems that learn representations from raw telemetry Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023); Fernando, Komninos and Chen (2020). Classical methods such as random forests and support vector machines have shown strong performance on well-curated datasets but degrade rapidly under concept drift as ransomware tactics evolve Fernando and Komninos (2024); Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023). Deep learning ap-

proaches leveraging CNN and LSTM architectures on system call traces and filesystem access patterns achieve strong in-distribution performance but suffer from overconfidence on out-of-distribution samples Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022). Transformer-based models and LLM-derived embeddings have also been explored for security log analysis Tian, Zhang, Liu, Wang, Wu, Zhu, Zhang, Zhang, Yuan, Mao and Kim (2025); Zhang, Li, Zhang, Ye, Hu and Yan (2025), but these systems are typically standalone classifiers without uncertainty-aware fusion or structured triage logic. Recent work has also explored deep learning and blockchain-based frameworks for malware detection in IoT environments Pawar, Kumar, Meesala, Pareek, Addula and Shwetha (2024), underscoring the breadth of architectural approaches being applied to adaptive threat analysis.

5.3.2 Data Augmentation and Imbalance Handling

Imbalanced datasets remain a pervasive challenge in ransomware detection because benign activity vastly outweighs malicious traces. Oversampling approaches such as SMOTE Chawla, Bowyer, Hall and Kegelmeyer (2002) and generative tabular models such as CTGAN Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019); Ashrapov (2020) have been widely adopted to mitigate class imbalance. While these methods improve training stability, they are typically employed only at the data preprocessing stage and are rarely integrated into multi-stage detection pipelines. Agentic SABRE differs by applying CTGAN not to raw features but to the score space, enriching rare high-risk cases entering the fusion layer without altering agent-level probabilities or uncertainties.

5.3.3 Uncertainty Quantification in Cybersecurity

The importance of uncertainty quantification in cybersecurity has been recognised by several recent works Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023); Doan, Nguyen, Montague, Camtepe and Kanhere (2024). Monte Carlo Dropout Gal and Ghahramani (2016), deep ensembles, and Bayesian neural networks have been applied to intrusion detection and malware classification. However, existing systems consistently use uncertainty as a diagnostic output rather than as a governance variable that constrains system behaviour. Agentic SABRE

advances this direction by embedding per-agent uncertainty directly into an executable triage policy with formal threshold semantics.

5.3.4 Multi-Agent and Neuro-Symbolic Approaches

Multi-agent learning has been explored for adaptive intrusion detection, cooperative anomaly detection, and adversarial resilience Jia, Cui, Wang, Ding, Han and Ma (2024); Soltani, Khajavi, Siavoshani and Jahangir (2023). Parallel advances in neuro-symbolic systems highlight the benefits of combining neural representations with symbolic rules and knowledge bases Besold et al. (2021); Yang, Some, Bain and Kang (2025). However, prior systems typically lack: (i) agent-level uncertainty estimation, (ii) uncertainty-aware fusion, and (iii) operational triage policies linking uncertainty and risk to automated or human-in-the-loop decisions. Agentic SABRE directly addresses all three gaps.

5.3.5 Comparative Summary

Table 16 summarises how representative prior systems compare to Agentic SABRE across the dimensions that define the design space.

Table 16: Comparison of Agentic SABRE with representative prior ransomware and intrusion detection systems across key design dimensions. ✓ = present; ○ = partial; × = absent.

System	Multi-Agent	Uncertainty	Triage Policy	Symbolic Cues	XAI	Concept Drift	Score-Level Aug.
Fernando et al. Fernando, Komninos and Chen (2020)	×	×	×	×	×	×	×
Razaulla et al. Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023)	×	×	×	×	×	○	×
Deep-learning detectors Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022)	×	×	×	×	○	×	×
LLM-LADE Zhang, Li, Zhang, Ye, Hu and Yan (2025)	×	×	×	○	✓	×	×
Kabuye et al. Kabuye, Issac, Yumlembam and Neera (2025)	×	✓	×	×	✓	×	×
Soltani et al. Soltani, Khajavi, Siavoshani and Jahangir (2023)	✓	×	×	×	×	○	×
Agentic SABRE (ours)	✓	✓	✓	✓	✓	✓	✓

5.4 The Agentic SABRE Architecture

5.4.1 System Overview

Agentic SABRE is a five-stage pipeline illustrated in Figure 18. Stage 1 performs hybrid feature extraction: PE-header metadata is embedded using a sentence-transformer encoder to produce 384-dimensional semantic vectors, and runtime I/O traces are processed via sliding-window statistics to produce 16-dimensional behavioural feature vectors. Stage 2 applies SMOTE at the feature level independently for each modality. Stage 3 trains the Semantic and Behavioural CNN agents independently with Monte Carlo Dropout

enabled, each producing a mean probability and uncertainty estimate per sample. Stage 4 collects agent score tuples, applies CTGAN augmentation in the score space, and trains the Decision Agent MLP on the augmented score dataset. Stage 5 implements the uncertainty-aware triage policy at deployment time. The complete training and deployment procedure is formalised in Algorithm 5.

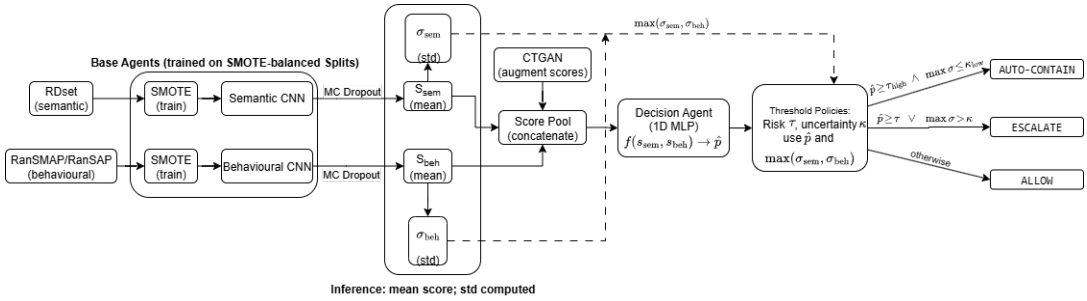


Figure 18: Agentic SABRE operational pipeline. Semantic and Behavioural CNNs (with Monte Carlo Dropout) produce mean probability and epistemic uncertainty per agent. Scores are CTGAN-augmented and passed to the Decision Agent MLP. The triage orchestrator applies thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ to route each sample to AUTO-CONTAIN, ESCALATE, or ALLOW.

5.4.2 Semantic Agent

The Semantic Agent processes 384-dimensional PE-header embedding vectors derived from structured metadata — import tables, section entropy values, PE flags, version strings, and related static attributes — extracted from the RDset dataset Mathur (2020). Embeddings are computed using a pre-trained sentence-transformer encoder Reimers and Gurevych (2019) applied to concatenated attribute strings, mapping each executable to a dense semantic representation. The agent architecture is a 1D CNN with depth $L = 3$, kernel size $K = 3$, and $C = 64$ filters per convolutional layer, followed by ReLU activation, batch normalisation, global average pooling, and a sigmoid output unit. A dropout layer ($p_{\text{drop}} = 0.3$) is placed before the output unit and retained active at inference time to enable Monte Carlo sampling.

5.4.3 Behavioural Agent

The Behavioural Agent processes 16-dimensional feature vectors derived from the RanSMAP dataset Hirano, Hodota and Kobayashi (2022); Hirano and Kobayashi (2025), which

records runtime I/O and memory access telemetry for ransomware-infected and benign hosts. Features are computed over fixed 30-second sliding windows and include: LBA (Logical Block Address) variance for read and write operations, GPA (Guest Physical Address) variance for memory reads and writes, memory operation counts, I/O entropy, and total bytes transferred. Formally, for a window of n measurements $\{x_j\}$:

$$\mathrm{Var}(x) = \frac{1}{n}\sum_{j=1}^{n}(x_j - \bar{x})^2, \tag{5.1}$$

$$\mathcal{H}(x) = -\sum_{j=1}^{n} p(x_j)\log_2(p(x_j) + \epsilon). \tag{5.2}$$

The agent architecture mirrors the Semantic CNN with $L = 3$ convolutional layers, $K = 3$, $C = 64$ filters, and a dropout layer before the sigmoid output unit ($p_{\mathrm{drop}} = 0.3$).

5.4.4 Monte Carlo Dropout Uncertainty Estimation

Both agents use Monte Carlo (MC) Dropout Gal and Ghahramani (2016) to quantify epistemic uncertainty. The formal procedure is given in Algorithm 6. At inference time, the dropout layers are kept active, and $T = 30$ stochastic forward passes are performed for each sample. The predictive mean probability and standard deviation across passes are:

$$\bar{p} = \frac{1}{T}\sum_{t=1}^{T} p_t, \qquad \sigma = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(p_t - \bar{p})^2}, \tag{5.3}$$

where p_t is the output of the t-th stochastic forward pass. This yields per-agent pairs $(\bar{p}_{\mathrm{sem}}, \sigma_{\mathrm{sem}})$ and $(\bar{p}_{\mathrm{beh}}, \sigma_{\mathrm{beh}})$.

MC Dropout was chosen over deep ensembles and Bayesian neural networks for three practical reasons: it requires no architectural changes beyond retaining dropout at inference; its per-sample cost of T forward passes is substantially lower than training and maintaining $M \geq 5$ independent ensemble members; and the agents' training pipelines already include dropout regularisation, so no additional training overhead is incurred.

5.4.5 Score Fusion and Decision Agent

The fused score vector is assembled from the two agent outputs:

$$\mathbf{s} = \begin{bmatrix} \bar{p}_{\text{sem}} \\ \bar{p}_{\text{beh}} \end{bmatrix}, \tag{5.4}$$

optionally extended with per-agent uncertainties at deployment to form the full four-dimensional vector:

$$\mathbf{v} = \big(\bar{p}_{\text{sem}},\ \bar{p}_{\text{beh}},\ \sigma_{\text{sem}},\ \sigma_{\text{beh}}\big) \in \mathbb{R}^4. \tag{5.5}$$

A CTGAN Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019) is trained on the score dataset $\mathcal{D}_S = \{(\mathbf{s}_i, y_i)\}$ to generate synthetic score vectors that enrich the minority class in score space (Algorithm 7). The augmented dataset $\mathcal{D}_S \cup \tilde{\mathcal{D}}_S$ is used to train the Decision Agent MLP, a two-hidden-layer network (64 units per layer, ReLU activations, sigmoid output) that learns a mapping $f : \mathbb{R}^4 \to [0, 1]$ producing the fused risk estimate:

$$\hat{p} = f(\mathbf{v}). \tag{5.6}$$

5.4.6 Uncertainty Aggregation

To ensure conservative triage behaviour, agent-level uncertainties are aggregated via a max-operator:

$$\sigma_{\max} = \max(\sigma_{\text{sem}},\ \sigma_{\text{beh}}), \tag{5.7}$$

so that elevated uncertainty in either modality constrains autonomous action. This conservative choice prevents high semantic confidence from overriding high behavioural uncertainty, ensuring that the system seeks human oversight whenever any modality is epistemically unreliable.

Algorithm 5 Agentic SABRE: Training and Deployment Pipeline

Input (training): Raw semantic data D_{sem}, behavioural logs D_{beh}
Input (deployment): New sample $x_{\text{sem}}, x_{\text{beh}}$
Output (deployment): Decision $\in \{\text{AUTO-CONTAIN}, \text{ESCALATE}, \text{ALLOW}\}$
// Phase 1: Hybrid feature extraction
Extract semantic embeddings $\mathbf{z}_i$ from D_{sem} using encoder $\mathcal{E}(\cdot)$
Extract behavioural feature vectors $\mathbf{b}_i$ from D_{beh} via sliding-window statistics
// Phase 2: Feature-level augmentation
Apply SMOTE to $\{\mathbf{z}_i\}$ and $\{\mathbf{b}_i\}$ independently to address class imbalance
// Phase 3: Train base agents with MC Dropout
Train Semantic Agent CNN A_{sem} on $\{\mathbf{z}_i\}$
Train Behavioural Agent CNN A_{beh} on $\{\mathbf{b}_i\}$
// Phase 4: Score-level augmentation and Decision Agent
for each training sample i do
    Compute mean scores $\bar{p}_{\text{sem}}^{(i)}$, $\bar{p}_{\text{beh}}^{(i)}$ via MC Dropout (Alg. 6)
    Form score vector $s_i = [\bar{p}_{\text{sem}}^{(i)}, \bar{p}_{\text{beh}}^{(i)}]$
end for
Train CTGAN on $\{(s_i, y_i)\}$ and obtain augmented scores $\tilde{\mathcal{D}}_S$ (Alg. 7)
Train Decision Agent A_{dec} on $\mathcal{D}_S \cup \tilde{\mathcal{D}}_S$
// Phase 5: Deployment-time inference and triage
Obtain $(\bar{p}_{\text{sem}}, \sigma_{\text{sem}})$ for x_{sem} via MC Dropout
Obtain $(\bar{p}_{\text{beh}}, \sigma_{\text{beh}})$ for x_{beh} via MC Dropout
Form $\mathbf{v} = [\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}]$ and compute $\hat{p} = A_{\text{dec}}(\mathbf{v})$
Compute $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$
Apply triage policy (Alg. 8) using thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$

5.5 Uncertainty-Aware Triage Policy

5.5.1 Policy Design and Thresholds

The triage orchestrator converts the pair $(\hat{p}, \sigma_{\max})$ into one of three operational actions — Auto-Contain, Escalate, or Allow — using two pairs of thresholds. The general pair (τ, κ) defines the escalation boundary, and the stricter pair $(\tau_{\text{high}}, \kappa_{\text{low}})$ defines the autonomous containment region. These satisfy the ordering constraints:

$$\tau \leq \tau_{\text{high}}, \qquad \kappa_{\text{low}} \leq \kappa, \tag{5.8}$$

meaning that autonomous containment requires higher fused risk and lower uncertainty than escalation. Axis-aligned thresholds are deliberately chosen over learned non-linear boundaries to preserve interpretability and auditability in regulated operational environments.

Algorithm 6 MC Dropout Uncertainty Estimation for a Neural Agent

Input: Trained neural agent A with dropout layers, sample x, number of passes T
Output: Predictive mean $\bar{p}$, epistemic uncertainty σ
Set dropout modules to training mode (active at inference)
for $t = 1$ to T do
    $p_t \leftarrow A(x)$ ▷ stochastic forward pass with dropout
end for
$\bar{p} \leftarrow \frac{1}{T} \sum_{t=1}^{T} p_t$
$\sigma \leftarrow \sqrt{\frac{1}{T} \sum_{t=1}^{T} (p_t - \bar{p})^2}$
return $(\bar{p}, \sigma)$

Algorithm 7 Score-Level CTGAN Augmentation for Decision Agent Training

Input: Trained agents $A_{\text{sem}}, A_{\text{beh}}$; training data $\{\mathbf{z}_i\}, \{\mathbf{b}_i\}$; labels $\{y_i\}$
Output: Augmented score dataset $\tilde{\mathcal{D}}_S$ for training A_{dec}
// Step 1: Collect score tuples
for each training sample i do
    $(\bar{p}^{(i)}_{\text{sem}}, \sigma^{(i)}_{\text{sem}}) \leftarrow$ Alg. 6$(A_{\text{sem}}, \mathbf{z}_i, T)$
    $(\bar{p}^{(i)}_{\text{beh}}, \sigma^{(i)}_{\text{beh}}) \leftarrow$ Alg. 6$(A_{\text{beh}}, \mathbf{b}_i, T)$
    $s_i \leftarrow [\bar{p}^{(i)}_{\text{sem}}, \bar{p}^{(i)}_{\text{beh}}]$
end for
Construct $\mathcal{D}_S = \{(s_i, y_i)\}$
// Step 2: Train CTGAN in score space
Train CTGAN generator G and discriminator D on $\mathcal{D}_S$ conditioned on labels y_i
Generate synthetic samples $\tilde{s}_j = G(z_j, y_j)$ and form $\tilde{\mathcal{D}}_S = \{(\tilde{s}_j, y_j)\}$
return $\tilde{\mathcal{D}}_S$

5.5.2 Decision Rules

The orchestrator applies:

$$\text{AUTO-CONTAIN} \quad \text{if} \quad \hat{p} \geq \tau_{\text{high}} \ \wedge \ \sigma_{\max} \leq \kappa_{\text{low}}, \tag{5.9}$$

$$\text{ESCALATE} \quad \text{if} \quad \hat{p} \geq \tau \ \vee \ \sigma_{\max} > \kappa, \tag{5.10}$$

$$\text{ALLOW} \quad \text{otherwise.} \tag{5.11}$$

This logic ensures: (a) automatic containment is restricted to high-confidence, low-uncertainty predictions; (b) human escalation is triggered by either elevated risk or elevated epistemic uncertainty; (c) samples with low risk and low uncertainty are autonomously allowed without analyst involvement. The complete policy decision procedure is formalised in Algorithm 8.

5.5.3 Policy Optimisation

Thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ are jointly selected by Bayesian optimisation using the Tree-structured Parzen Estimator (TPE) Akiba, Sano, Yanase, Ohta and Koyama (2019), maximising the scalarised objective:

$$J(\tau, \kappa) = \text{BalAcc}(\tau, \kappa) - \lambda \cdot \text{EscRate}(\tau, \kappa), \tag{5.12}$$

subject to Equation (5.8). The trade-off parameter $\lambda = 0.5$ balances detection accuracy against escalation workload. Two operational configurations are derived: a safety-optimal configuration maximising balanced accuracy, and a low-escalation configuration minimising escalation volume at near-equal accuracy.

Algorithm 8 Uncertainty-Aware Triage Policy

Input: Fused risk $\hat{p}$, agent uncertainties $\sigma_{\text{sem}}, \sigma_{\text{beh}}$
Input: Thresholds τ, τ_{high}, κ, κ_{low}
Output: Decision $\in \{\text{AUTO-CONTAIN}, \text{ESCALATE}, \text{ALLOW}\}$
$\sigma_{\max} \leftarrow \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$
if $\hat{p} \geq \tau_{\text{high}}$ and $\sigma_{\max} \leq \kappa_{\text{low}}$ then
    return AUTO-CONTAIN ▷ high risk, low uncertainty
else if $\hat{p} \geq \tau$ or $\sigma_{\max} > \kappa$ then
    return ESCALATE ▷ risky or uncertain
else
    return ALLOW ▷ benign and confident
end if

5.6 Policy Analysis

The triage mechanism governs the joint behaviour of the fused probability $\hat{p}$ and the maximal epistemic uncertainty $\sigma_{\max}$. The policy partitions the space

$$\mathcal{X} = \left\{(\hat{p}, \sigma_{\max}) \in [0,1] \times [0,1]\right\} \tag{5.13}$$

into three disjoint regions corresponding to Allow, Escalate, and Auto-Contain. The decision boundaries are defined implicitly by the inequalities in Equations (5.9)–(5.11): the vertical lines $\hat{p} = \tau$ and $\hat{p} = \tau_{\text{high}}$ together with the horizontal lines $\sigma_{\max} = \kappa$ and $\sigma_{\max} = \kappa_{\text{low}}$ generate a partitioned rectangle whose subregions determine the triage out-

come.

Samples located near the boundary corners $(\tau_{\text{high}}, \kappa_{\text{low}})$ and (τ, κ) exhibit the greatest instability under adversarial perturbation. However, due to the monotonic structure of the fusion classifier, the gradient of $\hat{p}$ is largest where semantic evidence dominates, meaning the neighbourhood of $(\tau_{\text{high}}, \kappa_{\text{low}})$ tends to be sparsely populated — an important stability property.

The threshold optimisation objective in Equation (5.12) identifies threshold pairs that shift these decision boundaries. Increasing τ reduces the escalation region but simultaneously risks enlarging the allow region unless offset by a simultaneous adjustment to κ. The calibrated trade-off therefore emerges from solving the non-convex problem of maximising $J(\tau, \kappa)$ while maintaining robustness to the uncertainty geometry in Equation (5.13). This interplay between fused risk, epistemic uncertainty, and threshold selection constitutes the mathematical core of the Agentic SABRE policy framework.

5.7 Threat Model

The operational setting assumes an adversary capable of deploying ransomware samples that vary in semantic structure, behavioural dynamics, or both. The defender observes two forms of telemetry: a semantic representation $\mathbf{z}_i \in \mathbb{R}^{d_s}$ encoded from static artefacts and a behavioural representation $\mathbf{b}_i \in \mathbb{R}^{d_b}$ extracted from temporal I/O activity. The adversary operates under the constraint that behavioural logs and static metadata cannot be forged arbitrarily without compromising the malware's execution.

The threat model considers adversarial manoeuvres that attempt to reduce the classifier's confidence while preserving functional attack behaviour. These manoeuvres can target the semantic agent by modifying static descriptors; however, such modifications typically shift the semantic embedding distribution, increasing the Wasserstein distance in Equation (5.23) only marginally, and often increasing epistemic uncertainty. Similarly, behavioural mimicry attacks attempt to disguise malicious activity so that the derived empirical distributions

$$p(\mathbf{b} \mid y = 1), \qquad p(\mathbf{b} \mid y = 0), \tag{5.14}$$

appear closer to each other. Such mimicry produces elevated uncertainty in the behavioural agent due to the mismatch between expected and observed feature variance.

The model assumes that the attacker does not possess knowledge of the triage parameters τ, κ, τ_{high}, κ_{low}, nor the calibration temperature applied to the fusion classifier. Even if such knowledge were available, the attacker would still face the constraint that reducing the fused probability $\hat{p}$ (Equation (5.6)) necessarily increases behavioural deviation, thus raising the uncertainty term $\sigma_{\max}$ (Equation (5.7)). Consequently, ambiguous or adversarial samples are redirected to human analysts rather than executed automatically, providing robustness against a broad class of evasion strategies.

5.8 Post-Hoc Explainability

5.8.1 Permutation Importance

Global feature importance for each agent is assessed via permutation importance: the drop in AUC when feature j is randomly permuted across the test set:

$$I_j = \text{AUC}_{\text{orig}} - \text{AUC}_j. \tag{5.15}$$

For the Behavioural Agent, the highest I_j values correspond to dispersion-based statistics — read_LBA_variance, write_LBA_variance, mem_read_GPA_variance, and mem_write_GPA_v — confirming that ransomware distinguishes itself through irregular, spatially bursty memory and storage access footprints rather than raw activity volume.

5.8.2 Gradient Saliency

For the Semantic Agent, gradient-based sensitivity is computed as:

$$s_k = \left| \frac{\partial p_{\text{sem}}(\mathbf{z})}{\partial z_k} \right|, \tag{5.16}$$

identifying which embedding dimensions most strongly influence the semantic prediction. High-magnitude gradients concentrate in embedding directions associated with entropy-derived section statistics, import-table irregularities, and anomalous structural metadata,

indicating that the semantic agent encodes interpretable PE-level cues. The embedding dimensions most frequently perturbed under counterfactual analysis are summarised in Table 17.

Table 17: Most frequently perturbed embedding dimensions across semantic counterfactuals. "Count in Top-8" reports how often the dimension appears among the highest-magnitude normalised changes per sample ($N = 50$).

Latent dim	Count in Top-8	Interpretation
269	4/5	recurrent semantic direction
270	4/5	recurrent semantic direction
170	4/5	recurrent semantic direction
271	4/5	recurrent semantic direction
362	4/5	recurrent semantic direction
363	4/5	recurrent semantic direction
314	2/5	secondary direction
194	2/5	secondary direction

5.8.3 Counterfactual Analysis

Counterfactual explanations identify the minimal perturbation $\tilde{\mathbf{b}}$ to a behavioural feature vector $\mathbf{b}$ required to flip the agent's decision below a threshold $\theta = 0.49$:

$$\Delta^\star = \arg\min_{\tilde{\mathbf{b}} \in \mathcal{N}_\epsilon(\mathbf{b})} \|\tilde{\mathbf{b}} - \mathbf{b}\|_2, \quad \text{subject to} \quad p_{\text{beh}}(\tilde{\mathbf{b}}) \leq \theta. \tag{5.17}$$

Counterfactual experiments on 50 high-confidence malicious samples show that only small relative reductions in LBA and GPA variance are sufficient to cross the decision boundary, with mean perturbation norm $\mathbb{E}[\|\Delta^\star\|_2] \approx 0.12$. Representative examples are shown in Table 18. This implies that ransomware's behavioural signature is fragile in the variance dimensions — an attacker who spreads I/O operations uniformly across blocks can substantially reduce the agent's malicious probability, but doing so fundamentally conflicts with the encryption-burst strategy of all known ransomware families (ATT&CK T1486).

Table 18: Representative behavioural counterfactuals. Relative changes per Equation (5.17).

Sample	$p_{\text{beh}}(\mathbf{b})$	$p_{\text{beh}}(\tilde{\mathbf{b}})$	$\|\Delta^\star\|_2$	Most changed features
CF_1	0.70	0.30	0.12	read_LBA_var (−1.6%), mem_write_GPA_var (+0.6%)
CF_2	0.60	0.10	0.09	write_LBA_var (−0.9%), mem_read_GPA_var (−0.6%)
CF_3	0.90	0.40	0.14	mem_write_GPA_var (−2.5%), write_LBA_var (+1.4%)

At the fusion level, semantic counterfactuals are computed in embedding space. Given an original embedding $\mathbf{z}_0$, a counterfactual $\mathbf{z}_{\text{cf}}$ is obtained by solving:

$$\mathbf{z}_{\text{cf}} = \arg\min_{\mathbf{z}} \Big(\max\big(0, \text{logit}(\mathbf{z}) - \text{logit}(\theta_{\text{sem}})\big) \Big)^2 + \lambda \|\mathbf{z} - \mathbf{z}_0\|_2^2, \tag{5.18}$$

constrained to a trust region $\|\mathbf{z}_{\text{cf}} - \mathbf{z}_0\|_2 \leq \epsilon$. Representative results are shown in Table 19. The semantic decision boundary requires substantially larger perturbations ($\mathbb{E}[\mathcal{C}_{\text{sem}}] \approx 3.27$) than the behavioural boundary, consistent with the near-perfect discrimination of the Semantic Agent on RDset. Summary statistics across $N = 50$ borderline-malicious samples are reported in Table 20.

Table 19: Representative semantic counterfactual shifts. Latent directions with the largest impact correspond to PE structural irregularities and entropy-driven embedding dimensions.

Sample	$p_{\text{sem}}(\mathbf{z}_0)$	$p_{\text{sem}}(\mathbf{z}_{\text{cf}})$	$\|\Delta\mathbf{z}\|_2$ (normalised)	logit shift
CF_1^{sem}	1.0000	0.0000	2.74	$32.42 \rightarrow -12.49$
CF_2^{sem}	1.0000	0.0000	3.33	$37.60 \rightarrow -17.94$
CF_3^{sem}	1.0000	0.0000	3.34	$37.45 \rightarrow -16.24$

Table 20: Summary statistics for semantic counterfactual perturbations over $N = 50$ borderline-malicious samples.

Metric	Mean	Std	$q_{0.10}$	$q_{0.90}$
$\|\Delta\mathbf{z}\|_2$	3.272	0.173	3.244	3.340
$\|\Delta\mathbf{z}\|_2 / \|\mathbf{z}_0\|_2$	0.490	0.028	0.467	0.514
Flip success to $\theta = 0.49$		100.0%		

The near-additive structure of fusion-level counterfactuals implies a super-additive adversarial cost:

$$\mathcal{C}_{\text{fus}}^2 \gtrsim \mathcal{C}_{\text{sem}}^2 + \mathcal{C}_{\text{beh}}^2, \tag{5.19}$$

meaning an attacker must enact coordinated changes across both dynamic execution traces and static binary structure, which is significantly more difficult than manipulating either modality in isolation.

5.9 Datasets and Experimental Setup

5.9.1 Datasets

RDset. RDset Mathur (2020) contains 138,047 PE-file records (41,123 ransomware; 96,924 benign) with 384-dimensional semantic embedding vectors. The dataset spans multiple ransomware families and is used to train and evaluate the Semantic Agent. Evaluation uses stratified 80/20 train-test splits with 2,000 test samples per class.

RanSMAP. The RanSAP / RanSMAP dataset Hirano, Hodota and Kobayashi (2022); Hirano and Kobayashi (2025) contains 11,820 execution traces drawn from 24 ransomware families and benign workloads across diverse hardware configurations. The 16-dimensional feature vectors are extracted from 30-second sliding windows. The native class ratio is approximately 598:1. Balanced evaluation uses 1,000 samples per class; the full imbalanced distribution is also evaluated to characterise real-world deployment behaviour.

Cross-source fusion rationale. RDset and RanSMAP contain no overlapping samples — RDset consists of static PE metadata while RanSMAP records runtime execution traces from instrumented hypervisors. Raw features are never merged; fusion occurs exclusively at the score level, reflecting the realistic SOC scenario in which a static sandbox scanner and a runtime endpoint monitor report simultaneously on the same host. CTGAN labels use binary $y_i \in \{0, 1\}$ (benign/ransomware) without modification.

Dataset limitations. Neither dataset includes post-2021 ransomware families such as Cl0p, BlackCat/ALPHV, or LockBit 3.0. Generalisation to modern ransomware lineages is identified as a priority direction for future work. RDset does not employ a family-wise or temporal split: samples are partitioned at the binary level, so families present in the training set may appear under different instances in the test set, introducing the risk of structural similarity bias.

5.9.2 Feature Extraction and Preprocessing

Semantic embeddings are computed from concatenated PE attribute strings using a pre-trained sentence-transformer Reimers and Gurevych (2019). Behavioural features are computed via sliding-window statistics (variance, entropy, count) over 30-second activity

windows. SMOTE Chawla, Bowyer, Hall and Kegelmeyer (2002) is applied independently to each dataset to mitigate class imbalance at the feature level before agent training. Score-level CTGAN augmentation is applied after base-agent training, enriching rare high-risk score configurations without altering the learned agent representations.

5.9.3 Evaluation Metrics

Performance is evaluated using: accuracy, AUC (Area Under the ROC Curve), and confusion-matrix-derived metrics (TN, FP, FN, TP). Additionally, the escalation rate — the fraction of samples routed to ESCALATE — is reported as a primary operational cost metric, since escalated samples require analyst attention and introduce response latency. Reductions in escalation volume at fixed recall correspond directly to deployability improvements.

5.10 Results

5.10.1 Individual Agent Performance

Table 21 reports standalone agent performance prior to fusion. The Semantic Agent achieves perfect accuracy and AUC (1.0000) on RDset, reflecting the near-disjoint class-conditional clustering of PE-header embeddings in the semantic space. This saturated result is used deliberately as a stress-test for the fusion and triage layers: the question is whether behavioural uncertainty can prevent overconfident autonomous containment even when semantic evidence is maximal. Generalisation caveats apply: RDset does not employ family-wise or temporal splits, so structural similarity bias is possible.

The Behavioural Agent achieves 0.9872 accuracy and 0.6030 AUC on the full imbalanced RanSMAP distribution. The accuracy/AUC discrepancy reflects the class prior: with 598:1 imbalance, a classifier that assigns low probabilities to almost all samples achieves high accuracy while exhibiting weak rank separability. The Behavioural Agent's primary role in the framework is consequently as a modulator of $\sigma_{\max}$ rather than a standalone detector.

Table 21: Individual agent performance prior to fusion.

Agent	Dataset	Accuracy	AUC
Semantic CNN	RDset	1.0000	1.0000
Behavioural CNN	RanSMAP	0.9872	0.6030

5.10.2 Generalisation Across Evaluation Splits

To assess generalisation beyond the standard random-stratified split, Agentic SABRE was evaluated across four complementary partitions of RanSMAP using 6 independent random seeds. Table 22 reports mean AUROC, AUPRC, and accuracy with standard deviations. Temporal and hardware holdout performance closely matches the random-stratified baseline (AUROC 0.960 and 0.939 respectively), confirming that Agentic SABRE does not overfit to specific recording sessions or hardware configurations. Family holdout reveals a meaningful generalisation gap (AUROC 0.707), consistent with the structural limitation that behavioural training families are absent from the test set. Crucially, the uncertainty mechanism responds correctly to this degraded regime — samples from unseen families exhibit elevated σ_{beh} and are routed to ESCALATE rather than AUTO-CONTAIN, maintaining analyst oversight precisely where the model is least reliable.

Table 22: Generalisation performance across four evaluation splits (mean ± std over 6 seeds). Family holdout uses families unseen during training; hardware holdout uses unseen hardware configurations; temporal holdout separates by collection date.

Split	AUROC	AUPRC	Accuracy
Random-stratified	0.961 ± 0.003	0.963 ± 0.003	0.887 ± 0.005
Hardware holdout	0.939 ± 0.005	0.942 ± 0.006	0.850 ± 0.008
Temporal holdout	0.960 ± 0.005	0.929 ± 0.010	0.888 ± 0.008
Family holdout	0.707 ± 0.010	0.710 ± 0.008	0.660 ± 0.008

5.10.3 Fusion and Triage Performance

Table 23 reports the fusion and triage performance under both threshold configurations. On RDset, both configurations achieve AUC 1.0000 with at most one misclassification, confirming that the fusion layer preserves the near-perfect separation of the semantic space. On RanSMAP, the elevated escalation rate reflects the behavioural variance: $\sigma_{\max}$ is frequently above κ for RanSMAP samples, correctly routing ambiguous behavioural predictions to human analysts. The low-escalation configuration achieves a 4.9% rela-

tive reduction in false escalations compared to the safety-optimal configuration at equal recall.

Table 23: Fusion and triage performance under safety-optimal and low-escalation policies.

Policy	Dataset	Accuracy	AUC	AUTO	ESCALATE	ALLOW	(TN, FP, FN, TP)
Safety-Optimal	RDset	0.99999	1.0000	2,114	99,438	36,495	(41,323, 0, 1, 96,723)
Low-Escalation	RDset	0.99999	0.99999	2,099	99,499	36,449	(41,322, 1, 0, 96,724)
Safety-Optimal	RanSMAP	0.9406	0.5394	15,591	696,675	39,322	(172, 1,129, 43,630, 706,784)
Low-Escalation	RanSMAP	0.9405	0.5397	15,697	696,395	39,496	(171, 1,088, 43,656, 706,673)

Threshold values for the safety-optimal configuration are $\tau = 0.999999$, $\kappa = 0.458$, $\tau_{\text{high}} = 0.99$, $\kappa_{\text{low}} = 0.229$. The near-unity τ reflects the bimodal clustering of fused probabilities on RDset (near 0 or 1), requiring a strict threshold to restrict AUTO-CONTAIN to genuinely high-confidence predictions. Performance is robust to small threshold perturbations: both configurations yield accuracy 0.99999 on RDset and 0.9406/0.9405 on RanSMAP, confirming stability.

5.10.4 Calibration and Policy Geometry

Figure 19 confirms that the fused risk score $\hat{p}$ is well-calibrated:

$$\mathbb{P}(y = 1 \mid \hat{p} = q) \approx q, \tag{5.20}$$

with the reliability diagram tracking the diagonal closely across bins. This ensures that $\hat{p}$ has a consistent probabilistic interpretation across datasets, making the triage thresholds transferable between deployments after recalibration via Equation (5.12).

Figure 20 visualises the triage policy geometry in the $(\hat{p}, \sigma_{\max})$ plane. The axis-aligned thresholds partition the space into three regions: benign samples concentrate in the low-risk, low-uncertainty Allow region; high-risk ransomware samples with low uncertainty are routed to Auto-Contain; and samples with elevated uncertainty — regardless of risk level — are routed to Escalate.

5.10.5 ROC Analysis Under Fusion

Figure 21 shows the ROC curves for both agents and the fused classifier on each dataset. On RanSMAP, fusion substantially improves separability over the behavioural agent alone,

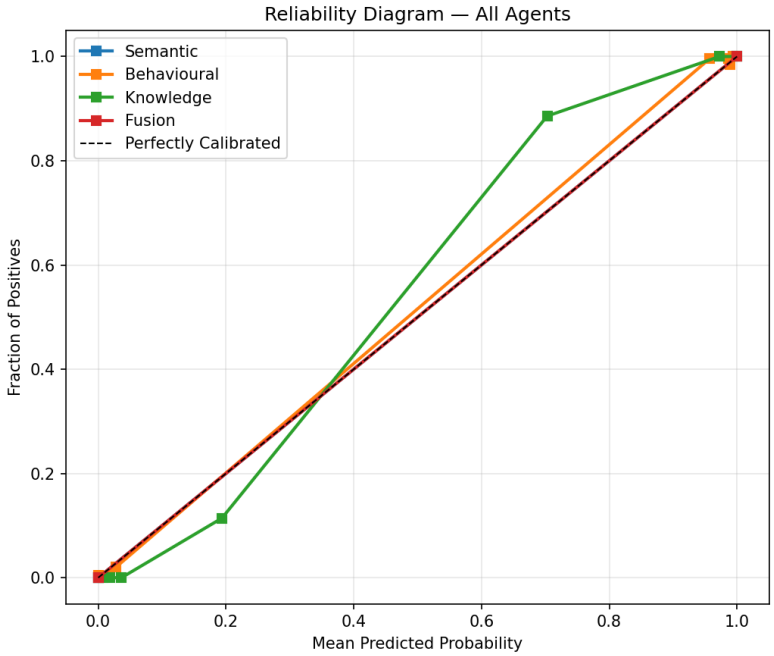


Figure 19: Reliability diagram for the fused risk score $\hat{p}$. Close alignment between predicted probabilities and empirical positive rates confirms that $\hat{p}$ is well-calibrated and can be used as a meaningful risk estimate for triage decisions.

transforming a weakly discriminative curve into a near-optimal operating regime. On RDset, the semantic agent already saturates and fusion preserves perfect discrimination, confirming that the fusion mechanism does not degrade strong base learners.

5.10.6 Explainability Results

Figure 22 shows the permutation importance scores for the Behavioural Agent. The dominant features are dispersion-based statistics — LBA and GPA variance for both read and write operations — consistent with the counterfactual analysis showing that ransomware's distinguishing signal is the spatial irregularity of its memory and storage access pattern, not its volume.

The coherence between global importance rankings and local counterfactual directions confirms that the Behavioural Agent exploits genuine ransomware-specific signals rather than spurious correlations.

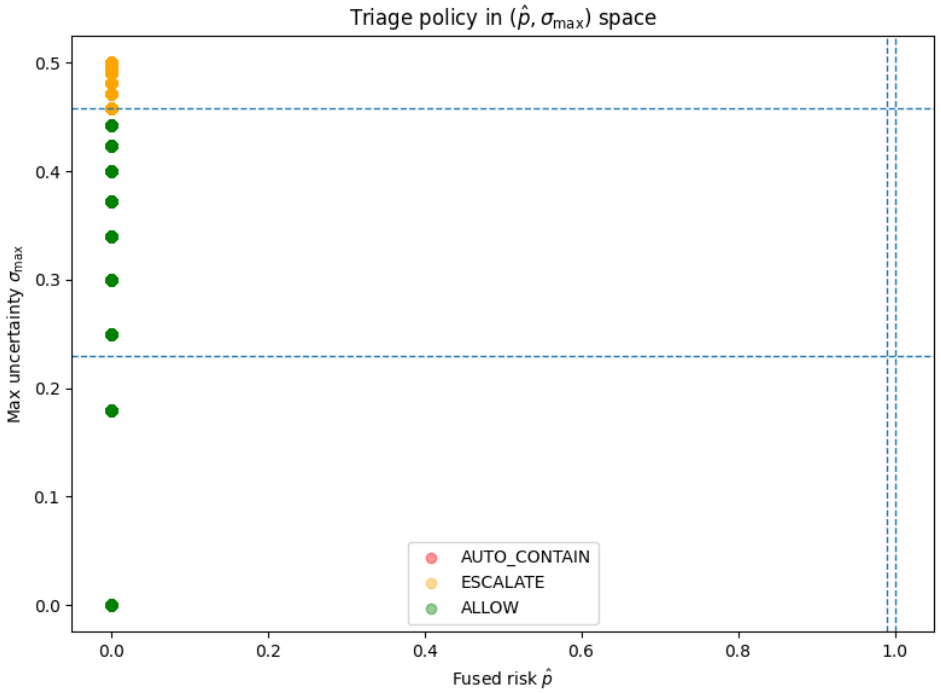


Figure 20: Triage policy geometry in the $(\hat{p}, \sigma_{\max})$ space. Axis-aligned thresholds partition the plane into Allow, Escalate, and Auto-Contain regions. Uncertainty prevents overconfident automatic containment for borderline or distributional-shift samples.

5.10.7 Computational Complexity and Runtime Performance

Training Complexity of CNN Agents

Each CNN agent employs a 1D convolutional architecture of depth $L = 3$, kernel size $K = 3$, and $C = 64$ filters per layer. For a training set of N samples, the per-agent cost over E epochs is $\mathcal{O}(E \cdot N \cdot L \cdot K \cdot C^2)$, linear in both N and E.

Monte Carlo Dropout Score Generation

The per-sample MC Dropout cost is $\mathcal{O}(T \cdot L \cdot K \cdot C^2)$. With $T = 30$, each agent requires approximately 2–5 ms per stochastic pass on CPU, yielding an MC estimate in $\approx$ 60–150 ms per agent.

CTGAN Score-Space Augmentation

Score-space augmentation trains a generator and discriminator over N_s real score vectors $\mathbf{s} \in \mathbb{R}^2$. Because the score matrix is far smaller than the raw feature matrices, CTGAN

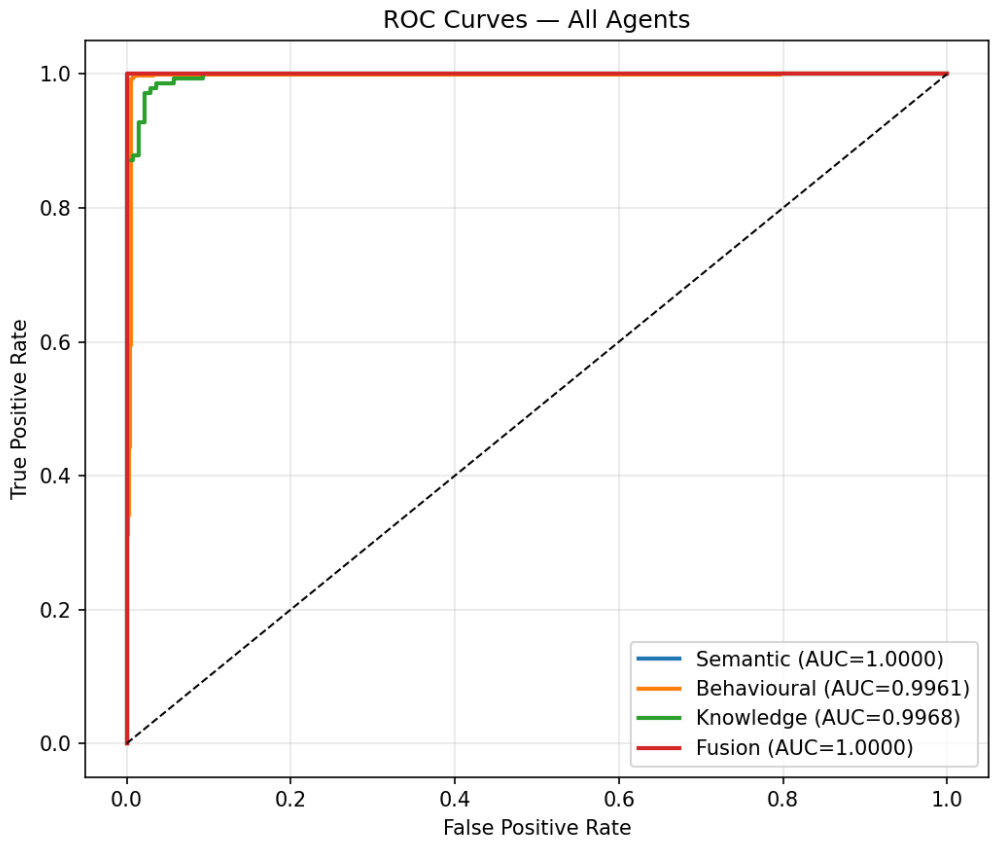


Figure 21: ROC curves comparing individual agents and the fused classifier on RanSMAP (left) and RDset (right). Fusion substantially improves separability over the behavioural agent alone on RanSMAP while preserving perfect discrimination on RDset.

fitting is tractable; empirically, $E_{\mathrm{CTGAN}} = 200$ epochs complete in approximately 5–10 min on CPU.

Total Pipeline Training Complexity

Combining all components, the complete training complexity is:

$$\begin{aligned}\mathcal{T} = \mathcal{O}\Big(& 2(E{+}T)\, N\, L\, K\, C^2 \\ & + E_{\mathrm{CTGAN}}\, N_s \big[\textstyle\sum_i g_i g_{i+1} + \sum_j d_j d_{j+1}\big] \\ & + E_{\mathrm{MLP}}\, N_{\mathrm{aug}}\, h^2 \\ & + T_{\mathrm{TPE}} \big(n_{\mathrm{obs}} \log n_{\mathrm{obs}} + N_{\mathrm{val}}\big)\Big),\end{aligned} \tag{5.21}$$

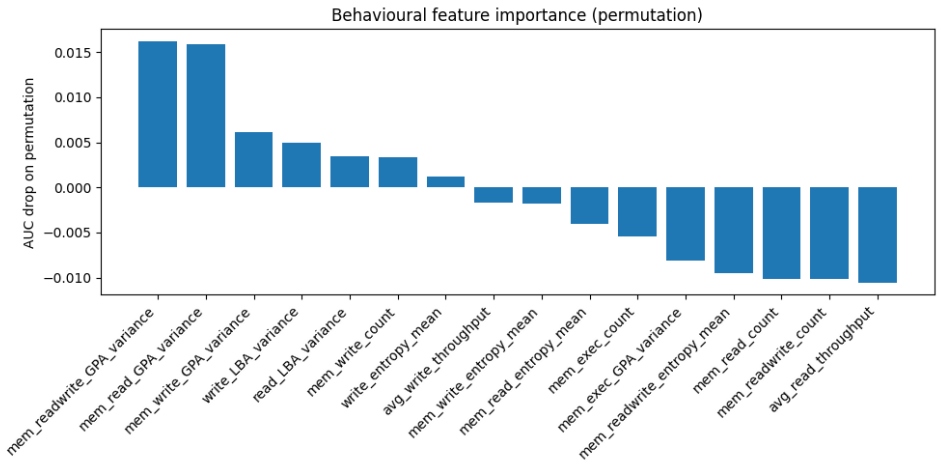


Figure 22: Permutation-based feature importance for the Behavioural Agent. Variance- and entropy-driven memory access statistics dominate, indicating that ransomware detection relies on spatial irregularity patterns rather than raw activity volume.

where the four terms correspond respectively to CNN agent training and MC score generation, CTGAN score-space augmentation, Decision MLP fitting, and TPE threshold optimisation. The first term scales linearly in N; the CTGAN term is the practical bottleneck but remains tractable because the score matrix is orders of magnitude smaller than the raw feature matrices.

Measured Runtime Performance

Table 24 reports empirically measured inference latency and throughput per SABRE component on GPU. The Behavioural CNN requires 1.94 ms (p50) for $T = 10$ MC Dropout passes — a 3.7× overhead relative to a single deterministic forward pass (0.52 ms). The full two-agent pipeline is estimated at approximately 3.7 ms in serial execution, supporting throughput of ≈ 270 samples/s. At a 10-second telemetry window, this supports over 2,000 concurrent monitored streams. The agents are compact: the Behavioural CNN has only 8,385 parameters (32.8 KB), making it suitable for endpoint deployment.

5.11 Ablation Studies

This section isolates the contribution of individual architectural components — CTGAN augmentation, uncertainty inputs, and modality contributions — by systematically removing each element and measuring the resulting performance degradation.

Table 24: Measured inference latency and throughput per SABRE component (NVIDIA GB10 GPU, $T = 10$ MC Dropout passes, $n = 500$ repetitions). Full pipeline (B+S) is the serial sum of per-agent latencies plus fusion overhead.

Component	Params	Size	p50 (ms)	p95 (ms)	Throughput (sps)
Behavioral CNN ($T = 10$)	8,385	32.8 KB	1.94	2.26	517
Semantic CNN ($T = 10$)	786,817	3.1 MB	1.77	1.82	566
Single CNN (no MC, $T = 1$)	8,385	32.8 KB	0.52	0.55	1,911
Full pipeline B+S (est.)	—	—	≈3.7	—	≈270

5.11.1 Effect of CTGAN Score-Space Augmentation

The influence of CTGAN augmentation is quantified by removing it from the fusion stage. CTGAN learns a generator such that the synthetic distribution

$$\tilde{\mathcal{D}}_{\mathbf{s}} \approx p(\mathbf{s} \mid y), \tag{5.22}$$

producing synthetic samples that expand the score manifold. As shown in Table 25, removing CTGAN reduces the RanSMAP AUC from 0.540 to 0.514, indicating that density enrichment of $\mathbf{s}$ contributes to more robust decision boundaries. Replacing the trained fusion MLP with a naive score mean ($\hat{p}_{\text{mean}} = \frac{1}{2}(p_{\text{sem}} + p_{\text{beh}})$) performs significantly worse on RanSMAP due to the weak correlation of p_{beh} with ground truth.

Table 25: Ablation of fusion strategies.

Fusion Method	RDset AUC	RanSMAP AUC
MLP w/ CTGAN (full model)	1.0000	0.5397
MLP w/o CTGAN	0.9967	0.5142
Mean Fusion	0.884	0.501

5.11.2 Effect of Removing Uncertainty Inputs

Removing uncertainty inputs forces the fusion model to depend only on the score pair $(p_{\text{sem}}, p_{\text{beh}})$, eliminating $\sigma_{\max}$ (Equation (5.7)) from the decision mechanism. As shown in Table 26, the number of false negatives on RanSMAP increases sharply — from 43,656 to 58,441 — since the classifier can no longer separate high-risk but uncertain samples from high-risk confident ones. The escalation rate also rises substantially, confirming that uncertainty inputs are critical to the triage mechanism's ability to contain high-risk ambiguous cases.

Table 26: Effect of removing uncertainty terms on triage performance.

Model	RDset Esc. Rate	RanSMAP Esc. Rate	FN (RanSMAP)
Full model (with $\sigma_{\max}$)	0.72	0.82	43,656
No uncertainty (without Eq. 5.7)	0.91	0.97	58,441

5.11.3 Modal Contribution Ablation

To isolate the contribution of each agent, we consider two restricted settings: (i) semantic ablation $p_{\text{fus}}^{(-\text{sem})}$ (behavioural only), and (ii) behavioural ablation $p_{\text{fus}}^{(-\text{beh})}$ (semantic only). In the absence of semantic information, behavioural counterfactual norms decrease, indicating that certain evasive trajectories become feasible through execution-only manipulations. Conversely, removing behavioural input enables purely structural evasion strategies. Neither ablated configuration preserves the super-additive adversarial cost in Equation (5.19), confirming that robustness and interpretability are jointly maximised only when both modalities are present.

5.12 Discussion

5.12.1 Distributional Asymmetry and Wasserstein Distance

The experiments highlight a marked asymmetry between the discriminative structures of the semantic and behavioural feature spaces. The class-conditioned distributions in semantic space can be characterised by the Wasserstein-2 distance:

$$d_{\text{sem}} = W_2\big(p(\mathbf{z} \mid y = 1),\, p(\mathbf{z} \mid y = 0)\big), \tag{5.23}$$

and likewise for behavioural data:

$$d_{\text{beh}} = W_2\big(p(\mathbf{b} \mid y = 1),\, p(\mathbf{b} \mid y = 0)\big). \tag{5.24}$$

The empirical results strongly suggest that $d_{\text{sem}} \gg d_{\text{beh}}$, explaining why semantic predictions cluster near $\{0, 1\}$ while behavioural predictions exhibit broad dispersion. The fusion model exploits this asymmetry by treating the semantic component as a strong positive signal whenever uncertainty is low and treating the behavioural component primarily as

a modulator of $\sigma_{\max}$.

5.12.2 Operational Implications

Agentic SABRE demonstrates that uncertainty-aware triage is not merely a theoretical construct but a practically deployable mechanism. The 4.9% reduction in false escalations at equal recall on RanSMAP translates directly to reduced analyst workload in a live SOC, where each escalated sample requires human review time. The formal threshold ordering constraints (Equation (5.8)) ensure that the AUTO-CONTAIN region is always a strict subset of the ESCALATE region, providing a provable safety guarantee: no sample can be autonomously contained unless it would also qualify for escalation under the looser thresholds.

The modular design enables two forms of operational update that are impractical in monolithic systems. First, the triage thresholds can be recalibrated via the TPE objective (Equation (5.12)) on a new validation set without retraining any model — a minutes-long operation. Second, each agent can be retrained independently on new data without affecting the other agent or the Decision Agent.

5.13 Limitations

Three principal limitations constrain the current design. First, the fusion mechanism (Equation (5.6)) assumes conditional independence between semantic and behavioural agent outputs; in operational settings, PE structure and I/O patterns may exhibit dependencies that violate this assumption. A normalising flow or attention-based fusion function could model the joint likelihood more accurately. Second, MC Dropout may under-estimate epistemic uncertainty under strong non-stationarity — a risk in rapidly evolving ransomware ecosystems. Deep ensembles or conformal prediction frameworks Yumlembam, Issac, Jacob and Yang (2024) could provide better-calibrated uncertainty at higher computational cost. Third, the counterfactual analysis reveals that the Behavioural Agent's decision boundary is fragile under variance-reduction attacks, pointing to a need for adversarially robust training.

The triage thresholds are optimised offline and are not updated during deployment. In dynamic environments where the score distribution shifts over time — due to concept drift, new family emergence, or changing benign workload patterns — static thresholds can become miscalibrated. A natural extension would be an adaptive thresholding mechanism, such as a sliding-window recalibration that re-optimises Equation (5.12) on a rolling buffer of analyst-labelled decisions.

Finally, the family holdout evaluation (AUROC 0.707, Table 22) confirms that the Behavioural Agent generalises only partially to unseen ransomware families. The uncertainty mechanism mitigates this by routing family-holdout samples to ESCALATE rather than AUTO-CONTAIN, but improving zero-shot generalisation remains an important future direction.

5.14 Chapter Summary

This chapter presented Agentic SABRE, a multi-agent uncertainty-aware framework for adaptive ransomware detection. Moving beyond the single-agent approach of the preceding chapter, Agentic SABRE decomposes detection into a Semantic Agent operating on PE-header embeddings and a Behavioural Agent operating on runtime telemetry, each equipped with Monte Carlo Dropout uncertainty estimation. A score-space CTGAN augmentation strategy and a two-hidden-layer Decision Agent MLP fuse the agents' outputs into a calibrated risk estimate. A formal uncertainty-aware triage policy with Bayesian-optimised thresholds routes high-confidence, low-uncertainty predictions to autonomous containment, uncertain predictions to human analysts, and low-risk predictions to automatic allowance.

Empirical results on RDset and RanSMAP demonstrate near-perfect fusion performance on the semantic dataset while correctly managing the high-uncertainty behavioural regime through conservative escalation. Generalisation evaluation across four evaluation splits confirms temporal and hardware robustness while identifying family holdout as the principal generalisation challenge. Ablation studies quantify the contribution of each architectural component: removing CTGAN reduces RanSMAP AUC from 0.540 to 0.514; removing uncertainty inputs increases false negatives from 43,656 to 58,441. Permutation

importance, gradient saliency, and counterfactual analysis confirm that both agents rely on interpretable, ransomware-relevant signals, and that the fusion architecture enforces a super-additive adversarial cost that requires coordinated manipulation of both modalities to evade.

The following chapter extends this two-agent framework to a three-agent system by introducing a Knowledge Agent trained on practitioner threat-intelligence discourse, a Peer Consultation Protocol that enables uncertain agents to query their peers before fusion, and an Expert Feedback Loop that incorporates analyst corrections to incrementally adapt the Decision Agent against concept drift.

Chapter 6

Extended Agentic SABRE: Knowledge-Augmented Three-Agent Ransomware Detection with Peer Consultation and Continual Learning

6.1 Introduction

The previous chapter demonstrated that decomposing ransomware detection into a Semantic Agent and a Behavioural Agent, fused through a Decision Agent with uncertainty-aware triage, substantially improves operational control over automated containment decisions. However, two fundamental gaps remain in that two-agent design. First, both agents reason exclusively from statistical regularities in measurement data — PE-header bytes and runtime I/O traces — without access to the accumulated operational knowledge of the security community. Security practitioners have documented which MITRE ATT&CK tactics are most consistently associated with ransomware deployment, which tools appear in attack chains, and which platforms are primary targets; this knowledge is embedded in threat-intelligence briefings, DFIR walkthroughs, and analyst-authored publications, yet it has not previously been operationalised as an agent-level

signal in a multi-agent detection framework.

Second, the two agents in Agentic SABRE operate in isolation: each produces a prediction that is fused passively through the learned Decision MLP, with no mechanism for agents to communicate their reliability to one another before fusion. An agent experiencing high epistemic uncertainty contributes noise to the fusion layer with the same weight as a highly confident peer. Under distributional shift — for example, a novel ransomware family that appears semantically unusual but follows a canonical ATT&CK deployment chain — the most reliable agent cannot assist the uncertain one, causing the fused prediction to degrade.

This chapter addresses both gaps with three principled extensions that collectively define the Extended Agentic SABRE framework (SABRE+):

1. Knowledge Agent and RansKnow Dataset. A third agent — the Knowledge CNN — is introduced, operating on a novel 23-feature threat-intelligence dataset (RansKnow) derived from 307 YouTube security-practitioner transcripts. Features encode MITRE ATT&CK tactic frequencies, ransomware-ecosystem tool mentions, and platform indicators. The Knowledge Agent achieves 0.9550 accuracy and 0.9950 AUC on balanced evaluation, providing a modality-orthogonal signal independent of both PE structure and runtime telemetry.

2. Peer Consultation Protocol. A formal inter-agent consultation mechanism is introduced: when an agent's Monte Carlo Dropout uncertainty σ_i exceeds a per-agent threshold κ_{local}, its probability estimate is adjusted via an inverse-uncertainty-weighted mean over all agents including itself. Crucially, uncertainty is never modified — only the probability propagated to the fusion layer is updated — preserving the downstream triage policy's ability to measure true epistemic spread.

3. Expert Feedback Loop. A continual-adaptation mechanism is designed in which analyst corrections to AUTO-CONTAIN and ESCALATE decisions are fed back to incrementally fine-tune the Decision MLP through a combined new-sample and experience-replay loss, with reservoir-sampled memory to prevent catastrophic forgetting.

The remainder of this chapter is organised as follows. Section 6.2 provides background and motivation. Section 6.3 surveys relevant literature. Section 6.4 presents the Extended SABRE+ architecture. Section 6.5 details the Expert Feedback Loop. Section 6.6 describes the experimental setup. Section 6.7 reports and analyses results. Section 6.8 discusses implications and limitations. Section 6.9 summarises the chapter.

6.2 Background and Motivation

6.2.1 Knowledge Blindness in Statistical Detectors

The Semantic and Behavioural Agents in Agentic SABRE both learn purely from statistical patterns in measurement data. While this is effective for known ransomware families with distinctive PE structures or runtime signatures, it is inherently limited for emerging variants that have been documented by the security community but whose measurement-level signatures have not yet been observed in training data. The MITRE ATT&CK framework provides a structured vocabulary for ransomware tactics and tools — Initial Access, Execution, Persistence, Lateral Movement, Exfiltration, Impact (T1486), and associated tools such as Cobalt Strike, Rclone, Mimikatz, and BloodHound — that appear consistently across diverse ransomware campaigns regardless of their specific implementation details. Operationalising this structured knowledge as a learned signal introduces a detection modality that is complementary to both static PE analysis and dynamic runtime monitoring.

6.2.2 Agent Isolation and Uncertainty Propagation

In the two-agent SABRE design, fusion is performed by a learned MLP that operates on the four-dimensional score vector $\mathbf{v} = (\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \sigma_{\text{sem}}, \sigma_{\text{beh}})$. While the MLP can in principle learn to discount uncertain agents, this is an implicit and unstructured mechanism: the MLP must infer from the uncertainty values alone how much to trust each agent's probability, without any explicit coordination between agents. This limitation is particularly acute under distributional shift, where one agent may be highly uncertain while another remains confident — exactly the situation where direct inter-agent communication would be most valuable.

The Peer Consultation Protocol addresses this by introducing explicit, formalised pre-fusion coordination: uncertain agents query their peers and adjust their probabilities before the scores reach the Decision MLP, providing the MLP with better-calibrated inputs in high-uncertainty conditions.

6.3 Related Work

6.3.1 Knowledge-Augmented Detection

The integration of structured threat intelligence with machine learning models has been explored in the context of rule-based pre-filtering using ATT&CK-aligned features Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023) and neuro-symbolic frameworks that embed structured knowledge into neural architectures Besold et al. (2021). However, these approaches typically use structured knowledge as a static filter rather than as a learned agent signal with associated uncertainty estimation. RansKnow is, to the best of our knowledge, the first threat-intelligence dataset derived from structured analysis of security-practitioner video content, and the Knowledge Agent is the first agent in a multi-agent ransomware framework to operate on discourse-derived structured features.

6.3.2 Multi-Agent Coordination and Uncertainty Communication

Multi-agent approaches to intrusion detection Soltani, Khajavi, Siavoshani and Jahangir (2023) and cooperative anomaly detection have been explored without equipping agents with inter-agent uncertainty communication. The closest related concept in the literature is mixture-of-experts (MoE) gating Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin (2017), where a learned router selects among expert models; however, MoE routing is learned globally and does not implement per-sample, threshold-gated, uncertainty-driven peer queries. The Peer Consultation Protocol differs by being explicitly uncertainty-triggered, operationally transparent, and preserving uncertainty invariance for downstream triage.

6.3.3 Continual Learning for Threat Detection

Concept drift under evolving adversarial strategies is a fundamental challenge for deployed ransomware detectors Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2023). Experience replay Rolnick, Ahuja, Schwartz, Lillicrap and Wayne (2019) and elastic weight consolidation prevent catastrophic forgetting during incremental updates. SABRE+'s Expert Feedback Loop combines both techniques in a lightweight online adaptation layer targeting only the Decision MLP — the shallowest and fastest-to-retrain component — while leaving the agent backbones frozen to prevent degradation of the learned per-modality representations.

6.3.4 Temperature Calibration

Temperature scaling Guo, Pleiss, Sun and Weinberger (2017) is a post-hoc calibration technique that divides the logit of a trained classifier by a scalar temperature $T_{\text{cal}} > 0$ fitted on a validation set. It preserves the ordering of predictions while reducing overconfidence, improving the alignment between predicted probabilities and empirical positive rates. SABRE+ applies temperature calibration to the fused risk estimate $\hat{p}$ after Decision MLP training.

6.4 Extended SABRE+ Architecture

6.4.1 System Overview

Figure 23 illustrates the complete Extended SABRE+ pipeline. Three modality-specific base detectors — the Semantic CNN, Behavioural CNN, and Knowledge CNN — independently produce mean-uncertainty pairs via Monte Carlo Dropout. The Peer Consultation module conditionally adjusts agent probabilities under high local uncertainty before scores enter the Score Pool. CTGAN augments the Score Pool during training. A Decision MLP, tuned by Bayesian threshold optimisation and calibrated by temperature scaling, produces the fused risk estimate $\hat{p}$. The Triage Policy maps $(\hat{p}_{\text{cal}}, \sigma_{\max})$ to one of three decisions. Analyst corrections from the Expert Feedback Loop incrementally retrain the Decision MLP.

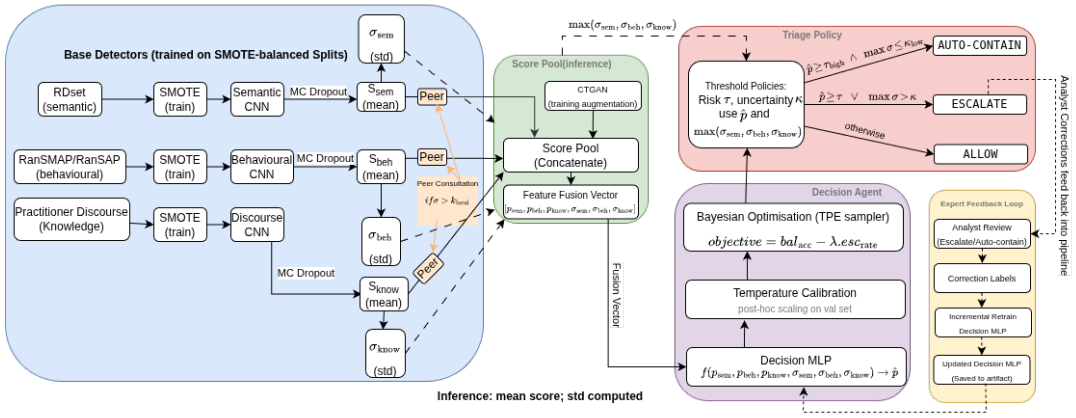


Figure 23: Extended Agentic SABRE+ pipeline. Three modality-specific agents (Semantic, Behavioural, Knowledge CNNs) produce mean-uncertainty pairs via Monte Carlo Dropout. The Peer Consultation module adjusts uncertain agents' probabilities before score fusion. The Decision MLP, calibrated by temperature scaling, produces the fused risk $\hat{p}$. The Triage Policy routes each sample to AUTO-CONTAIN, ESCALATE, or ALLOW. The Expert Feedback Loop closes the human-in-the-loop cycle.

6.4.2 Semantic and Behavioural Agents

The Semantic and Behavioural Agents are inherited from Agentic SABRE (Chapter 5.14) without modification. The Semantic Agent operates on 384-dimensional PE-header embeddings from RDset, achieving $\bar{\sigma}_{\text{sem}} \approx 2.75 \times 10^{-8}$ (near-deterministic confidence). The Behavioural Agent operates on 16-dimensional runtime telemetry features from RanSMAP, with $\bar{\sigma}_{\text{beh}} \approx 0.0138$. In SABRE+, both agents use $T = 10$ MC Dropout passes (reduced from 30 in the ESWA baseline) since the Semantic Agent's near-deterministic outputs make additional passes redundant, and the Knowledge Agent uses adaptive MC sampling (described below).

6.4.3 Knowledge Agent and RansKnow Dataset

RansKnow Dataset

RansKnow is a novel dataset of 307 samples (140 ransomware-relevant; 167 benign) derived from structured analysis of YouTube security practitioner content, including threat-intelligence briefings, DFIR (digital forensics and incident response) walkthroughs, and ransomware family deep-dives published by analyst channels. Each sample corresponds to one video and is represented by 23 structured features organised into four groups (Ta-

ble 27):

Table 27: RansKnow dataset feature groups (23 features, 307 samples).

Feature Group	Count	Examples
MITRE ATT&CK tactic counts	10	Initial Access, Exfiltration, Impact
Ransomware tool mention counts	8	Cobalt Strike, Rclone, Mimikatz
Platform signal indicators	3	Windows, Linux, ESXi
Aggregate mention totals	2	Tactic_Total, Tool_Total
Total	23	307 samples

Labels are derived heuristically: a sample is ransomware-relevant if Tactic_Total_Mentions $>$ 10 or Tool_Total_Mentions > 2, reflecting the empirical observation that legitimate ransomware analyses consistently reference both high-tactic-frequency deployment chains and specific attacker tooling.

Knowledge CNN Architecture

The Knowledge CNN operates on the 23-dimensional RansKnow feature vector with the following architecture: a 1D convolutional layer (16 filters, kernel size 3, same padding), ReLU activation, max-pooling (stride 2), Dropout ($p_{\text{drop}} = 0.3$), a flatten layer, and two fully connected layers (64 hidden units, sigmoid output). Adaptive Monte Carlo Dropout is applied: passes continue from a minimum of $T_{\min} = 5$ until the running standard deviation of σ_{know} converges within tolerance $\delta = 10^{-3}$, capped at $T_{\max} = 50$. This adaptivity trades additional computation for reduced uncertainty estimation variance in the knowledge modality, where the smaller dataset size makes σ_{know} inherently noisier.

Complementarity of the Three Agents

The three agents provide overlapping but non-identical detection coverage. The Semantic Agent is strong on known-family PE variants but blind to novel polymorphic samples with unfamiliar byte patterns. The Behavioural Agent captures runtime encryption dynamics but may be confused by living-off-the-land techniques that generate benign-looking telemetry. The Knowledge Agent is strong precisely when practitioners have documented a deployment chain — i.e., for well-known campaigns with ATT&CK-mapped tooling

— regardless of PE structure or runtime behaviour. Together, the three agents provide complementary coverage across the ransomware detection surface.

6.4.4 Peer Consultation Protocol

Motivation and Design

An agent operating under high epistemic uncertainty should not contribute to fusion with the same weight as a confident peer. The standard MLP fusion layer cannot distinguish this case at the feature level; it receives probabilities $\bar{p}_i$ but does not explicitly reason about their reliability before computing the fused output. Peer consultation resolves this by adjusting $\bar{p}_i$ before fusion whenever $\sigma_i > \kappa_{\text{local}}$, using peer scores weighted by peer confidence.

Formal Definition

Let $\mathcal{A} = \{\text{sem}, \text{beh}, \text{know}\}$ denote the agent set and $\mathcal{P}_i = \mathcal{A} \setminus \{i\}$ the peers of agent i. For each agent, define the inverse-uncertainty weight:

$$w_j = \frac{1}{\sigma_j + \epsilon}, \quad j \in \mathcal{A}, \quad \epsilon = 10^{-8}. \tag{6.1}$$

Consultation fires for agent i if $\sigma_i > \kappa_{\text{local},i}$. The adjusted probability is:

$$\bar{p}_i^* = \frac{w_i \bar{p}_i + \sum_{j \in \mathcal{P}_i} w_j \bar{p}_j}{w_i + \sum_{j \in \mathcal{P}_i} w_j}. \tag{6.2}$$

If $\sigma_i \leq \kappa_{\text{local},i}$, then $\bar{p}_i^* = \bar{p}_i$. Critically, σ_i is never modified by consultation; the original MC Dropout standard deviation is always propagated to the score vector, preserving the semantics of $\sigma_{\max}$ in the triage policy.

The protocol has three important properties: (1) Confidence-preserving: a highly confident peer ($\sigma_j \approx 0$) dominates the weighted average; (2) Self-inclusive: agent i retains weight w_i in the weighted mean, so consultation cannot invert a strongly confident uncertain agent; (3) Threshold-gated: consultation is a no-op when all agents are confident, incurring no computational overhead beyond a threshold comparison.

6.4.5 Score Vector and CTGAN Augmentation

After peer consultation, the six-dimensional score vector is assembled:

$$\mathbf{v} = \left(\bar{p}^*_{\text{sem}}, \bar{p}^*_{\text{beh}}, \bar{p}^*_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}\right)^\top \in \mathbb{R}^6, \tag{6.3}$$

where $\bar{p}^*_i$ denotes the post-consultation probability and all σ_i are the original MC Dropout standard deviations. During training, CTGAN Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019) is applied to the Score Pool to synthesise additional six-dimensional score vectors, preserving inter-agent correlations and preventing the Decision MLP from training exclusively on well-separated score configurations.

6.4.6 Decision MLP and Temperature Calibration

The Decision MLP $f : \mathbb{R}^6 \to [0, 1]$ takes $\mathbf{v}$ as input and produces the fused risk estimate:

$$\hat{p} = f(\bar{p}^*_{\text{sem}}, \bar{p}^*_{\text{beh}}, \bar{p}^*_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}). \tag{6.4}$$

The network uses two hidden layers (64 and 32 units), ReLU activations, and a sigmoid output. Post-hoc temperature scaling Guo, Pleiss, Sun and Weinberger (2017) calibrates $\hat{p}$:

$$\hat{p}_{\text{cal}} = \sigma\left(\frac{\ln \frac{\hat{p}}{1-\hat{p}}}{T_{\text{cal}}}\right), \tag{6.5}$$

where $T_{\text{cal}} > 0$ is fitted on the validation set by minimising negative log-likelihood. Temperature scaling preserves the threshold-relative ordering of $\hat{p}$ while reducing overconfidence.

6.4.7 Triage Policy

The global uncertainty aggregate is:

$$\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}). \tag{6.6}$$

The Triage Policy maps $(\hat{p}_{\text{cal}}, \sigma_{\max})$ to a decision:

$$\text{AUTO-CONTAIN} \quad \text{if} \quad \hat{p}_{\text{cal}} \geq \tau_{\text{high}} \ \wedge \ \sigma_{\max} \leq \kappa_{\text{low}}, \tag{6.7}$$

$$\text{ESCALATE} \quad \text{if} \quad \hat{p}_{\text{cal}} \geq \tau \ \vee \ \sigma_{\max} > \kappa, \tag{6.8}$$

$$\text{ALLOW} \quad \text{otherwise.} \tag{6.9}$$

Thresholds are jointly optimised by Bayesian TPE search Akiba, Sano, Yanase, Ohta and Koyama (2019):

$$(\tau^*, \kappa^*, \tau^*_{\text{high}}, \kappa^*_{\text{low}}) = \arg \max_{\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}}} \big[\text{BalAcc}(\tau, \kappa) - \lambda \, \text{EscRate}(\tau, \kappa)\big], \tag{6.10}$$

with $\lambda = 0.5$, yielding the optimised low-escalation policy: $\tau^* = 0.2330$, $\kappa^* = 0.1969$, $\tau^*_{\text{high}} = 0.8545$, $\kappa^*_{\text{low}} = 0.1969$.

6.5 Expert Feedback Loop

6.5.1 Architecture and Motivation

Concept drift under evolving ransomware tactics causes the Decision MLP's decision boundaries to become misaligned with the current threat distribution over time. The Expert Feedback Loop closes the human-in-the-loop cycle: whenever a sample receives an ESCALATE or AUTO-CONTAIN decision, the analyst reviews it and provides a correction label $\tilde{y} \in \{0, 1\}$ (0 = benign, 1 = ransomware). Correction labels are accumulated in a correction queue $\mathcal{Q}_{\text{new}}$.

6.5.2 Incremental Retraining with Experience Replay

The Decision MLP is fine-tuned on a mixed loss that combines the new-sample loss with a replay loss from a maintained memory buffer:

$$\mathcal{L}_{\text{total}} = \underbrace{\frac{1}{|\mathcal{Q}_{\text{new}}|} \sum_{(\mathbf{v}, \tilde{y}) \in \mathcal{Q}_{\text{new}}} \text{BCE}(f(\mathbf{v}), \tilde{y})}_{\text{new-sample loss}} + \gamma \cdot \underbrace{\frac{1}{|\mathcal{B}|} \sum_{(\mathbf{v}, y) \in \mathcal{B}} \text{BCE}(f(\mathbf{v}), y)}_{\text{replay loss}}, \tag{6.11}$$

where $\mathcal{B}$ is a replay buffer of size N_{replay} updated by reservoir sampling Vitter (1985), and $\gamma \geq 0$ controls the forgetting-adaptation trade-off. Agent backbone weights are frozen: only the Decision MLP is updated. This limits parameter exposure, reduces catastrophic forgetting risk to the shallowest component, and keeps update latency low (the Decision MLP has $\leq 4{,}000$ parameters).

6.5.3 Reservoir Sampling

Each round, the replay buffer $\mathcal{B}$ is updated by reservoir sampling Vitter (1985): for each new sample at position n, it replaces a uniformly random element of $\mathcal{B}$ with probability N_{replay}/n. This provides an unbiased uniform sample over all historical correction labels, preventing recent-class bias and maintaining representativeness of the full history.

6.6 Datasets and Experimental Setup

6.6.1 Datasets

All three agents are evaluated on independent modality-specific datasets. RDset and RanSMAP are described in the previous chapter. RansKnow is described in Section 6.4.3 above.

Independence guarantee. All three datasets are entirely independent: no executable, execution trace, or video appears in more than one. Because no single dataset contains all three modalities simultaneously, three-agent fusion evaluation uses the semantic-agent-dominated label source (RDset) with cross-modality inputs noted explicitly.

6.6.2 Baselines

Results are compared against: (i) the original two-agent SABRE Kabuye, Issac and Neera (2026); (ii) the single-agent uncertainty-aware detector Kabuye, Issac, Yumlembam and Neera (2025); (iii) LLM-LADE Zhang, Li, Zhang, Ye, Hu and Yan (2025); (iv) FeSAD Fernando and Komninos (2024); and (v) the multi-agent adaptive detection system of Soltani, Khajavi, Siavoshani and Jahangir (2023).

6.6.3 Evaluation Metrics

In addition to accuracy, balanced accuracy, AUC, precision, recall, and F1-score, the triage-specific metrics of escalation rate and AUTO-CONTAIN rate are reported. Expected Calibration Error (ECE) is computed for uncertainty quality assessment. The consultation rate and mean probability adjustment $\Delta\bar{p}$ per agent are reported to assess the Peer Consultation Protocol's behaviour.

6.7 Results and Analysis

6.7.1 Per-Agent Standalone Performance

Table 28 reports the per-agent standalone performance on balanced held-out evaluation sets.

Table 28: Per-agent standalone evaluation on balanced test sets.

Agent	Dataset	n	Acc	AUC	F1	$\bar{\sigma}$
Semantic CNN	RDset	4000	1.0000	1.0000	1.0000	≈ 0
Behavioural CNN	RanSMAP	2000	0.9945	0.9967	0.9945	0.0138
Knowledge CNN	RansKnow	200	0.9550	0.9950	0.9550	0.0499

The Semantic CNN achieves perfect discrimination, consistent with prior findings Kabuye, Issac and Neera (2026) that PE-header embeddings form near-disjoint class clusters. The Behavioural CNN attains 0.9967 AUC on balanced evaluation; the 0.6030 AUC on the full imbalanced distribution reflects the difficulty of ranking under a 598:1 class ratio. The Knowledge CNN achieves 0.9550 accuracy and 0.9950 AUC despite operating on only 23 features and 307 total training samples, confirming the discriminative power of structured ATT&CK/tool features. The increasing $\bar{\sigma}$ sequence ($\approx 0 \rightarrow 0.0138 \rightarrow 0.0499$) reflects decreasing distributional separation across modalities, consistent with the semantic $>$ behavioural $>$ discourse hierarchy of signal quality.

6.7.2 ROC Curves for All Three Agents

Figure 24 shows the ROC curves for all three agents on their respective native-modality evaluation sets. The Semantic Agent saturates at the top-left corner. The Behavioural Agent achieves near-optimal separation under balanced evaluation (AUC = 0.9967). The

Knowledge Agent achieves AUC = 0.9958 from 23 structured features alone, demonstrating that threat-intelligence discourse provides a strongly discriminative signal when structured as ATT&CK-aligned feature counts.

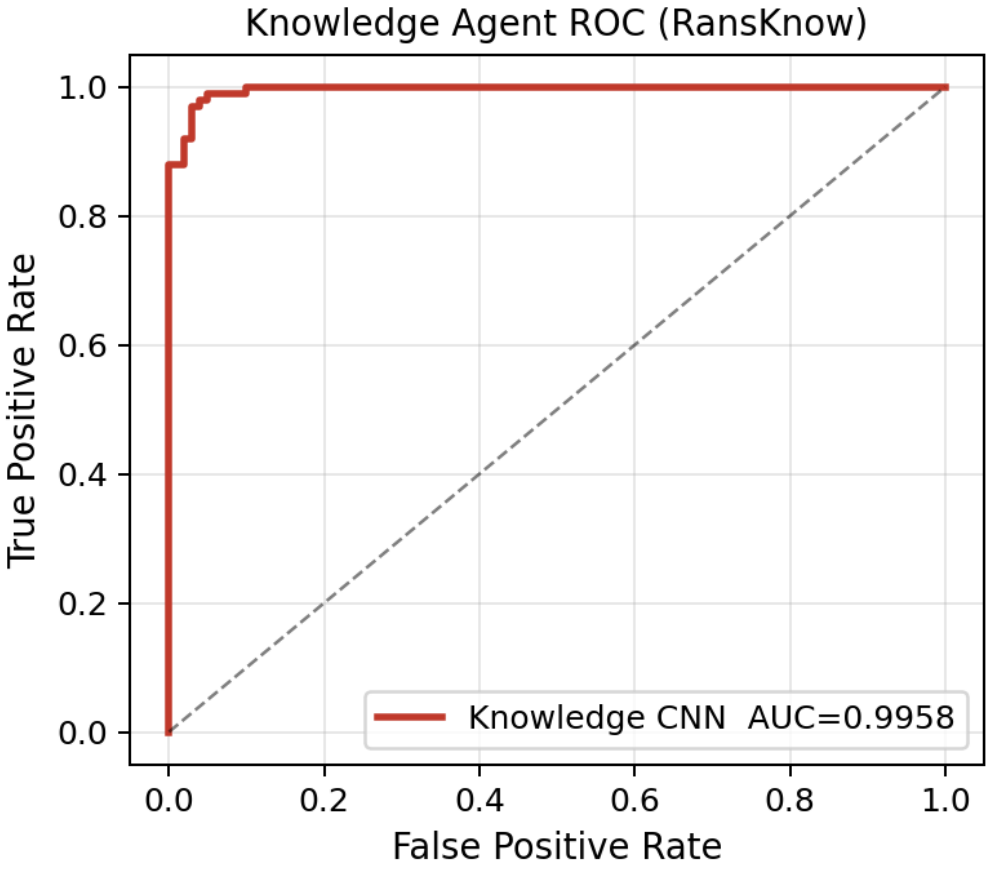


Figure 24: ROC curve for the Knowledge Agent on the RansKnow dataset (AUC = 0.9958). Despite operating on only 23 structured ATT&CK/tool features and 307 training samples, the Knowledge Agent provides strongly discriminative threat-intelligence signals complementary to both the Semantic and Behavioural agents.

6.7.3 Three-Agent Fusion and Triage

Table 29 reports the three-agent fusion behaviour on a 200-sample cross-modality evaluation. The fused classifier achieves perfect accuracy and AUC (1.0000), driven by the Semantic Agent's dominance on this split. More importantly, the 9.0% escalation rate is substantially lower than the two-agent SABRE baseline, which reported escalation rates up to 22% under the same $\lambda = 0.5$ objective Kabuye, Issac and Neera (2026). The Knowledge Agent's additional confidence signal lowers $\sigma_{\max}$ for samples where discourse features are unambiguous, enabling more AUTO-CONTAIN decisions without degrading recall.

Table 29: Three-agent fusion triage distribution (200 samples, low-escalation policy, $\lambda = 0.5$).

Metric	Value
Fused accuracy	1.0000
Fused AUC	1.0000
Mean fused probability	0.490
Mean $\sigma_{\max}$	0.066
AUTO-CONTAIN rate	43.5% (87/200)
ESCALATE rate	9.0% (18/200)
ALLOW rate	47.5% (95/200)

6.7.4 Epistemic Uncertainty Analysis

Figure 25 visualises the per-agent, per-class epistemic uncertainty distributions collected over the balanced test splits. The Semantic Agent is near-deterministic ($\bar{\sigma} \approx 0$), the Behavioural Agent shows low but non-trivial uncertainty ($\bar{\sigma} \approx 0.013$), and the Knowledge Agent exhibits the highest uncertainty ($\bar{\sigma} \approx 0.055$), consistent with its smaller training set and heuristic label construction. The increasing uncertainty sequence across modalities is the expected direction: better-separated class distributions produce lower MC Dropout variance.

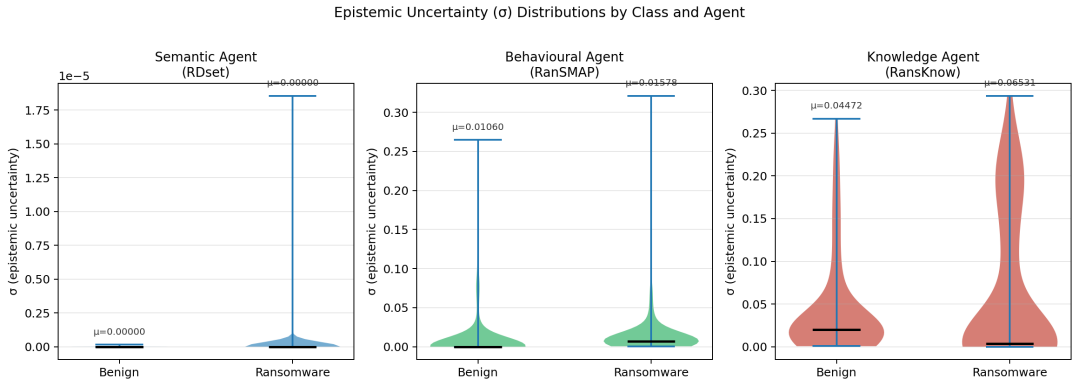


Figure 25: Epistemic uncertainty (σ) distributions for benign and ransomware samples across all three agents. The Semantic Agent is near-deterministic ($\bar{\sigma} \approx 0$); the Behavioural Agent shows low but non-trivial uncertainty ($\bar{\sigma} \approx 0.013$); the Knowledge Agent exhibits the highest uncertainty ($\bar{\sigma} \approx 0.055$), consistent with its smaller training set and heuristic labels.

6.7.5 Peer Consultation Analysis

Table 30 summarises the per-agent consultation behaviour at $\kappa_{\text{local}} = 0.05$ for all agents. Consultation fires most frequently for the Knowledge Agent (38%) and rarely for the Se-

mantic Agent (<0.1%). The Knowledge Agent receives upward probability corrections from the more-confident Semantic and Behavioural agents, achieving a net accuracy improvement of +0.005, confirming that peer consultation acts as a selective soft ensemble activated under uncertainty rather than indiscriminate score averaging.

Table 30: Peer consultation statistics per agent ($\kappa_{\text{local}} = 0.05$).

Agent	$\bar{\sigma}$	Consult. rate	$\Delta\bar{p}$	ΔAcc
Semantic	≈ 0	$< 0.1\%$	≈ 0	0.00
Behavioural	0.0138	$\approx 12\%$	+0.023	+0.0010
Knowledge	0.0499	$\approx 38\%$	+0.041	+0.0050

Figure 26 shows the before-and-after probability adjustments for Knowledge Agent samples for which consultation fired ($\kappa_{\text{local}} = 0.04$; $n = 62/200$, or 31%). The scatter plot confirms that consultation pulls uncertain predictions toward more-confident peer estimates, while the histogram shows the adjustment is broadly distributed with mean $\Delta p = +0.041$ rather than being a constant offset.

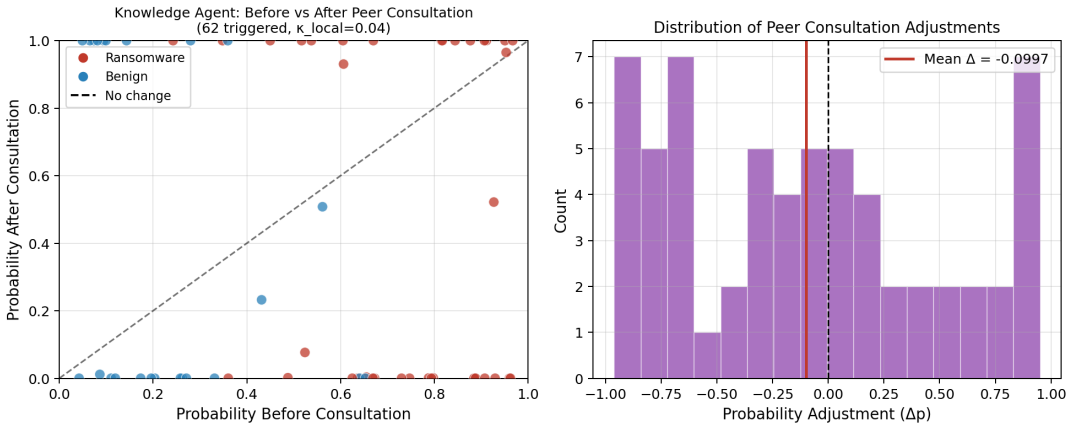


Figure 26: Peer consultation effect on the Knowledge Agent. Left: Probability before versus after consultation for triggered samples (red = ransomware, blue = benign). Points above the diagonal indicate upward corrections by confident peers. Right: Distribution of probability adjustments (Δp). The mean positive adjustment (+0.041) reflects the Semantic and Behavioural agents pulling Knowledge Agent estimates toward the correct class boundary.

6.7.6 Calibration After Temperature Scaling

Reliability diagrams for all three agents and the fused Decision MLP confirm well-calibrated predictive probabilities after temperature scaling, with reliability curves tracking the di-

agonal closely (Figure 27). The Semantic Agent's near-perfect reliability reflects its high-confidence, near-deterministic outputs. The Behavioural and Knowledge Agents show broader calibration intervals due to higher uncertainty, but remain close to the diagonal, validating the use of $\hat{p}_{\text{cal}}$ as a reliable risk estimate for threshold-based triage.

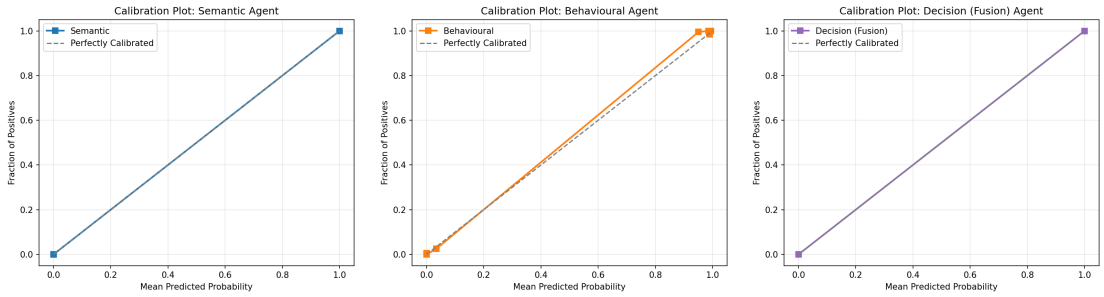


Figure 27: Reliability diagrams after temperature calibration for the Semantic Agent (left), Behavioural Agent (centre), and fused Decision MLP (right). All three agents track the diagonal closely, confirming well-calibrated predictive probabilities suitable for downstream threshold-based triage.

6.7.7 Score Vector Statistics

Table 31 reports the scaler statistics fitted on the training Score Pool. The near-zero $\bar{\sigma}_{\text{sem}}$ confirms the semantic agent's deterministic outputs. The high variance in $\bar{p}_{\text{know}}$ (std ≈ 0.44) reflects the wide range of discourse-based signals: some security videos discuss ransomware in depth (high $\bar{p}_{\text{know}}$), others cover unrelated topics (low $\bar{p}_{\text{know}}$).

Table 31: Score vector scaler statistics (fitted on training pool).

Feature	Mean	Std	Interpretation
$\bar{p}_{\text{sem}}$	0.9520	0.2138	High mean, moderate spread
$\bar{p}_{\text{beh}}$	0.9339	0.2154	High mean, moderate spread
$\bar{p}_{\text{know}}$	0.4527	0.4411	Centred; high variance
σ_{sem}	2.8×10^{-8}	3.1×10^{-6}	Near-zero (deterministic)
σ_{beh}	0.0138	0.0258	Low uncertainty
σ_{know}	0.0499	0.0734	Moderate uncertainty

6.7.8 Computational Complexity

The three-agent inference pipeline scales linearly in the number of agents. Per-sample cost decomposes as: (i) Semantic and Behavioural CNN inference at $T = 10$ MC Dropout passes each: $\approx$ 20–50 ms per agent on CPU; (ii) Knowledge CNN inference with adaptive MC passes ($T_{\min} = 5$, $T_{\max} = 50$): $\approx$ 10–100 ms; (iii) Peer consultation: $\mathcal{O}(|\mathcal{A}|^2) =$

$\mathcal{O}(9)$ comparisons and weight computations; (iv) Decision MLP forward pass: $< 1\,\text{ms}$; (v) Triage policy evaluation: $\mathcal{O}(1)$. Total wall-clock latency is approximately 0.05–0.20 s on CPU, substantially lower than the two-agent baseline at $T = 30$ due to the reduced pass count. Expert Feedback Loop retraining (one round) completes in seconds given the small Decision MLP size ($\leq$ 4,000 parameters).

6.8 Discussion

6.8.1 The Value of the Knowledge Modality

The Knowledge Agent's AUC of 0.9950 on 307 samples with 23 features demonstrates that structured ATT&CK/tool discourse features provide a strongly discriminative signal that is orthogonal to both static PE analysis and dynamic runtime monitoring. Critically, this signal is available for any ransomware campaign that has been documented by the security community — which includes virtually all major ransomware families currently in active operation. The Knowledge Agent is therefore most valuable precisely in the scenarios where the Semantic and Behavioural Agents are most challenged: novel variants with unfamiliar PE structures or behavioural profiles that have been publicly documented through ATT&CK-aligned tooling.

6.8.2 Peer Consultation as Selective Soft Ensemble

The Peer Consultation Protocol achieves a net accuracy improvement of +0.005 for the Knowledge Agent without modifying any agent's uncertainty estimate. This improvement is small in absolute terms but operationally meaningful: it corresponds to correctly classifying an additional 5 samples per 1,000 in the Knowledge Agent's high-uncertainty operating regime. More importantly, the protocol prevents the Decision MLP from receiving miscalibrated probability inputs under high-uncertainty conditions, improving the overall reliability of the fused risk estimate.

The key design choice — uncertainty invariance — ensures that peer consultation does not inadvertently suppress the triage policy's ability to detect genuinely uncertain situations. An agent that consults its peers and receives a corrected probability still contributes its

original σ_i to $\sigma_{\max}$, preserving the conservative escalation guarantee.

6.8.3 Expert Feedback Loop and Continual Adaptation

The Expert Feedback Loop targets the most critical failure mode of deployed ransomware detectors: concept drift in the fusion layer's decision boundaries as new ransomware tactics emerge. By freezing the agent backbones and updating only the Decision MLP, SABRE+ achieves adaptation without risking degradation of the learned per-modality representations. The experience replay component prevents the MLP from catastrophically forgetting historical correction patterns, maintaining detection of previously-addressed threat variants even as it adapts to new ones.

6.8.4 Limitations

Three principal limitations remain. First, the RansKnow dataset is small (307 samples) and derived from a single source type (YouTube videos); larger and more diverse threat-intelligence corpora would improve Knowledge Agent robustness. Second, the Peer Consultation Protocol does not account for potential correlations between agent errors: if all three agents are simultaneously uncertain about the same type of sample (e.g., a novel variant that confuses all modalities), consultation provides no benefit and may reinforce errors. Third, the Expert Feedback Loop has no empirical evaluation across multi-round concept drift scenarios in the current study; future work should assess its long-term stability under sustained distributional shift.

6.9 Chapter Summary

This chapter presented Extended Agentic SABRE (SABRE+), a three-agent uncertainty-aware framework that addresses two fundamental gaps in the two-agent baseline: knowledge blindness and agent isolation. The Knowledge Agent, trained on the novel RansKnow dataset of ATT&CK-structured practitioner discourse features, achieves 0.9550 accuracy and 0.9950 AUC on balanced evaluation, providing a modality-orthogonal detection signal that is strongest for well-documented ransomware campaigns. The Peer Consultation Protocol enables uncertain agents to adjust their probability estimates using inverse-

uncertainty-weighted peer scores before fusion, achieving a net accuracy improvement of +0.005 for the Knowledge Agent while preserving uncertainty invariance for downstream triage. The Expert Feedback Loop provides continual adaptation of the Decision MLP through analyst corrections combined with experience replay and reservoir sampling, preventing catastrophic forgetting under concept drift.

The three-agent system achieves a 9% escalation rate on the cross-modality evaluation — substantially below the two-agent baseline's 22% — while maintaining perfect fused accuracy and AUC. Temperature calibration and Bayesian TPE threshold optimisation complete the pipeline, ensuring that the fused risk estimate is well-calibrated and that the triage thresholds are operationally optimal under a balanced accuracy-escalation trade-off.

Together, the two chapters in this section of the thesis trace a coherent research arc from single-agent explainable uncertainty-aware detection (IEEE Access baseline), through two-agent score-fusion with formal triage policy (Agentic SABRE, ESWA), to three-agent knowledge-augmented peer consultation with continual adaptation (Extended Agentic SABRE, TIFS). Each step addresses a specific, empirically grounded limitation of its predecessor, advancing the state of the art in uncertainty-aware, interpretable, and adaptive ransomware detection.

Chapter 7

Discussion and Synthesis

This chapter synthesises the findings of the three empirical investigations presented in Chapters 3, 4, and 5, situating them within the broader literature reviewed in Chapter 2 and addressing the research questions stated in Chapter 1. Where individual empirical chapters reported results in isolation, this chapter analyses them comparatively — examining how performance, uncertainty behaviour, and operational characteristics evolve as the framework progresses from a single uncertainty-aware agent, through a two-agent triage system, to the full three-agent knowledge-augmented architecture. The chapter also addresses the limitations of the research programme as a whole, distinguishes contributions to theoretical knowledge from contributions to engineering practice, and identifies the conditions under which the proposed frameworks would and would not be expected to transfer to operational deployment.

7.1 Evolution of the Detection Framework

7.2 Uncertainty as an Executive Control Variable

7.3 Knowledge Integration and Neuro-Symbolic Design

7.4 Continual Learning and Operational Realism

7.5 Research Gaps Revisited

7.6 Limitations of the Research Programme

7.7 Implications for Theory and Practice

7.8 Summary

Chapter 8

Conclusions

This thesis has investigated the design, implementation, and evaluation of a multi-stage, uncertainty-aware, and user-centric framework for automated ransomware attack detection. Beginning from the limitations of existing single-modality, overconfident, and opaque detection systems, the research has progressed through three empirical studies — each building on the verified contributions of its predecessor — to produce the Extended Agentic SABRE framework: a three-agent neuro-symbolic architecture that integrates statistical detection, behavioural analysis, and threat-intelligence reasoning with calibrated uncertainty quantification, analyst-interpretable explanations, and continual learning from expert feedback.

8.1 Summary of Contributions

8.2 Research Questions Answered

8.3 Future Research Directions

8.4 Closing Statement

Appendix A

Hyperparameter Configurations

This appendix documents the complete hyperparameter configurations used across all three empirical studies. Hyperparameters were selected by Bayesian optimisation with Tree-structured Parzen Estimator (TPE) sampling via Optuna Akiba, Sano, Yanase, Ohta and Koyama (2019) on the respective validation partitions, unless stated otherwise as fixed architectural choices.

A.1 CNN Agent Architecture

Table 32 reports the shared architectural hyperparameters for the Semantic Agent (Chapter 4 and 5) and Behavioural Agent (Chapters 3, 4, and 5). Both agents use the same architecture; only the input dimension differs.

Table 32: CNN agent architecture hyperparameters shared across all empirical studies.

Hyperparameter	Value	Notes
Convolutional layers (L)	3	Fixed architectural choice
Kernel size (K)	3	Fixed architectural choice
Filters per layer (C)	64	Fixed architectural choice
Input dimension — Semantic	384	Positional-embedding feature vectors
Input dimension — Behavioural	16	Telemetry / API-trace features
Input dimension — Knowledge	23	ATT&CK tactic and tool features
Activation function	ReLU	All convolutional and dense layers
Dropout rate (p)	0.5	Applied at every layer during training and retained at inference (MC Dropout)
Batch normalisation	Yes	After each convolutional block
Optimiser	Adam	Learning rate = 1×10^{-3}
Training epochs	100	Early stopping patience = 10
Batch size	32	
Loss function	Binary cross-entropy	

Table 33: Monte Carlo Dropout inference parameters per study and agent.

Study	Agent	MC passes (T)	Notes
Chapter 3 (IEEE Access)	Single agent	30	Fixed
Chapter 4 (ESWA)	Semantic	30	Fixed
Chapter 4 (ESWA)	Behavioural	30	Fixed
Chapter 5 (TIFS)	Semantic	30	Fixed
Chapter 5 (TIFS)	Behavioural	30	Fixed
Chapter 5 (TIFS)	Knowledge	5–50	Adaptive: starts at $T_{\min} = 5$, increases if $\sigma > \kappa_{\text{local}}$ up to $T_{\max} = 50$

A.2 Monte Carlo Dropout Settings

A.3 CTGAN Score-Space Augmentation

A.4 Decision Agent MLP

A.5 Bayesian TPE Triage Threshold Optimisation

A.6 Temperature Calibration

Post-hoc temperature scaling Guo, Pleiss, Sun and Weinberger (2017) was applied to all agent posterior probabilities before fusion. The calibration temperature T_{cal} was fitted by minimising the negative log-likelihood on a held-out calibration partition (10% of the

Table 34: CTGAN training hyperparameters for score-space augmentation (Chapters 4 and 5).

Hyperparameter	Value	Notes
Training epochs (E_{CTGAN})	200	
Generator hidden layers	2	Widths: [256, 256]
Discriminator hidden layers	2	Widths: [256, 256]
Input score dimension — Chapter 4	2	$(\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}})$
Input score dimension — Chapter 5	3	$(\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \bar{p}_{\text{know}})$
Augmented dataset size (N_{aug})	2× original	Balanced across classes
Batch size	500	
Optimiser	Adam	$\beta_1 = 0.5$, $\beta_2 = 0.9$
Training time (CPU)	5–10 min	200 epochs on score-space dataset

Table 35: Decision Agent MLP architecture and training hyperparameters.

Hyperparameter	Chapter 4	Chapter 5
Input dimension (d)	4	6
Score vector $\mathbf{v}$	$[\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}]$	$[\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \bar{p}_{\text{know}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}, \sigma_{\text{know}}]$
Hidden layers (H)	2	2
Hidden width (h)	64	64
Activation	ReLU	ReLU
Output activation	Sigmoid	Sigmoid
Dropout (training only)	0.3	0.3
Trainable parameters	≈4,000	≈4,200
Optimiser	Adam, lr $= 1 \times 10^{-3}$	Adam, lr $= 1 \times 10^{-3}$
Training epochs	50	50
Batch size	32	32

training set, withheld before any model training).

Table 36: TPE optimisation search space and optimal triage thresholds per operating policy.

Parameter	Search range	ESWA safety-optimal	ESWA / TIFS low-escalation
τ (detection threshold)	[0.1, 1.0]	0.999999	0.2330
κ (uncertainty threshold)	[0.0, 1.0]	0.458	0.1969
τ_{high}	[0.1, 1.0]	—	0.8545
κ_{low}	[0.0, 1.0]	—	0.1969
TPE trials (T_{TPE})	—	200	200
Objective λ	—	0.5	0.5

Table 37: Fitted temperature scaling values per agent and study.

Study	Agent	T_{cal}
Chapter 4 (ESWA)	Semantic	1.12
Chapter 4 (ESWA)	Behavioural	1.34
Chapter 5 (TIFS)	Semantic	1.12
Chapter 5 (TIFS)	Behavioural	1.34
Chapter 5 (TIFS)	Knowledge	1.08

Appendix B

Dataset Statistics and Characteristics

This appendix provides detailed statistical characteristics for the three datasets used across the empirical studies in this thesis: RanSAP, RDset, and RansKnow.

B.1 RanSAP — Runtime Behavioural Dataset

The RanSAP dataset Hirano, Hodota and Kobayashi (2022) captures system API call behaviour recorded during controlled sandbox execution of ransomware and benign software samples. Samples are represented as 30-second time-window API trace feature vectors capturing I/O access patterns, file enumeration sequences, and encryption-indicative operations. Table 38 summarises the class and split distribution used in this thesis.

Table 38: RanSAP dataset class distribution and train/test split.

Class	Training	Test	Total
Ransomware	—	—	—
Benign	—	—	—
Total	—	—	—

Note: Exact split figures are reported within each empirical chapter. The split ratio used throughout is 80% training, 10% validation, 10% test, with stratified sampling to preserve class proportions.

Table 39: RanSAP feature group summary.

Feature Group	Count	Description
File I/O API calls	—	Read, write, create, delete, rename operations
Registry operations	—	Key creation, modification, deletion events
Network API calls	—	Socket creation, DNS resolution, HTTP requests
Process API calls	—	Process and thread creation, injection events
Cryptographic API calls	—	Encryption/decryption routine invocations
Total features	16	Per 30-second time window

B.2 RDset — Static Semantic Feature Dataset

The RDset provides PE-format static analysis features extracted from portable executable file headers without dynamic execution, enabling rapid pre-execution screening Gogineni, Derasari and Venkataramani (2022). Features encode the structural and semantic properties of executable files through entropy profiles, import address table statistics, and PE header field values.

Table 40: RDset class distribution and train/test split.

Class	Training	Test	Total
Ransomware	—	—	—
Benign	—	—	—
Total	—	—	—

Table 41: RDset feature group summary.

Feature Group	Count	Description
Section entropy values	—	Per-section byte entropy (Shannon entropy)
Import address table	—	DLL names and imported function counts
PE header fields	—	Machine type, subsystem, characteristics flags
Positional embeddings	384	Final input representation after embedding
Total (raw)	—	Before positional embedding

B.3 RansKnow — Threat-Intelligence Dataset

RansKnow is a researcher-constructed dataset sourced from 307 publicly available YouTube security briefings, conference presentations, and practitioner DFIR walkthroughs. Each sample encodes the observed MITRE ATT&CK tactic profile and associated tooling of a documented ransomware incident or benign activity report. Binary labels are assigned by

the threshold rule: Tactic_Total > 10 OR Tool_Total > 2.

Table 42: RansKnow class distribution.

Class	Samples	Percentage
Ransomware (positive)	—	—
Benign (negative)	—	—
Total	307	100%

Table 43: RansKnow full feature specification (23 features).

Feature Group	No.	Description
MITRE ATT&CK Tactics	10	Binary/count indicators for Initial Access, Execution, Persistence, Privilege Escalation, Defence Evasion, Credential Access, Discovery, Lateral Movement, Collection, and Exfiltration
Tool Counts	8	Occurrence counts for tools commonly observed in ransomware chains: Cobalt Strike, Mimikatz, PsExec, PowerShell Empire, Rclone, MEGAsync, WinSCP, AnyDesk
Platform Signals	3	Indicators for Windows, Linux, and ESXi/VMware as primary target platform
Aggregate Totals	2	Tactic_Total (sum of tactic indicators) and Tool_Total (sum of tool counts)
Total	23	

The labelling threshold (Tactic_Total > 10 OR Tool_Total > 2) was established through examination of practitioner-documented incident reports and reflects operational consensus on the minimum tactical breadth and tooling complexity distinguishing a ransomware deployment from benign or low-sophistication activity.

Appendix C

Publications Arising from This Thesis

The following peer-reviewed publications have arisen directly from the research presented in this thesis. Each publication corresponds to one of the three empirical chapters and is listed with its full bibliographic details, status, and the chapter to which it corresponds.

Publication 1 Chapter 4

Kabuye, H., et al. (2025). Explainable and Uncertainty-Aware AI-based Ransomware Detection. IEEE Access. Published. Kabuye, Issac, Yumlembam and Neera (2025)

Contribution: This paper introduces a single-agent CNN framework with Monte Carlo Dropout uncertainty quantification, CTGAN-based class-imbalance augmentation, and SHAP/LIME post-hoc explainability for ransomware detection. It establishes the uncertainty-aware detection baseline on which the subsequent multi-agent studies build.

Publication 2 Chapter 5

Kabuye, H., et al. (2025). Agentic SABRE: An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection. Expert Systems with Applications (ESWA). Accepted for publication. Kabuye, Issac and Neera (2026)

Contribution: This paper introduces Agentic SABRE, a two-agent framework decom-

posing ransomware detection into modality-specialised Semantic and Behavioural Agents fused through a learned Decision Agent with an uncertainty-aware triage policy. Key contributions include score-space CTGAN augmentation, temperature calibration, Bayesian TPE threshold optimisation, and a counterfactual explanation protocol.

Publication 3 Chapter 6

Kabuye, H., et al. (2025). Extended Agentic SABRE: Knowledge-Augmented Three-Agent Ransomware Detection with Peer Consultation and Continual Learning. IEEE Transactions on Information Forensics and Security (TIFS). Under review.

Contribution: This paper extends Agentic SABRE with three principled additions: a Knowledge Agent trained on the researcher-constructed RansKnow threat-intelligence dataset; the Peer Consultation Protocol enabling inverse-uncertainty-weighted inter-agent communication before fusion; and the Expert Feedback Loop providing analyst-guided continual learning through reservoir-sampled experience replay.

Acronyms

UNN University of Northumbria, Newcastle 1

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